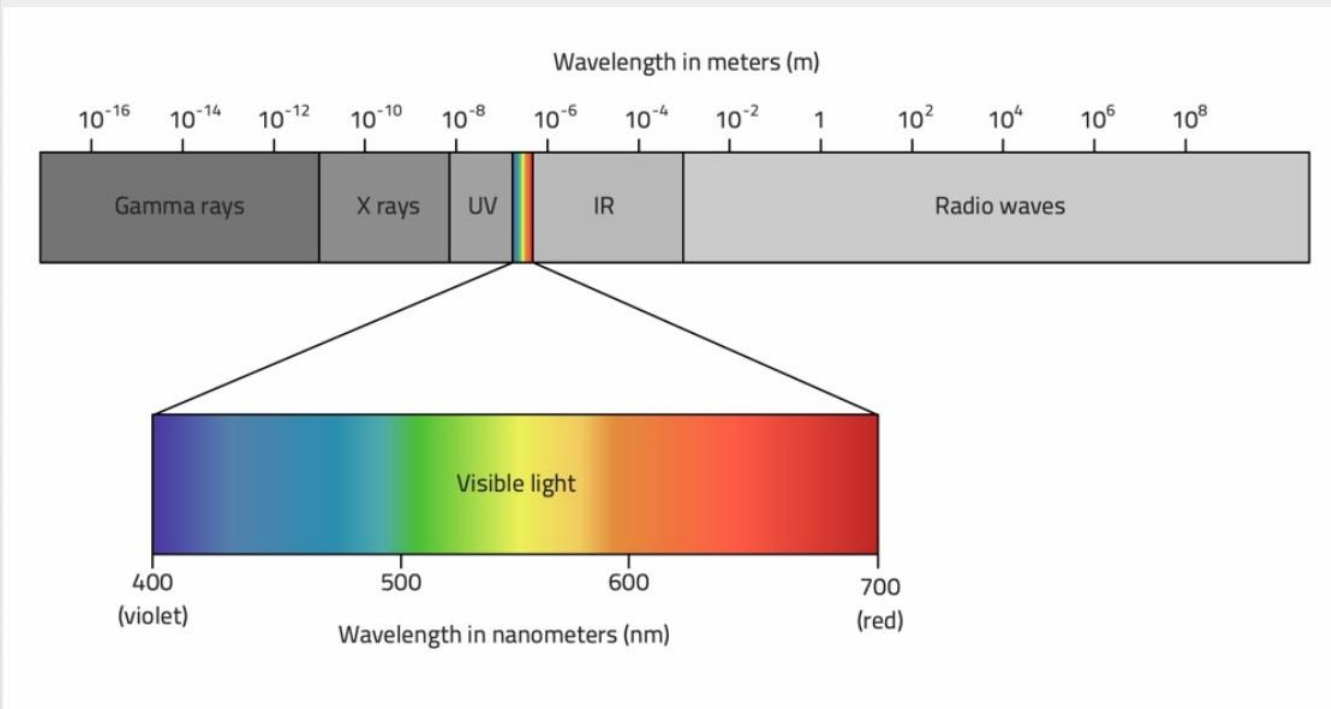
Week 10.

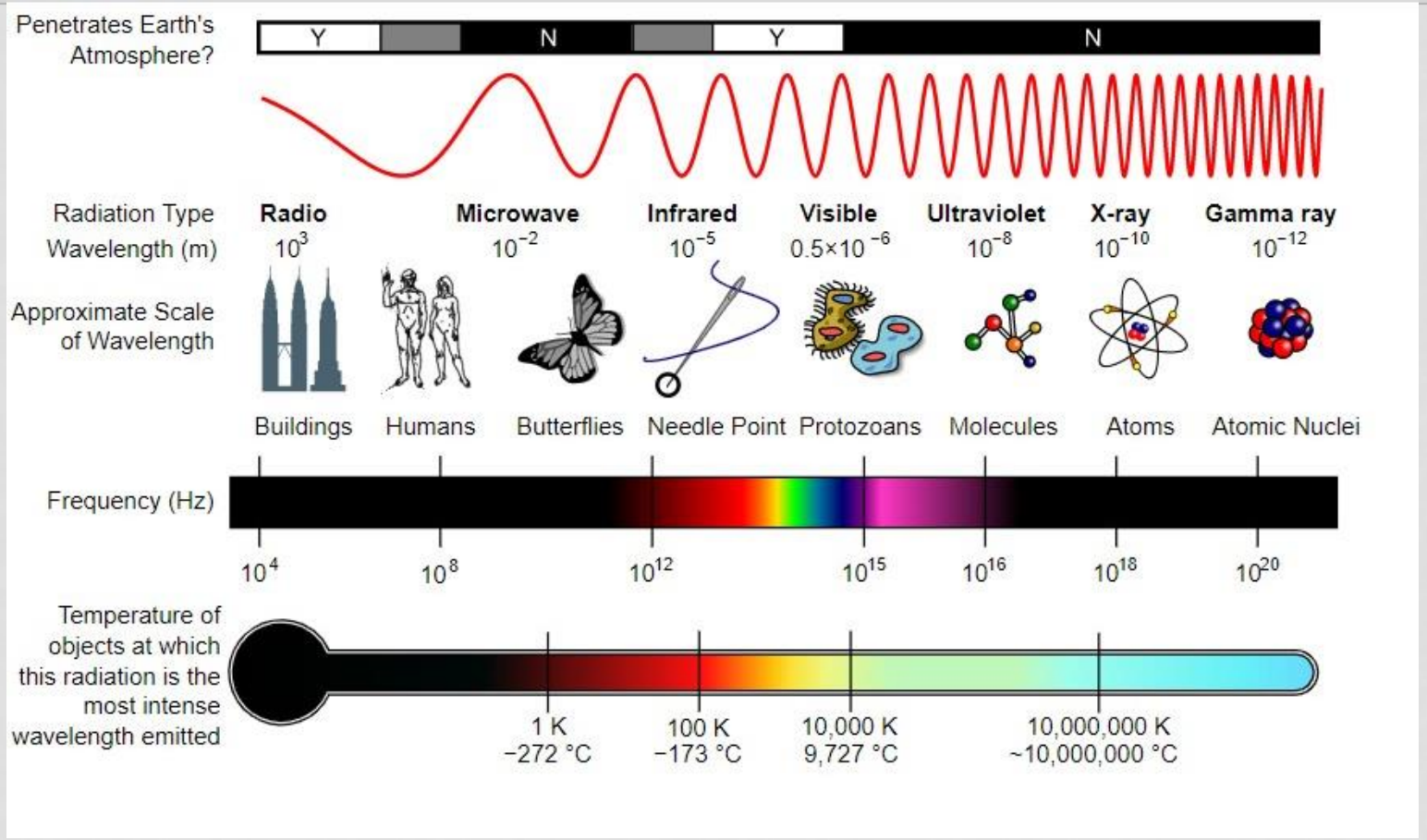
ATOMIC PHYSICS: (SOME) APPLICATIONS

ANASTASIA BORSCHEVSKY

EM SPECTRUM


Wavelength in meters (m)
10^{-16} 10^{-14} 10^{-12} 10^{-10} 10^{-8} 10^{-6} 10^{-4} 10^{-2} 1 10^{2} 10^{4} 10^{6} 10^{8}
Gamma rays
X rays
UV
IR
Radio waves
Visible light
400
(violet)
500
600
700
(red)
Wavelength in nanometers (nm)

EM SPECTRUM



By Inductiveload, NASA - self-made, information by NASABased off of File:EM Spectrum3-new.jpg by NASAThe butterfly icon is from the P icon set, File:P biology.svgThe humans are from the Pioneer plaque, File:Human.svgThe buildings are the Petronas towers and the Empire State Buildings, both from File:Skyscrapercompare.svg, CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=2974242

UNITS

Units:

- λ: usually [nm]
- $f=\upsilon=c/\lambda$ [Hz] $(\omega=2\pi\upsilon)$
- Wavenumber: $\tilde{\nu}=1/\lambda$, m^{-1}/**cm^{-1}** (also used as energy unit ($E=hc\tilde{\nu}$); 1 cm^{-1}=1/219474 a.u= 1/8066 eV)

- For example, 1s→2p transition in H

$\lambda = 121.567$ nm

$\tilde{\nu} = 82259\ cm^{-1}$

$\nu = 246604$ GHz

$E = 10.198753$ eV

EM SPECTRUM

	Excitations	λ	f [Hz]	E [eV]
γ-rays	Nuclear	< 1pm	> 10^{18}	> 1 MeV
X-rays	Atomic core	<10 nm	> 10^{16}	> KeV
UV (EUV, VUV)	Electronic	10-400 nm	10^{15}-10^{16}	3-100 eV
Visible	Electronic	400-750 nm	10^{14}	1.5-3 eV
IR	Vibrational	750 nm-25μm	10^{12}-10^{14}	0.001-1.5 eV
microwave	Rotational	25 μm – 1 mm	10^{10}-10^{12}	1 μeV-1 meV
radio	Hyperfine	> 1mm	KHz-MHz	< μeV

Units:

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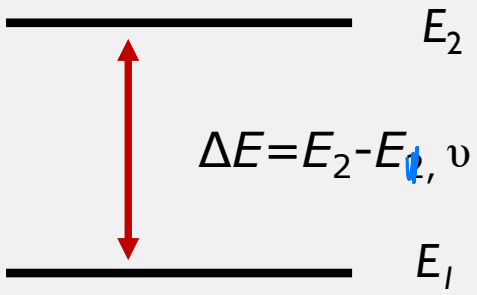
EM SPECTRUM

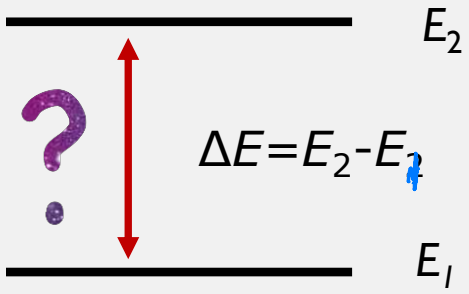
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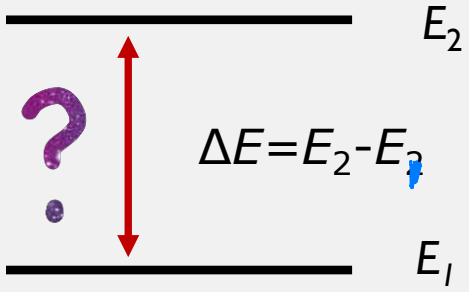
- λ: usually [nm]
- f=c/λ [Hz]
- Wavenumber: v=1/λ, m^{-1}/**cm^{-1}** (also used as energy unit (E=hc v); 1 cm^{-1}=1/219474 a.u= 1/8066 eV)

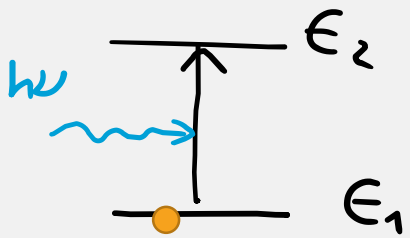
TRANSITIONS IN ATOMS


E_2
$\Delta E = E_2 - E_1, \upsilon$
E_1


E_2
$\Delta E = E_2 - E_1$
E_1

1). Absorption

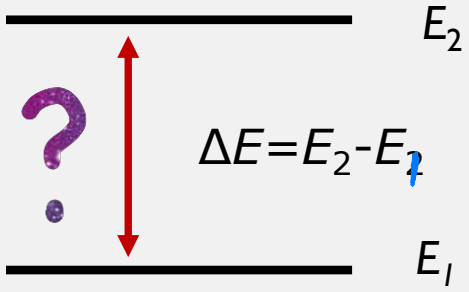


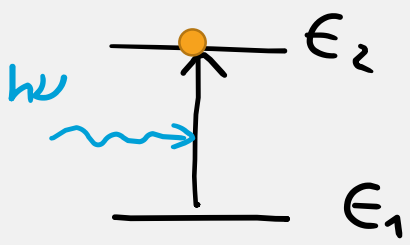


$h\nu = \Delta\epsilon$

Generally, the frequency of the radiation (laser) has to be very close to the frequency of the transition (corresponding to ΔE) in order for absorption to happen.
The closer, the more likely the transition.

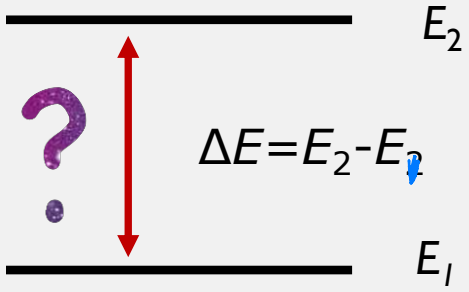
1). Absorption



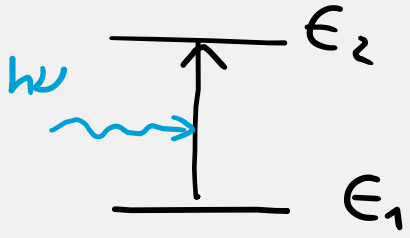


$h\nu = \Delta\epsilon$

Generally, the frequency of the radiation (laser) has to be very close to the frequency of the transition (corresponding to ΔE) in order for absorption to happen.
The closer, the more likely the transition.



1). Absorption



$h\nu = \Delta\epsilon$

$$\frac{dP_{12}^{abs}}{dt} = B_{12}\rho(\nu)$$

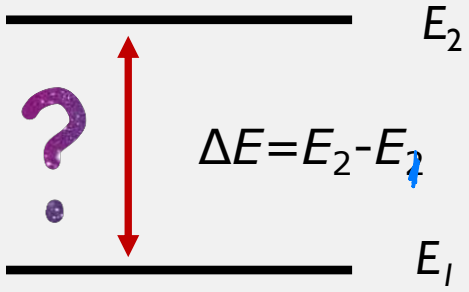
$$D_{12} = \langle \psi_1 | \vec{r} | \psi_2 \rangle$$

$$B_{12} = \frac{\pi e^2 |D_{12}|^2}{3 \epsilon_0 \hbar^2}$$

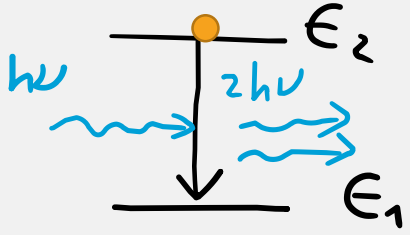
$$\rho(\nu) = n(\nu) \cdot h\nu$$

e- charge of the electron
E_0- electric field amplitude
$\rho(\upsilon)$ – radiation density
We assume that we are on resonance, i.e. the laser frequency is equal to the frequency of the transition

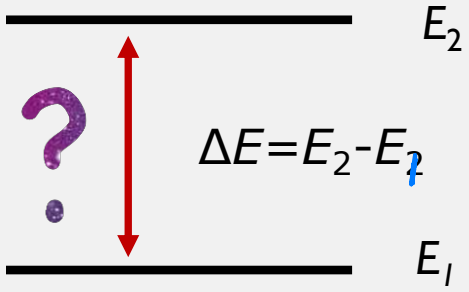
B_{12} – Einstein coefficient for absorption



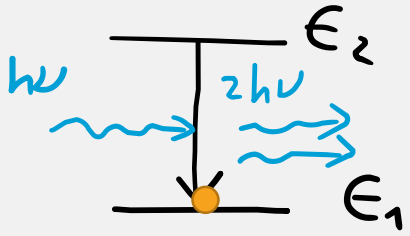
2). Stimulated emission



$h\nu = \Delta\epsilon$

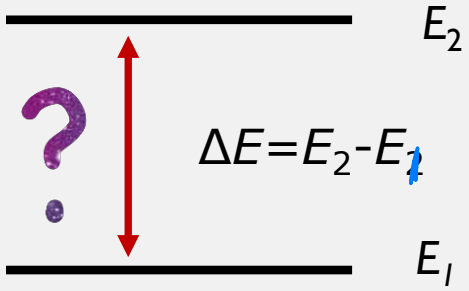


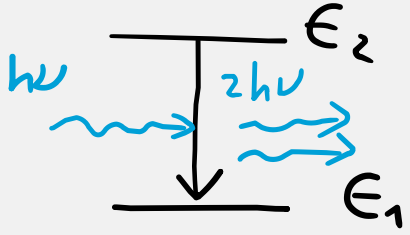
2). Stimulated emission



$h\nu = \Delta\epsilon$

2). Stimulated emission





$h\nu = \Delta\epsilon$

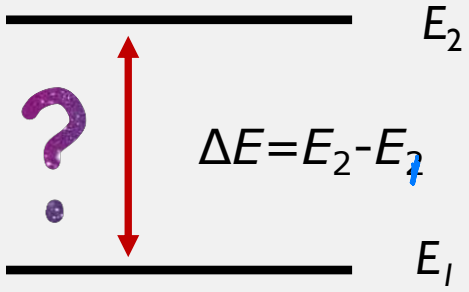
$$\frac{dP_{21}^{ind}}{dt} = B_{21}\rho(\nu)$$

$$D_{12} = \langle \psi_1 | \vec{r} | \psi_2 \rangle$$

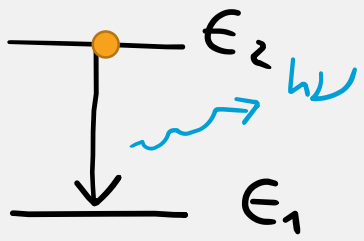
$$B_{21} = \frac{\pi e^2 |D_{12}|^2}{3\epsilon_0 \hbar^2} \cdot \frac{g_1}{g_2} = B_{12} \cdot \frac{g_1}{g_2}$$

e- charge of the electron
E_0- electric field amplitude
$g_i=2J_i+1$ – degeneracy of state I
We assume that we are on resonance, i.e. the laser frequency is equal to the frequency of the transition

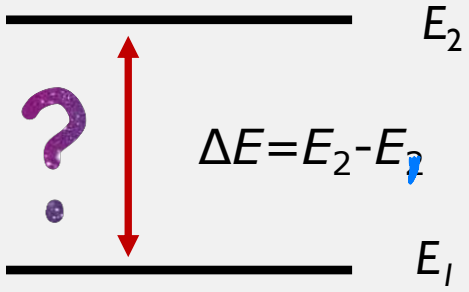
B_{21} – Einstein coefficient for stimulated emission



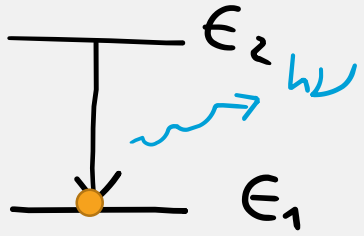
3). Spontaneous emission



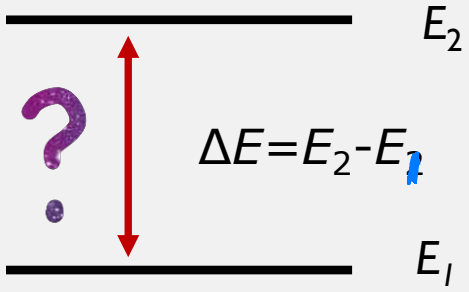
$h\nu = \Delta\epsilon$



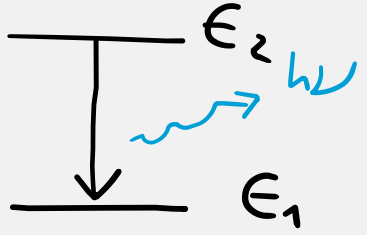
3). Spontaneous emission



$h\nu = \Delta\epsilon$



3). Spontaneous emission



$h\nu = \Delta\epsilon$

$$\frac{dP_{21}^{sp}}{dt} = A_{21}$$

$$D_{12} = \langle \Psi_1 | \vec{r} | \Psi_2 \rangle$$

$$A_{21} = \frac{4\alpha}{3c^2}\omega^3 |D_{12}|^2 \cdot \frac{g_1}{g_2} = \frac{\hbar\omega^3}{\pi^2 c^3} B_{21}$$

e- charge of the electron
E_0- electric field amplitude
ω- radial frequency of the transition, $\omega=2\pi\upsilon$
$g_i=2J_i+1$ – degeneracy of state i
c-speed of light
α-fine structure constant (1/137)

A_{12} – Einstein coefficient for spontaneous emission

$$B_{12} = \frac{\pi e^2 |D_{12}|^2}{3 \epsilon_0 \hbar^2}$$

$$B_{21} = \frac{\pi e^2 |D_{12}|^2}{3 \epsilon_0 \hbar^2} \cdot \frac{g_1}{g_2}$$

$$\left[\frac{m^3}{J \cdot sec^2}\right]$$

$$A_{21} = \frac{4\alpha}{3c^2} \omega^3 |D_{12}|^2 \cdot \frac{g_1}{g_2}$$

$$\left[\frac{1}{sec}\right]$$

$$B_{12} = \frac{\pi e^2 |D_{12}|^2}{3 \epsilon_0 \hbar^2}$$

$$B_{21} = \frac{\pi e^2 |D_{12}|^2}{3 \epsilon_0 \hbar^2} \cdot \frac{g_1}{g_2}$$

$$\left[\frac{m^3}{J \cdot sec^2}\right]$$

$$A_{21} = \frac{4\alpha}{3c^2} \omega^3 |D_{12}|^2 \cdot \frac{g_1}{g_2}$$

$$\left[\frac{1}{sec}\right]$$

$$\frac{\mathrm{d}N_2}{\mathrm{d}t} = N_1 B_{12} \rho(\omega_{12}) - N_2 B_{21} \rho(\omega_{12}) - N_2 A_{21}$$

$$\frac{\mathrm{d}N_1}{\mathrm{d}t} = -\frac{\mathrm{d}N_2}{\mathrm{d}t}$$

(for 2-state systems only)

$$A_{21} = \frac{4\alpha}{3c^2}\omega^3 |D_{12}|^2 \cdot \frac{g_1}{g_2}$$

$$\left[\frac{1}{sec}\right]$$

$$\frac{\mathrm{d}N_2}{\mathrm{d}t} = N_1 B_{12}\rho(\omega_{12}) - N_2 B_{21}\rho(\omega_{12}) - N_2 A_{21}$$

$$N_2(t) = N_2(0)\mathrm{e}^{-A_{21}t}$$

$$A_{21} = \frac{4\alpha}{3c^2}\omega^3 |D_{12}|^2 \cdot \frac{g_1}{g_2}$$

$$\left[\frac{1}{\text{sec}}\right]$$

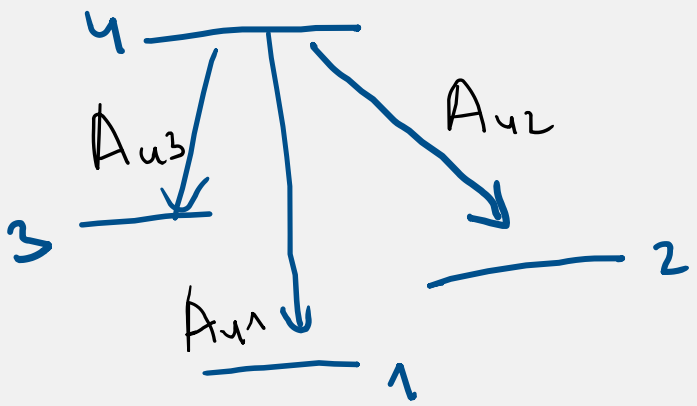
$$\frac{\mathrm{d}N_2}{\mathrm{d}t} = N_1 B_{12}\rho(\omega_{12}) - N_2 B_{21}\rho(\omega_{12}) - N_2 A_{21}$$

$$N_2(t) = N_2(0)\mathrm{e}^{-A_{21}t}$$

Excited state lifetime $\tau = 1/A_{21}$

$$A_{21} = \frac{4\alpha}{3c^2}\omega^3 |D_{12}|^2 \cdot \frac{g_1}{g_2}$$

$$\left[\frac{1}{\text{sec}}\right]$$



$$\frac{dN_2}{dt} = N_1 B_{12}\rho(\omega_{12}) - N_2 B_{21}\rho(\omega_{12}) - N_2 A_{21}$$

$$N_2(t) = N_2(0)e^{-A_{21}t}$$

$$\tau = \frac{1}{A_{41} + A_{42} + A_{43}} = \frac{1}{\sum_i A_{4i}}$$

Excited state lifetime $\tau = 1/A_{21}$

$$A_{21} = \frac{4\alpha}{3c^2}\omega^3 |D_{12}|^2 \cdot \frac{g_1}{g_2}$$

$$\left[\frac{1}{\text{sec}}\right]$$

$$\frac{\mathrm{d}N_2}{\mathrm{d}t} = N_1 B_{12}\rho(\omega_{12}) - N_2 B_{21}\rho(\omega_{12}) - N_2 A_{21}$$

$$N_2(t) = N_2(0)\mathrm{e}^{-A_{21}t}$$

Excited state lifetime $\tau = 1/A_{21}$

Lifetime of excited state with:

$|D_{12}|=10^{-10}$ m

$\lambda_{21}=500$ nm

$g_1=g_2$

$$A_{21} = \frac{4\alpha}{3c^2}\omega^3|D_{12}|^2 \cdot \frac{g_1}{g_2}$$

$$\left[\frac{1}{sec}\right]$$

$$\frac{dN_2}{dt} = N_1 B_{12}\rho(\omega_{12}) - N_2 B_{21}\rho(\omega_{12}) - N_2 A_{21}$$

$$N_2(t) = N_2(0)e^{-A_{21}t}$$

Excited state lifetime $\tau=1/A_{21}$

Excited state lifetime $\tau = \frac{1}{A_{21}}$

$$A_{21} = \frac{4\alpha}{3c^2} \omega^3 |D_{12}|^2 \cdot \frac{g_1}{g_2}$$

Lifetime of excited state with:

$|D_{12}| = 10^{-10}$ m

$\lambda_{21} = 500$ nm

$g_1 = g_2$

$$A_{21} = \frac{4 \cdot \frac{1}{137}}{3 \cdot (3 \cdot 10^8)^2 \frac{m^2}{sec^2}} \cdot (3,4 \times 10^{15})^3 \frac{1}{sec^3} (10^{-10})^2 m^2 =$$

$$= 4,3 \times 10^7 \frac{1}{sec}$$

$\omega = \nu \cdot 2\pi$

f=υ=c/λ [Hz]

$\nu = 6 \times 10^{14}$ Hz

$\omega = 3,4 \times 10^{15}$ Hz

$$\tau = \frac{1}{A_{21}} = 2,4 \times 10^{-8} \text{ sec}$$

Excited state lifetime $\tau = \frac{1}{A_{21}}$

$$A_{21} = \frac{4\alpha}{3c^2} \omega^3 |D_{12}|^2 \cdot \frac{g_1}{g_2}$$

Lifetime of excited state with:

$|D_{12}| = 10^{-10}$ m

$\lambda_{21} = 5$ nm

$g_1 = g_2$

$$A_{21} = \frac{4 \cdot \frac{1}{137}}{3 \cdot (3 \cdot 10^8)^2 \frac{m^2}{sec^2}} \cdot (3,4 \times 10^{17})^3 \frac{1}{sec^3} (10^{-10})^2 m^2 =$$

$$= 4,3 \times 10^{13} \frac{1}{sec}$$

$\omega = \nu \cdot 2\pi$

$\nu = 6 \times 10^{16}$ Hz

$\omega = 3,4 \times 10^{17}$ Hz

$$\tau = \frac{1}{A_{21}} = 2,4 \times 10^{-14} \text{ sec}$$

SELECTION RULES

$$A_{21} = \frac{4\alpha}{3c^2}\omega_{12}^3 |D_{12}|^2 \cdot \frac{g_1}{g_2}$$

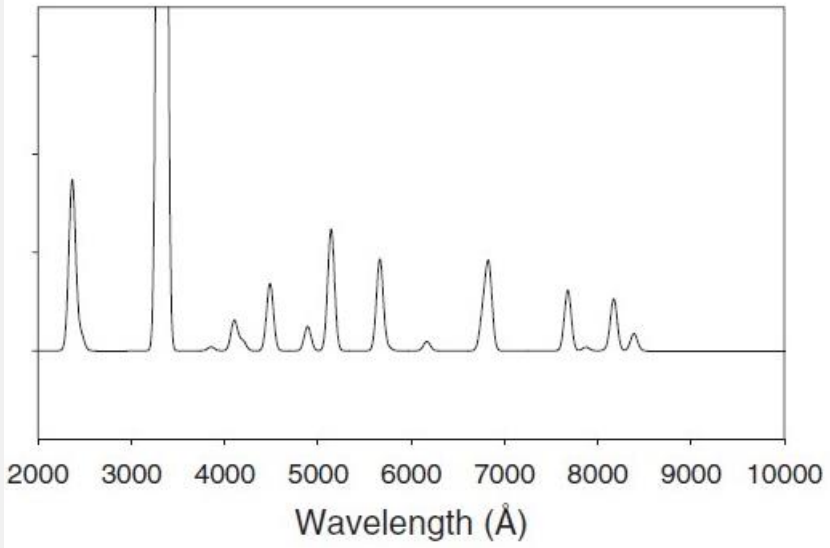
$$\tau = \frac{1}{A_{21}}$$

TABLE 1. *Radiative lifetimes for energy levels in neutral iron*

Configuration	Term	J	Level cm^{-1}	Lifetime ns	Number of transitions
$3d^6\,4s(a\,^6D)4p$	$z\,^7D^\circ$	5	19351	182000.	2
		4	19562	73700.	2
		3	19757	75600.	4
		2	19913	89200.	2
		1	20020	137000.	2
$3d^6\,4s(a\,^6D)4p$	$z\,^7F^\circ$	6	22650	788000.	1
		5	22846	12000.	2
		4	22997	8370.	3
		3	23111	10500.	2
		2	23193	9540.	5
		1	23245	15700.	4
$3d^6\,4s(a\,^6D)4p$	$z\,^7P^\circ$	4	23711	17200.	3
		3	24181	26900.	4
		2	24507	85100.	3
$3d^6\,4s(a\,^6D)4p$	$z\,^5D^\circ$	4	25900	59.8	6
		3	26140	60.6	6
		2	26340	59.1	6
		1	26479	54.4	5
		0	26550	54.4	2
$3d^6\,4s(a\,^6D)4p$	$z\,^5F^\circ$	5	26875	47.4	4
		4	27167	50.8	6
		3	27395	45.8	6
		2	27560	43.2	6
		1	27666	39.2	5
$3d^6\,4s(a\,^4D)4p$	$y\,^5P^\circ$	3	36767	4.09	10
		2	37158	3.58	9
		1	37410	3.15	8
$3d^7\,(b\,^2F)4s$	$d\,^3F$	2	36941	<4320.	2
$3d^7\,(a\,^4F)4p$	$y\,^3D^\circ$	3	38175	2.92	16
		2	38678	2.43	14
		1	38996	2.09	16
$3d^6 4s(a^4D)4p$	$x\,^5D^\circ$	4	39626	1.64	7
		3	39970	1.35	13
		2	40231	1.23	13
		1	40405	1.00	9
		0	40491	1.03	3
$3d^5\,4s^2\,(a\,^6S)4p$	$y\,^7P^\circ$	2	40052	119.	6
		3	40207	21.8	7
		4	40422	33.1	4
$3d^6\,4s(a\,^4D)4p$	$x\,^5F^\circ$	5	40257	2.65	3
		4	40594	1.81	7
		3	40842	1.49	9
		2	41018	1.28	9
		1	41131	.86	5
$3d^6\,4s(b\,^4P)4p$	$z\,^5S^\circ$	2	40895	8.04	9
$3d^6\,4s(b\,^4P)4p$	$x\,^5P^\circ$	3	42533	7.87	11

TABLE II. Ionization potential (top row) and excitation energies of Lr^+. The CI+MBPT and FSCC columns include the Breit and the QED corrections, the latter of which is shown in a separate column for comparison between the two calculations. The recommended values are obtained as the mean of FSCC and CI+MBPT results. Lifetimes and and g-factors derived from CI+MBPT calculations are also included.

State		g factor CI+MBPT	Energy (cm^{-1}) FSCC	ΔQED	CI+MBPT	ΔQED	Recommended	Lifetime (s)
$7s^2$	$^1S_0$		116949	−219			116949 ± 389	
$6d7s$	$^3D_1$	0.5	20265	−342	21426	−374	20846 ± 1161	2.23×10^{6}
	$^3D_2$	1.15	21623	−344	22507	−373	22065 ± 884	8.26×10^{-2}
	$^3D_3$	1.33	26210	−326	26313	−355	26262 ± 389	2.97×10^{-2}
	$^1D_2$	1.02	31200	−373	30942	−397	31071 ± 389	1.53×10^{-3}
$7s7p$	$^3P_0$		29487	−167	29059	−306	29273 ± 428	2.56×10^{-7}
	$^3P_1$	1.42	31610	−179	31470	−314	31540 ± 389	1.45×10^{-8}
	$^3P_2$	1.50	43513	−240	42860	−308	43186 ± 653	2.43×10^{-8}
	$^1P_1$	1.08	47819	−260	46771	−376	47295 ± 1048	1.11×10^{-9}



Why do the different levels have such different lifetimes? Why are some transitions stronger than others? And why do not all possible transitions take place?

$$A_{21} = \frac{4\alpha}{3c^2}\omega_{12}^3 |D_{12}|^2 \cdot \frac{g_1}{g_2}$$

$$\tau = \frac{1}{A_{21}}$$

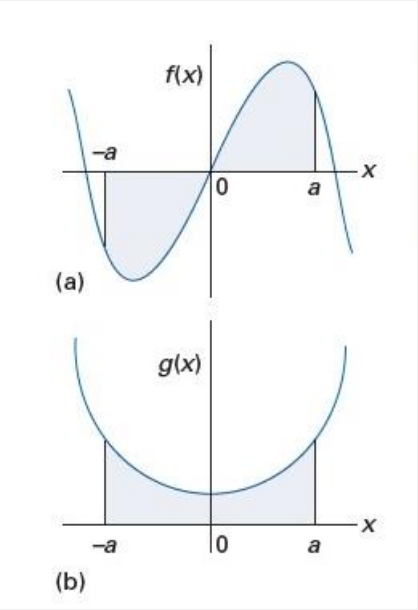
SELECTIONS RULES: E1 TRANSITIONS

- Parity:

$$D_{12} = \langle \psi_1 | \vec{r} | \psi_2 \rangle$$

odd

$$A_{21} = \frac{4\alpha}{3c^2} \omega^3 |D_{12}|^2 \cdot \frac{g_1}{g_2}$$



f(x) antisymmetric: integral 0

g(x) symmetric: integral can be nonzero

When is D_{12} very small? These are **forbidden** transitions.

- In electric dipole (E1) transitions:

1. Parity must change. We can have transitions from odd to even configurations or vice versa, but not even-even, odd-odd.

How do we know the parity? Sum over **valence** electrons:

For a many electron atom:

$$(-1)^{l_1}(-1)^{l_2}\cdots(-1)^{l_n} = (-1)^{\sum_{i=1}^{n} l_i}$$

Say 6s5d → $-1^{0+2}=1$, even. 6s6p → $-1^{0+1}=-1$, odd

Configuration	Term	J	Level (cm^{-1})	Ref.
$5s^2$	^1S	0	0.000	M52
$5s5p$	$^3P^\circ$	0	14317.520	M52
		1	14504.351	M52
		2	14898.563	M52
$5s4d$	^3D	1	18159.056	M52
		2	18218.795	M52
		3	18319.267	M52
$5s4d$	^1D	2	20149.700	M52
$5s5p$	$^1P^\circ$	1	21698.482	M52
$5s6s$	^3S	1	29038.795	M52
$5s6s$	^1S	0	30591.80	M52
$4d5p$	$^3F^\circ$	2	33266.872	M52
		3	33589.724	M52
		4	33919.330	M52
$4d5p$	$^1D^\circ$	2	33826.927	M52
$5s5d$	^1D	2	34727.483	M52
$5s5d$	^3D	1	35006.943	M52
		2	35022.015	M52
		3	35045.055	M52
$5p^2$	^3P	0	35193.47	M52
		1	35400.138	M52
		2	35674.668	M52
$4d5p$	$^3D^\circ$	1	36264.181	M52

Neutral Sr

Parity is determined by configuration, not by term!

ΔJ= +/-1, 0: Conservation of total angular momentum

But: we will not have a transition from J_1=0 to J_2=0

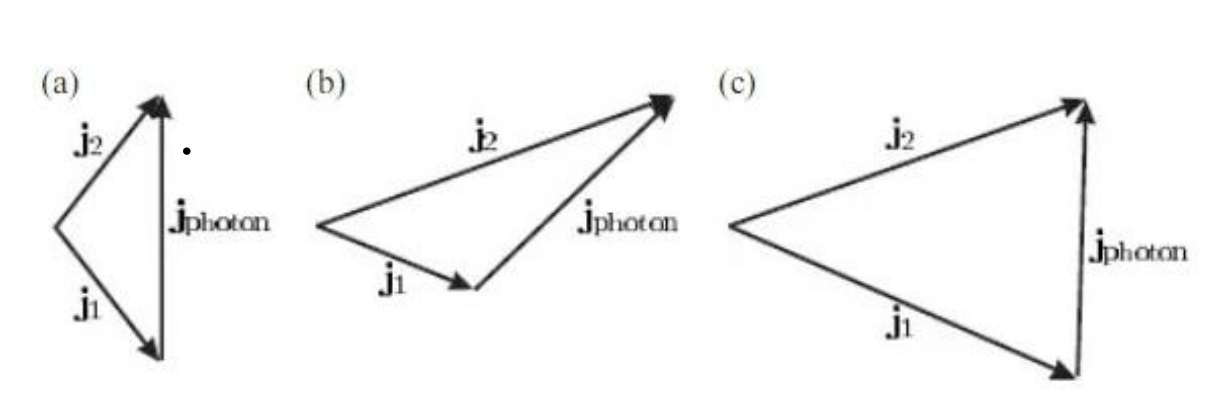


Fig. 2.8 The conservation of total angular momentum in electric dipole transitions that gives the selection rule in eqn 2.59 can be represented as vector addition. The photon has one unit of angular momentum, and so to go from level j_1 to j_2 the vectors must form a triangle, as shown for the case of (a) $j_1 = 1/2$ to $j_2 = 1/2$, (b) $j_1 = 1/2$ to $j_2 = 3/2$ and (c) $j_1 = 3/2$ to $j_2 = 3/2$.

- In electric dipole (E1) transitions:

1. Parity must change.
2. $\Delta J=0, +/-1$ (but not $J=0 \rightarrow J=0$)
3. $\Delta M_J= 0, +/-1$

- In electric dipole (E1) transitions:

1. Parity must change.
2. $\Delta J=0, +/-1$ (but not $J=0 \rightarrow J=0$)
3. $\Delta M_J= 0, +/-1$
4. $\Delta L=0, +/-1$ (but not $L=0 \rightarrow L=0$)
5. $\Delta M_L= 0, +/-1$

$\Delta S = 0$

Electric dipole operator does not act upon spin and thus can not change it

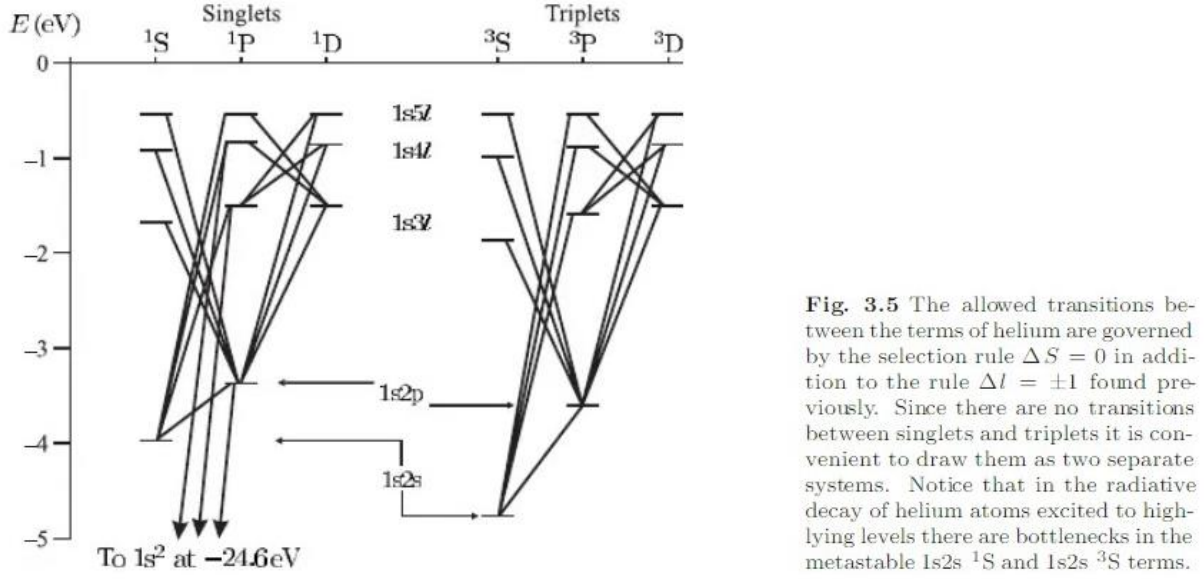


Fig. 3.5 The allowed transitions between the terms of helium are governed by the selection rule $\Delta S = 0$ in addition to the rule $\Delta l = \pm 1$ found previously. Since there are no transitions between singlets and triplets it is convenient to draw them as two separate systems. Notice that in the radiative decay of helium atoms excited to high-lying levels there are bottlenecks in the metastable 1s2s ^{1}S and 1s2s ^{3}S terms.

$^3S \rightarrow {}^3P$

$^3S \not\rightarrow {}^1P$

Table 5.1 Selection rules for electric dipole (E1) transitions in the *LS*-coupling scheme. Rules 1–4 apply to all electric dipole transitions; rules 5 and 6 are obeyed only when L and S are good quantum numbers. The right-hand column gives the structure to which the rule applies.

1	$\Delta J = 0, \pm 1$	$(J = 0 \nleftrightarrow J' = 0)$	Level
2	$\Delta M_J = 0, \pm 1$	$(M_J = 0 \nleftrightarrow M_{J'} = 0$ if $\Delta J = 0)$	State
3	Parity changes		Configuration
4	$\Delta l = \pm 1$	One electron jump	Configuration
5	$\Delta L = 0, \pm 1$	$(L = 0 \nleftrightarrow L' = 0)$	Term
6	$\Delta S = 0$		Term

- In electric dipole (E1) transitions:

1. Parity must change.
2. ΔJ=0, +/-1 (but not J=0→J=0)
3. ΔM_J= 0, +/-1
4. ΔL=0, +/-1 (but not L=0→L=0)
5. ΔM_L= 0, +/-1
6. ΔS=0

- In electric dipole (E1) transitions:

1. **Parity must change.**
2. **ΔJ=0, +/-1 (but not J=0→J=0)**
3. **ΔM_J= 0, +/-1**

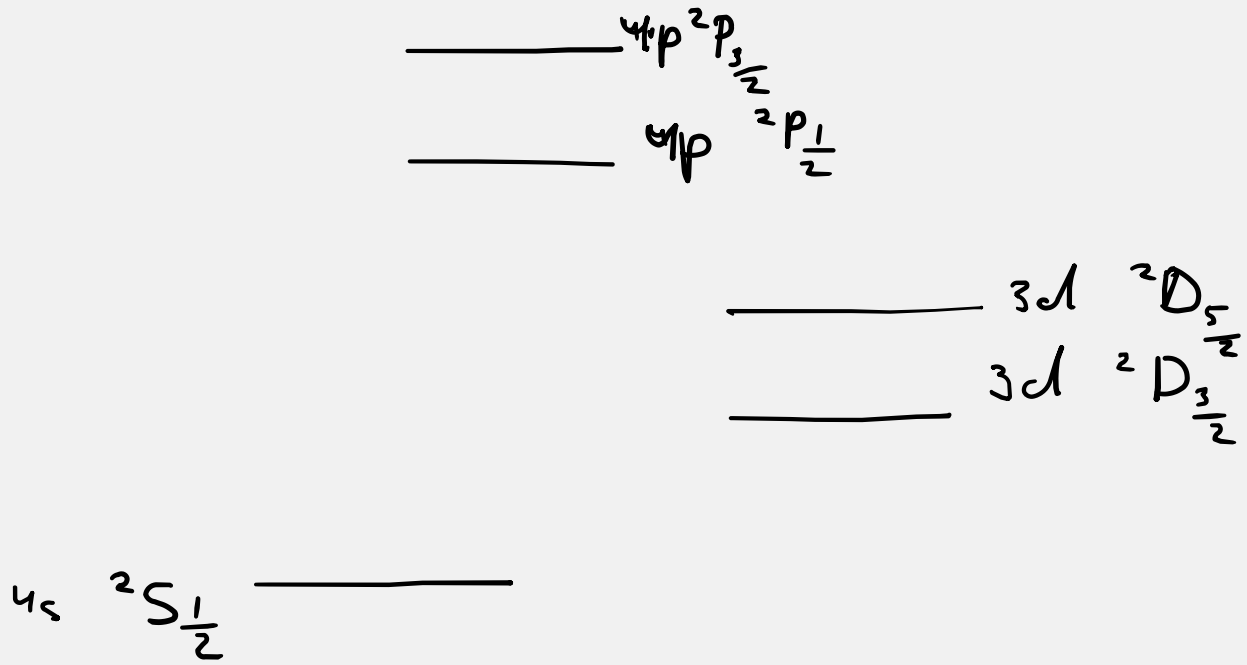
Transitions that violate these rules are **forbidden**

4. ΔL=0, +/-1 (but not L=0→L=0)
5. ΔM_L= 0, +/-1
6. ΔS=0

Depends whether LS-coupling holds: **semiforbidden**

If a transition is (semi)forbidden, D_{12} is very small, and thus A_{21} is very small, and thus τ (lifetime) is very large.

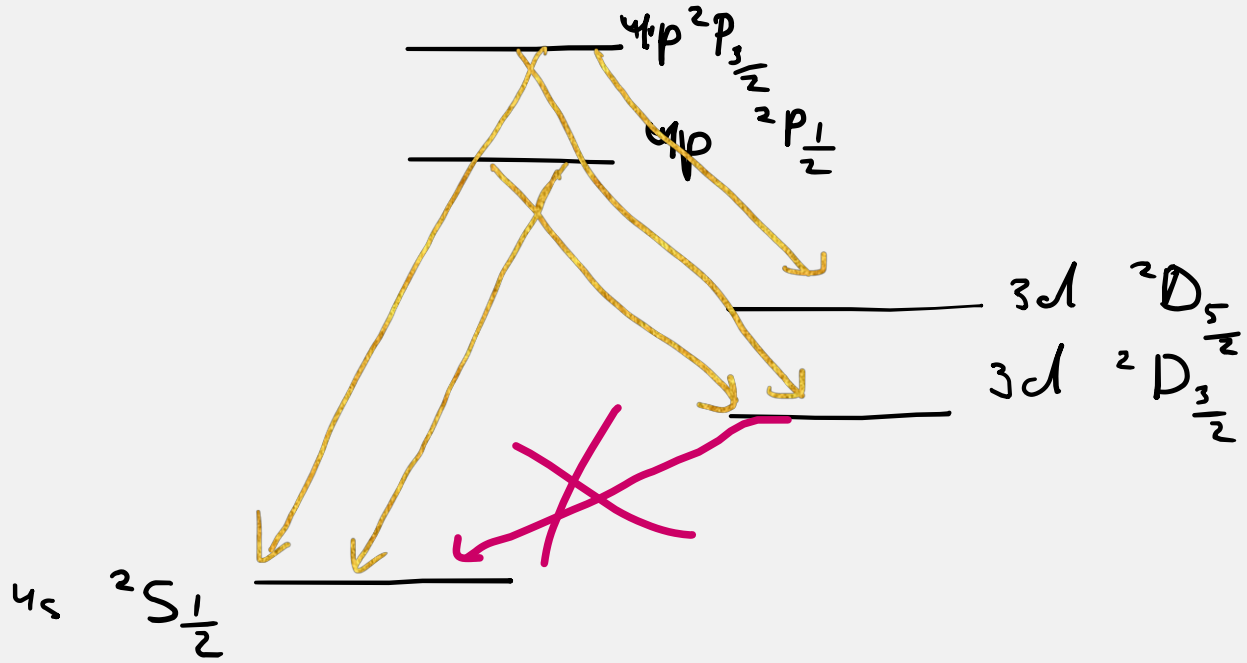
- Which electric dipole (E1) transitions are allowed between the Ca^+ ([Ar]4s) atomic levels?



- In electric dipole (E1) transitions:

1. Parity must change.
2. $\Delta J=0, +/-1$ (but not $J=0 \rightarrow J=0$)
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- Which electric dipole (E1) transitions are allowed between the Ca^+ atomic levels?



- E1 forbidden transitions can still take place via other mechanisms, but at a much slower rate (M1, E2).

- In electric dipole (E1) transitions:

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6. $\Delta S=0$

For each of the following atomic radiative transitions, indicate whether the transition is allowed or forbidden under the electric-dipole radiation selection rules. For the forbidden transitions, cite the selection rules which are violated.

(a) He: $(1s)(1p)\ ^1P_1 \rightarrow (1s)^2\ ^1S_0$

(b) C: $(1s)^2(2s)^2(2p)(3s)\ ^3P_1 \rightarrow (1s)^2(2s)^2(2p)^2\ ^3P_0$

(c) C: $(1s)^2(2s)^2(2p)(3s)\ ^3P_0 \rightarrow (1s)^2(2s)^2(2p)^2\ ^3P_0$

(d) Na: $(1s)^2(2s)^2(2p)^6(4d)\ ^2D_{5/2} \rightarrow (1s)^2(2s)^2(2p)^6(3p)\ ^2P_{1/2}$

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(c) C: $(1s)^2(2s)^2(2p)(3s)\ ^3P_0 \rightarrow (1s)^2(2s)^2(2p)^2\ ^3P_0$ $J=0 \rightarrow J=0$ ✗

(d) Na: $(1s)^2(2s)^2(2p)^6(4d)\ ^2D_{5/2} \rightarrow (1s)^2(2s)^2(2p)^6(3p)\ ^2P_{1/2}$ ✗, $\Delta J=2$

(e) He: $(1s)(2p)\ ^3P_1 \rightarrow (1s)^2\ ^1S_0$

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(c) C: $(1s)^2(2s)^2(2p)(3s)\ ^3P_0 \rightarrow (1s)^2(2s)^2(2p)^2\ ^3P_0$ $J=0 \rightarrow J=0$ ×

(d) Na: $(1s)^2(2s)^2(2p)^6(4d)\ ^2D_{5/2} \rightarrow (1s)^2(2s)^2(2p)^6(3p)\ ^2P_{1/2}$ ×, $\Delta J=2$

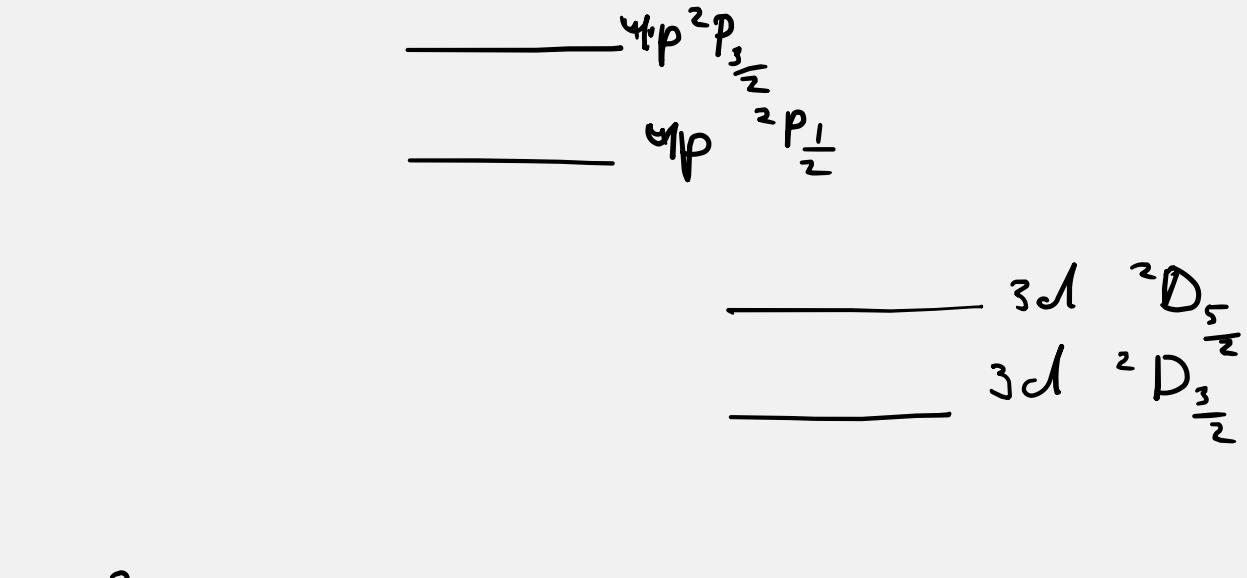
(e) He: $(1s)(2p)\ ^3P_1 \rightarrow (1s)^2\ ^1S_0$ × $\Delta S \neq 0$

- E1 forbidden transitions can still take place via other mechanisms (M1, E2, etc), but at a much slower rate.

SELECTION RULES: M1 TRANSITIONS

1. Parity is conserved
2. $\Delta J=0, +/-1$ (but not $J=0 \rightarrow J=0$)
3. $\Delta M_J= 0, +/-1$
4. *n* is conserved
5. $\Delta L=0$
6. $\Delta S=0$
7. $\Delta F=0,+/-1$ (but not $F=0 \rightarrow J=0$)
8. $\Delta M_F=0,+/-1$

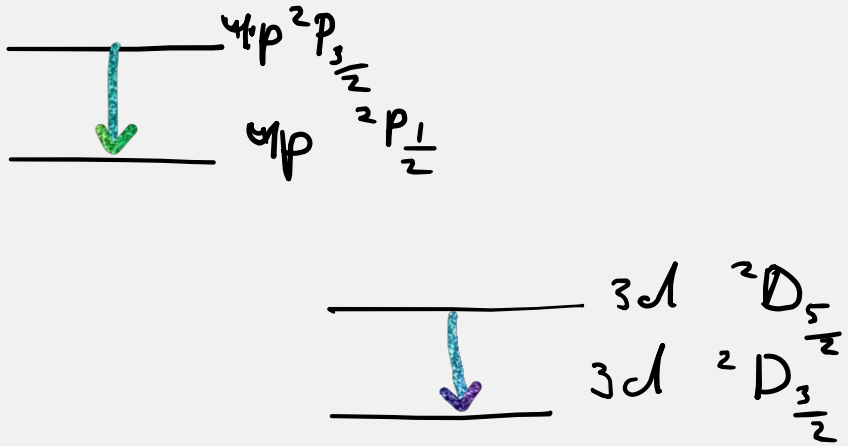
- Which magnetic dipole (M1) transitions are allowed between the Ca^+ atomic levels?



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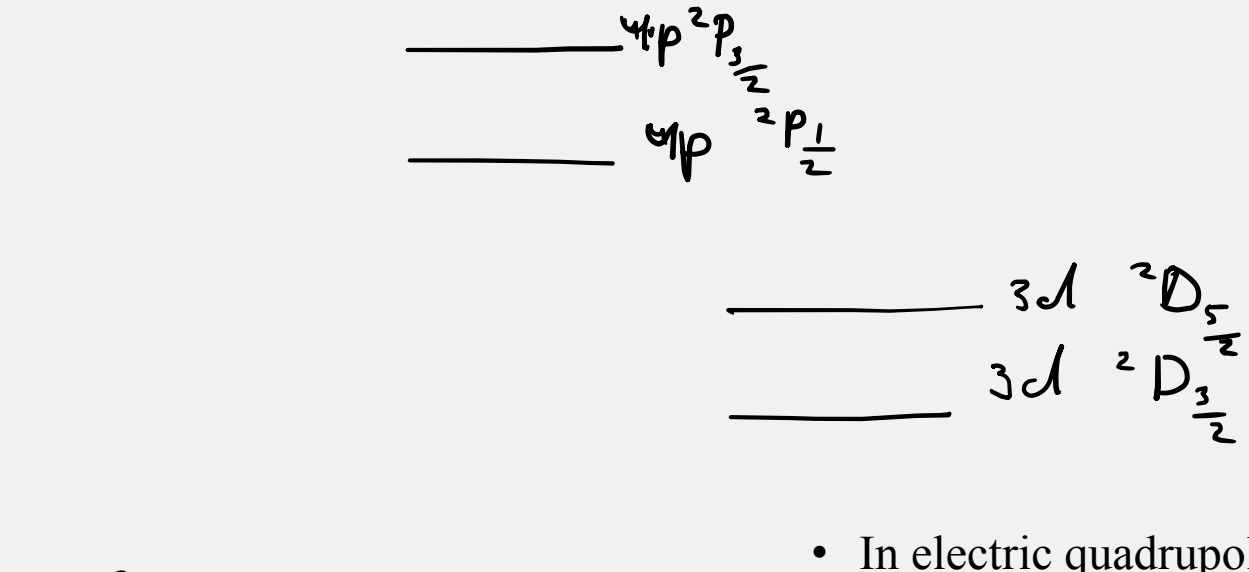
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4. *n* is conserved
5. $\Delta L=0$
6. $\Delta S=0$
7. $\Delta F=0,+/-1$ (but not $F=0 \rightarrow J=0$)
8. $\Delta M_F=0,+/-1$

SELECTION RULES: E2 TRANSITIONS

- In electric quadrupole (E2) transitions:

1. Parity is conserved
2. $\Delta J=0, +/-1, +/-2$ (but not $J=0 \rightarrow J=0$, or $J=1 \leftarrow \rightarrow J=0$, or $J=1/2 \leftarrow \rightarrow J=1/2$)
3. $\Delta M_J= 0, +/-1, +/-2$
4. $\Delta L=0, +/-1, +/-2$ (but not $L=0 \rightarrow J=0$ or $L=1 \leftarrow \rightarrow J=0$)
5. $\Delta S=0$

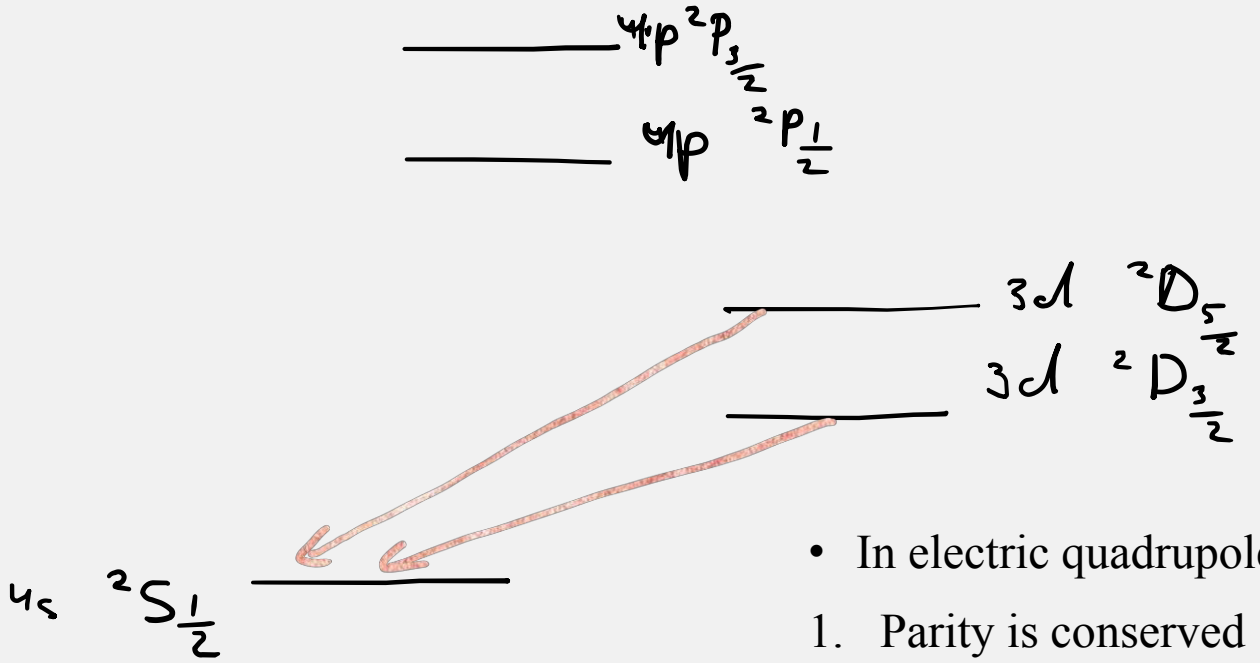
- Which electric quadrupole (E2) transitions are allowed between the Ca^+ atomic levels?



- In electric quadrupole (E2) transitions:

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2. $\Delta J=0, +/-1, +/-2$ (but not $J=0 \rightarrow J=0$, or $J=1 \leftarrow\rightarrow J=0$, or $J=1/2 \leftarrow\rightarrow J=1/2$)
3. $\Delta M_J= 0, +/-1, +/-2$
4. $\Delta L=0, +/-1, +/-2$ (but not $L=0 \rightarrow J=0$ or $L=1 \leftarrow\rightarrow J=0$)
5. $\Delta S=0$

- Which electric quadrupole (E2) transitions are allowed between the Ca^+ atomic levels?



- In electric quadrupole (E2) transitions:

1. Parity is conserved
2. $\Delta J=0, +/-1, +/-2$ (but not $J=0 \rightarrow J=0$, or $J=1 \leftarrow\rightarrow J=0$, or $J=1/2 \leftarrow\rightarrow J=1/2$)
3. $\Delta M_J= 0, +/-1, +/-2$
4. $\Delta L=0, +/-1, +/-2$ (but not $L=0 \rightarrow J=0$ or $L=1 \leftarrow\rightarrow J=0$)
5. $\Delta S=0$

Transition rates:

$$A_{E1} = \frac{2.03 \times 10^{18}}{(2J+1)\lambda^3} S(E1) \qquad S(E1) = |\langle 1 \| D \| 2 \rangle|^2$$

$$A_{M1} = \frac{2.7 \times 10^{13}}{(2J+1)\lambda^3} S(M1) \qquad S(M1) = |\langle 1 \| M \| 2 \rangle|^2$$

$$A_{E2} = \frac{1.1 \times 10^{18}}{(2J+1)\lambda^5} S(E2) \qquad S(E2) = |\langle 1 \| Q \| 2 \rangle|^2$$

J – total angular momentum of <u>upper</u> state

λ [A]

S – reduced matrix elements

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$$A_{E2} = \frac{1.1 \times 10^{18}}{(2J+1)\lambda^5} S(E2) \qquad S(E2) = |\langle 1 \| Q \| 2 \rangle|^2$$

10^5

λ^3 vs λ^5

J – total angular momentum of upper state

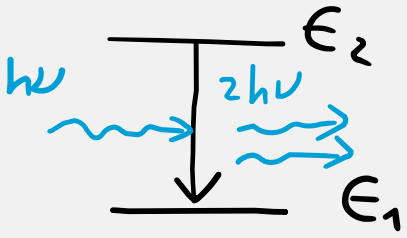
λ [A]

S – reduced matrix elements

LASERS



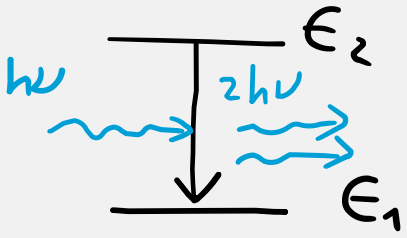
- Light Amplification by Stimulated Emission of Radiation
- Based on the stimulated emission process



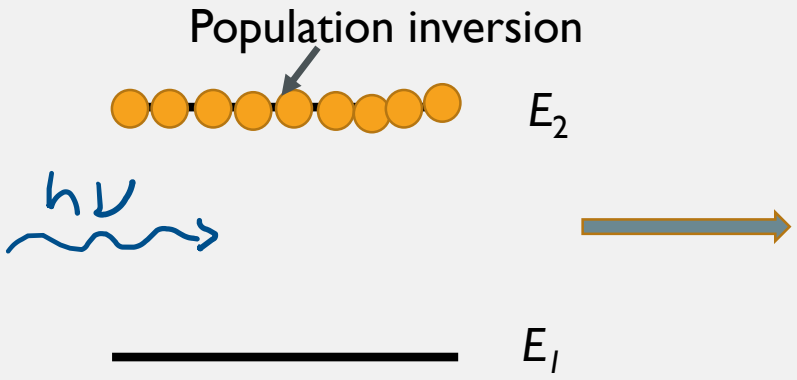
$h\nu = \Delta\epsilon$

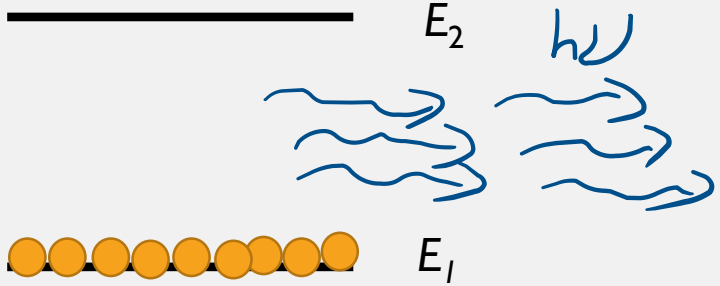


- Light Amplification by Stimulated Emission of Radiation
- Based on the stimulated emission process

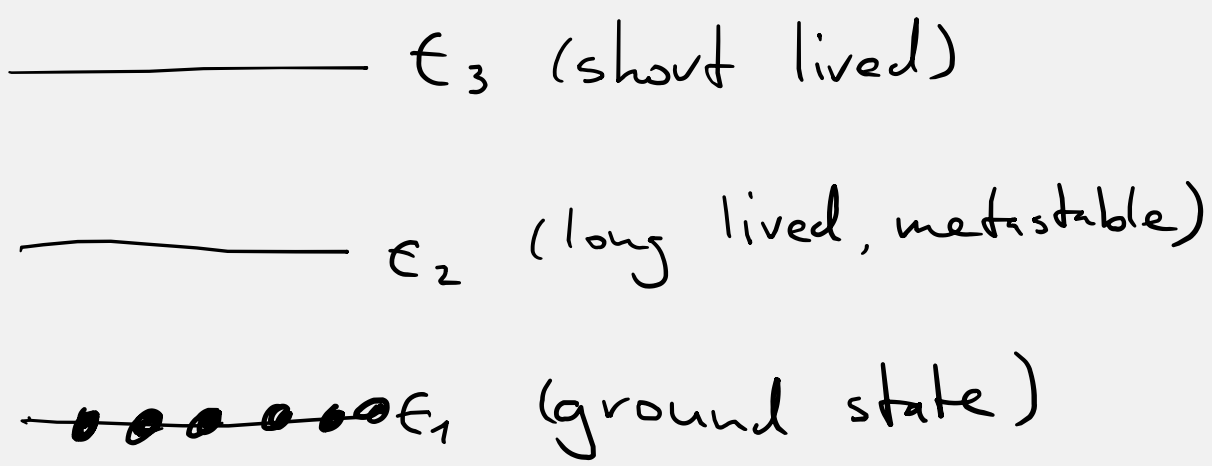


$h\nu = \Delta\epsilon$

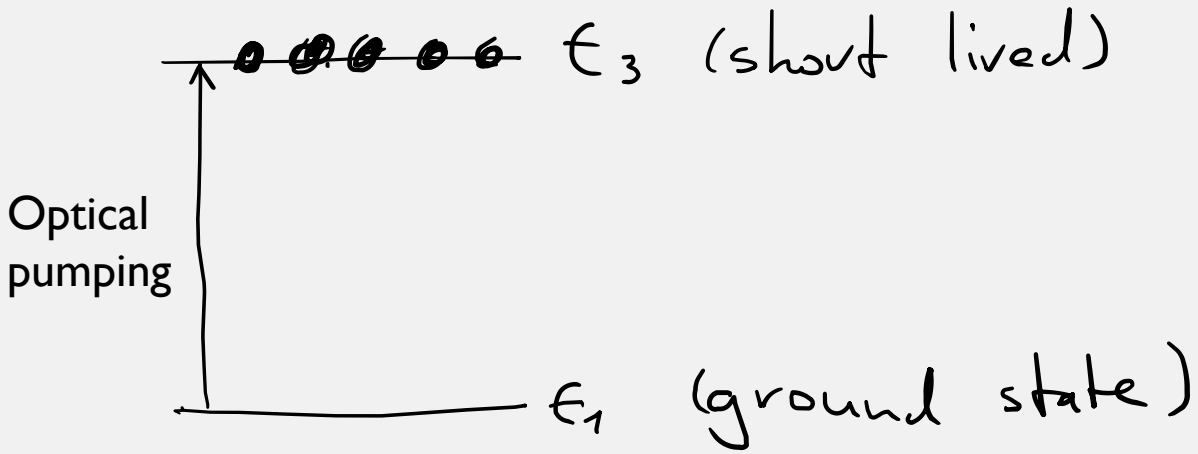




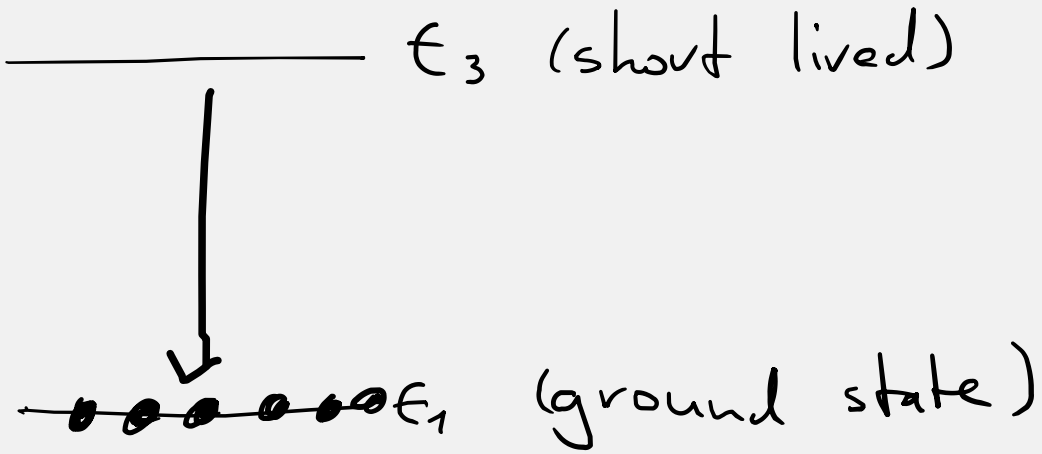
- If we want to have a population inversion, we need at least 3 levels:



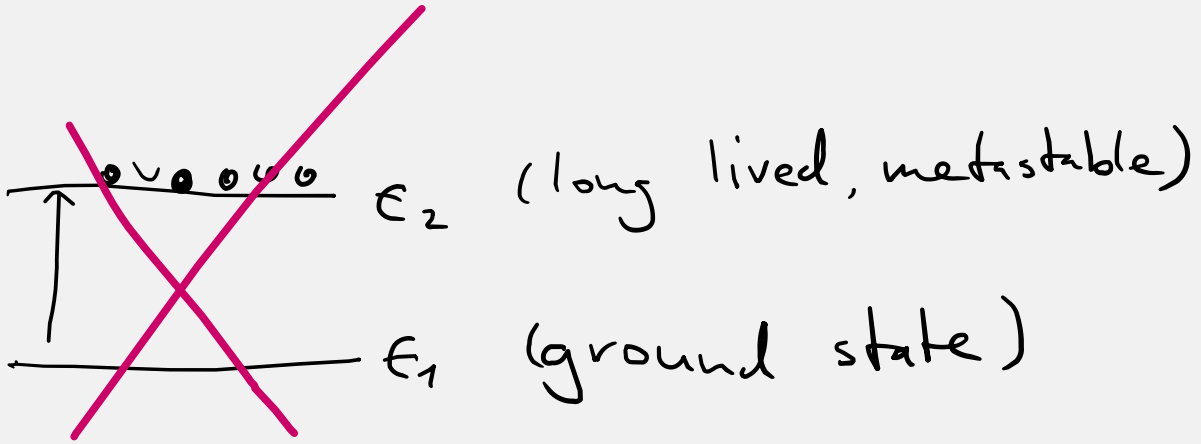
- If we have two levels only:



- If we have two levels only, and the excited state is short lived, the excited state will immediately decay to the ground state though spontaneous emission. No population inversion possible.

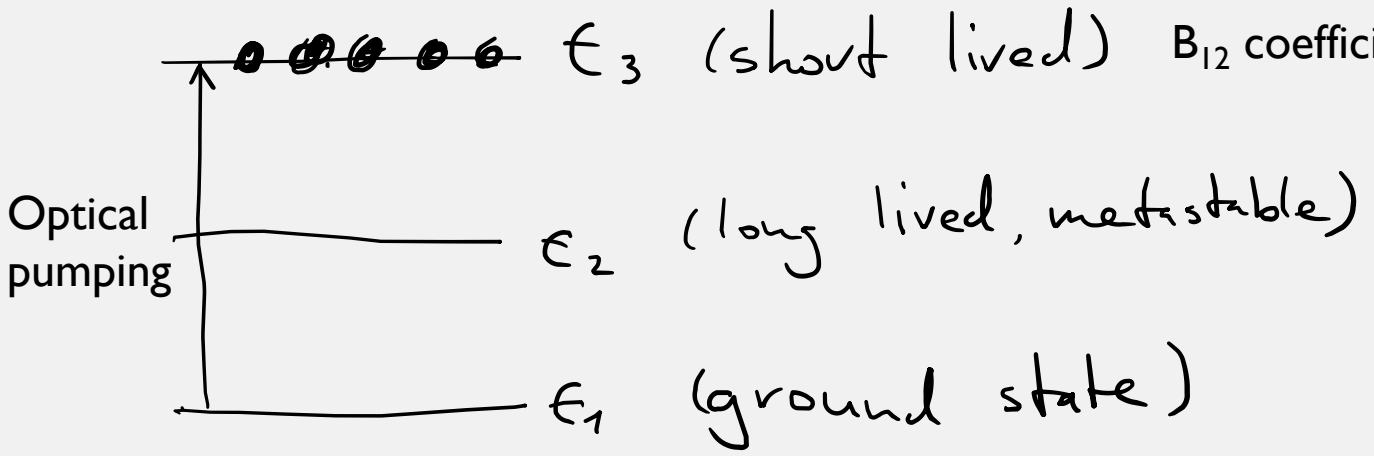


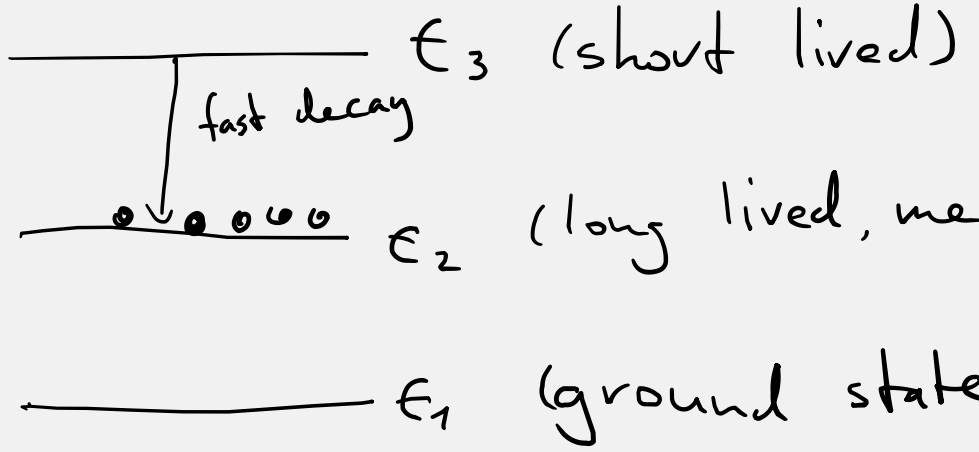
- If we have two levels only, and the excited state is long lived, the pumping will not be efficient. No population inversion possible.



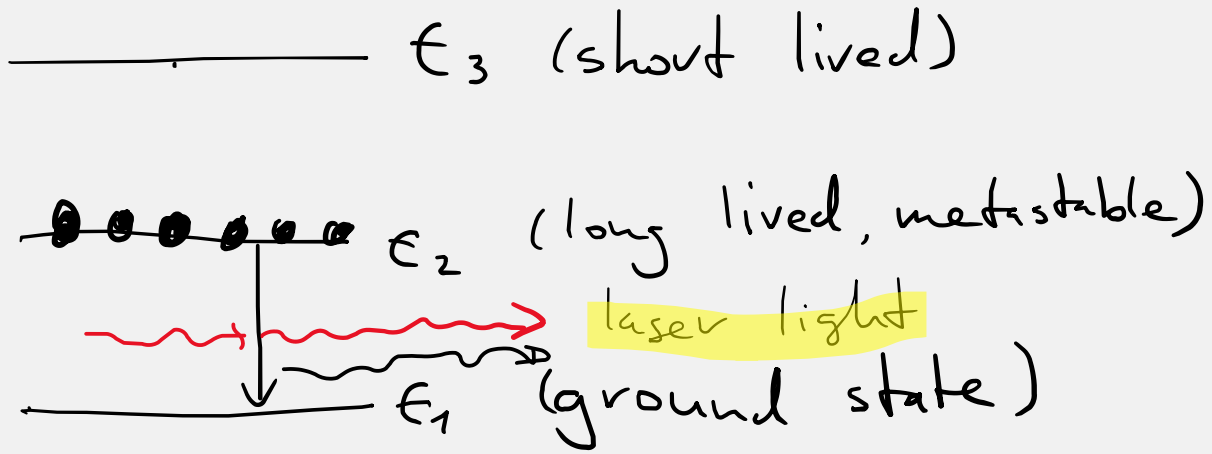
$(A_{21} \propto B_{12})$

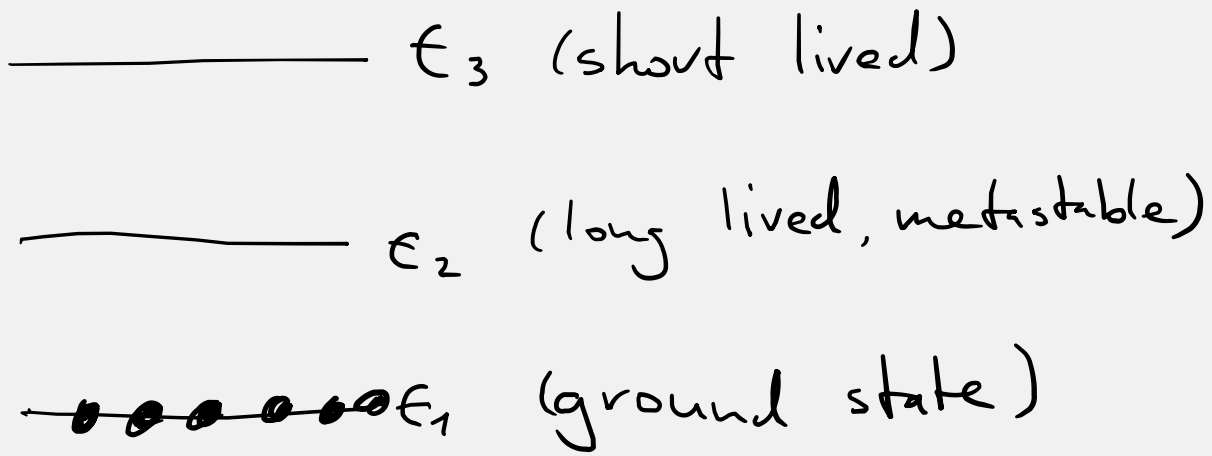
- If we want to have a population inversion, we need at least 3 levels:

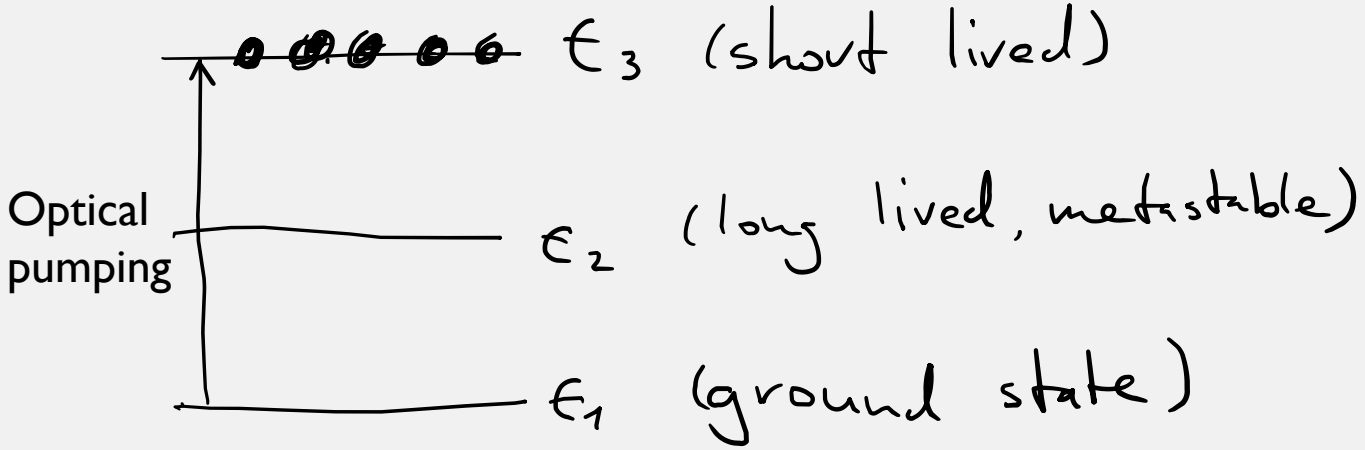




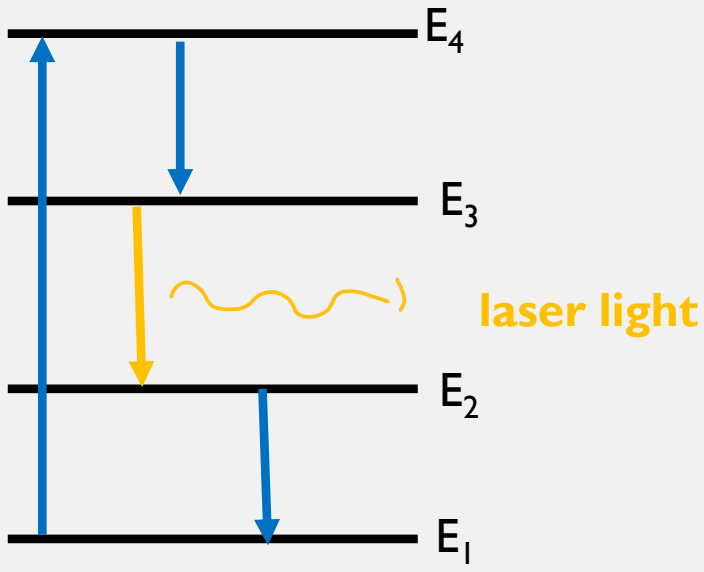
population inversion possible


ϵ_3 (short lived)
ϵ_2 (long lived, metastable)
laser light
ϵ_1 (ground state)


ϵ_3 (short lived)
ϵ_2 (long lived, metastable)
ϵ_1 (ground state)

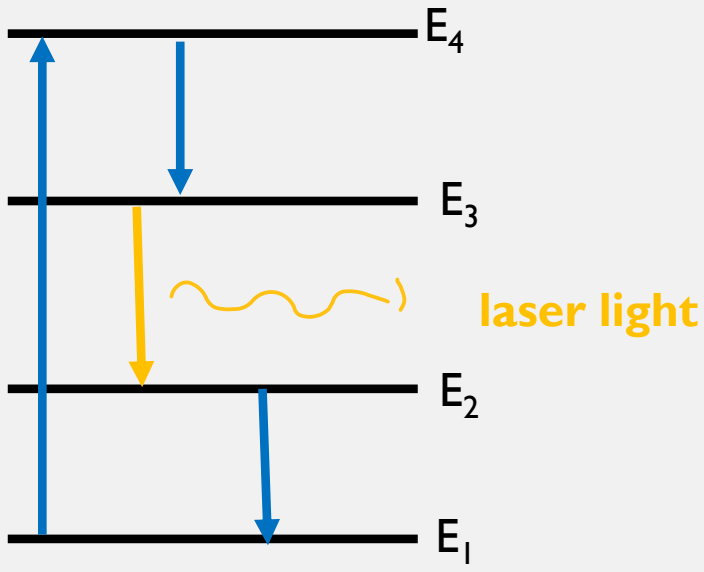

ϵ_3 (short lived)
Optical pumping
ϵ_2 (long lived, metastable)
ϵ_1 (ground state)

- What are the requirements for the 4→1, 4→3, 3→2, and 2→1 decays to make a laser?



4→1 slow/fast
4→3 slow/fast
3→2 slow/fast
2→1 slow/fast

- What are the requirements for the 4→1, 4→3, 3→2, and 2→1 decays to make a laser?



4→1 slow/fast
4→3 slow/fast
3→2 slow/fast
2→1 slow/fast

Excited state lifetime $\tau = 1/A_{21}$ is connected to uncertainty in energy through the uncertainty principle:

$\Delta t \Delta E \geq \hbar/2$

Short lived excited state: large uncertainty; long-lived: small.

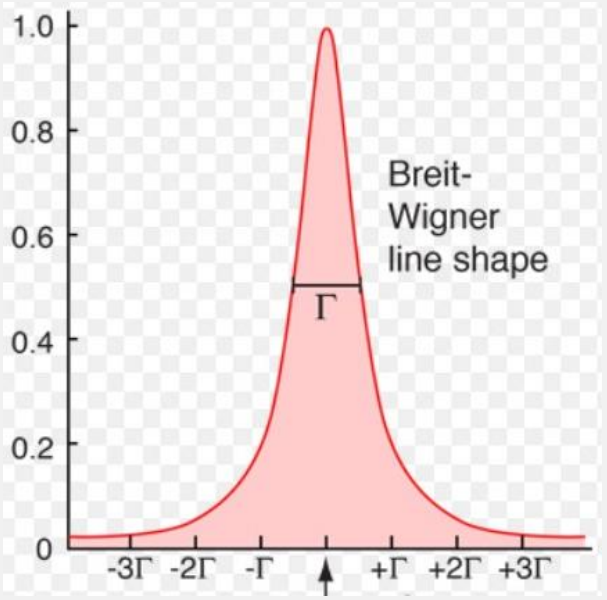
Example: excited ^{2}P state in hydrogen

$\epsilon = 10{,}198\ eV$

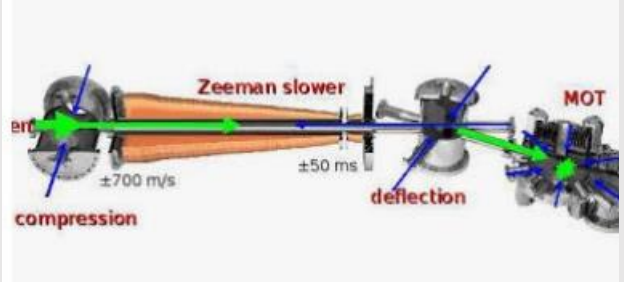
$\tau = 1{,}8\ ns$

$2 \cdot \Delta\epsilon = \Gamma = \frac{\hbar}{\tau}$

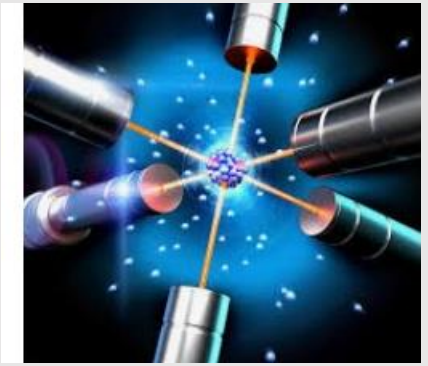
$\Delta\epsilon = 2 \cdot 10^{-7}\ eV$

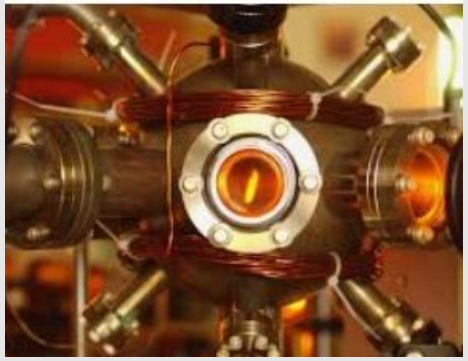


SLOWING, COOLING, AND TRAPPING OF ATOMS



- Slowing
- Cooling
- Trapping



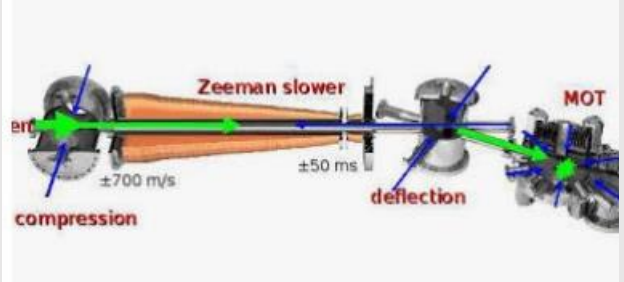


Why trap atoms?

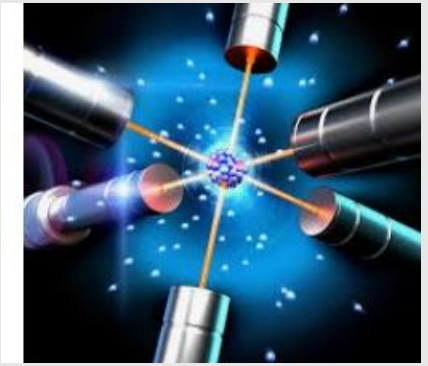
- Possibility to manipulate atoms
- Long interactions times

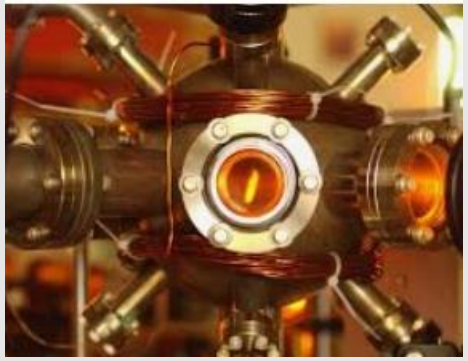


- Improving applications (i.e. atomic clocks, qubits)
- Reaching higher precision in experiments (in particular fundamental physics)

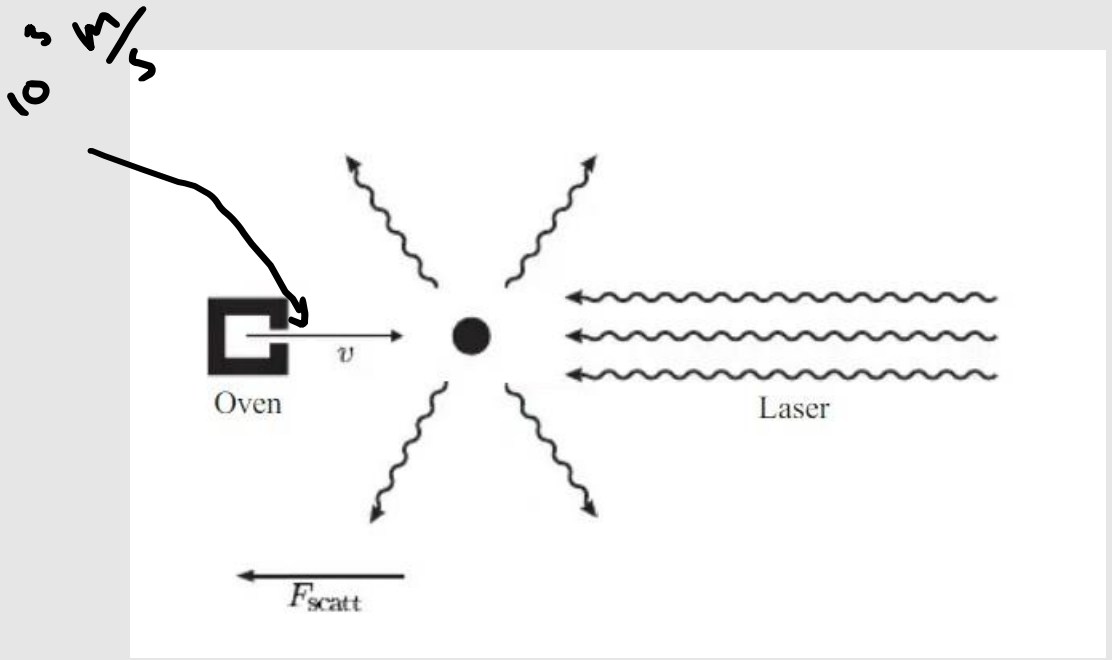


- Slowing
- Cooling
- Trapping

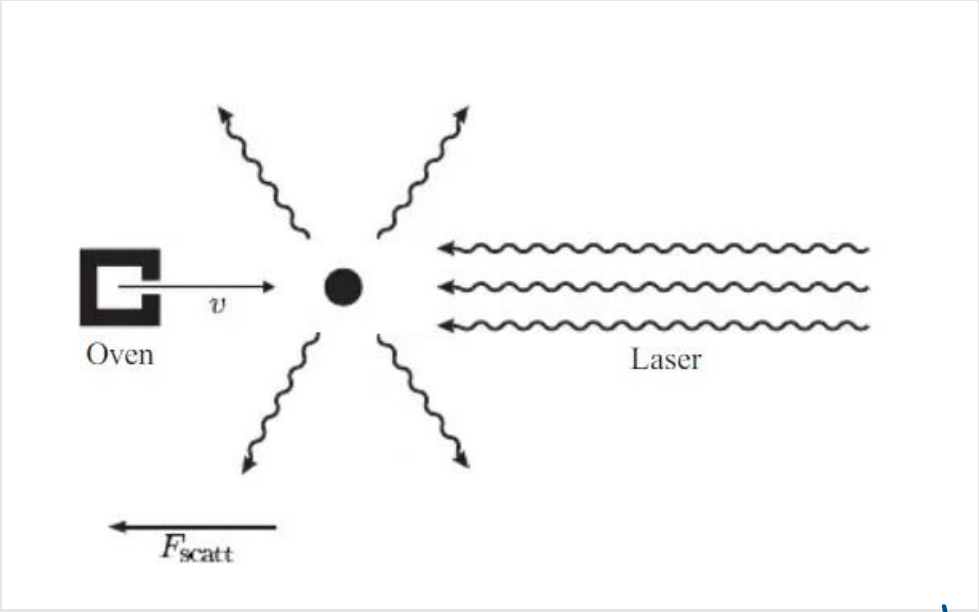




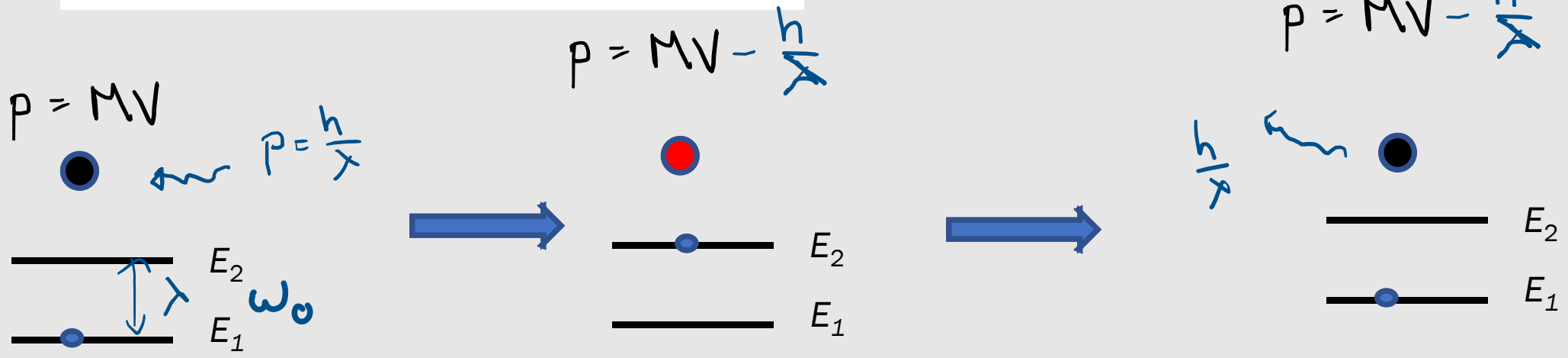
Deceleration by scattering force



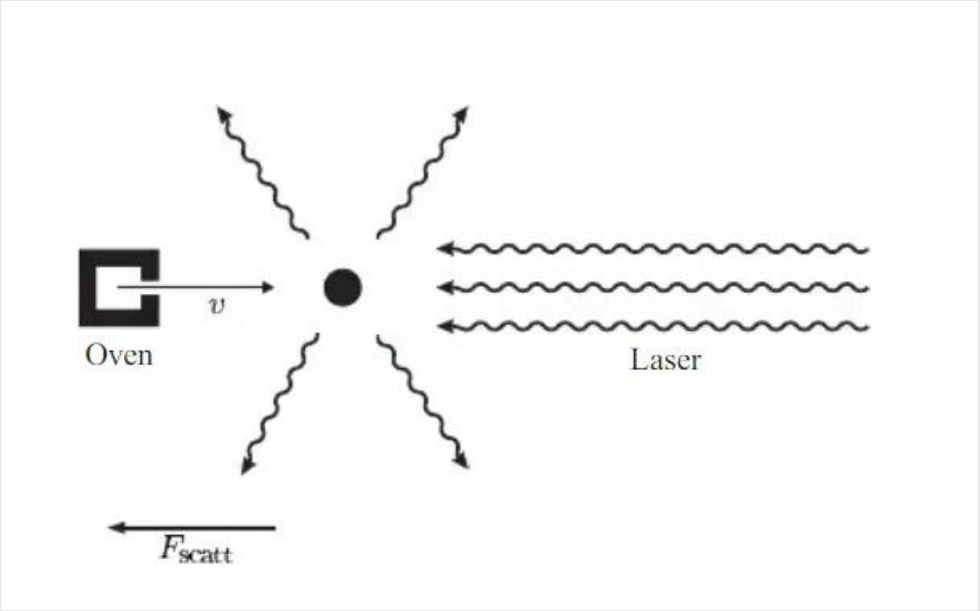
Deceleration by scattering force



Requires radiation on resonance with (very close to) the wavelength of an allowed transition in the atom. Why?



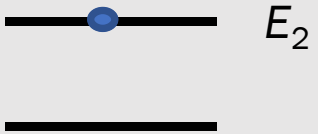
Deceleration by scattering force



$F_{scatt}=\Delta P/\Delta t = h/\lambda R_{scatt}$

$R_{scatt}=\rho_2/\tau= \rho_2 \Gamma \quad (\Gamma=1/\tau)$

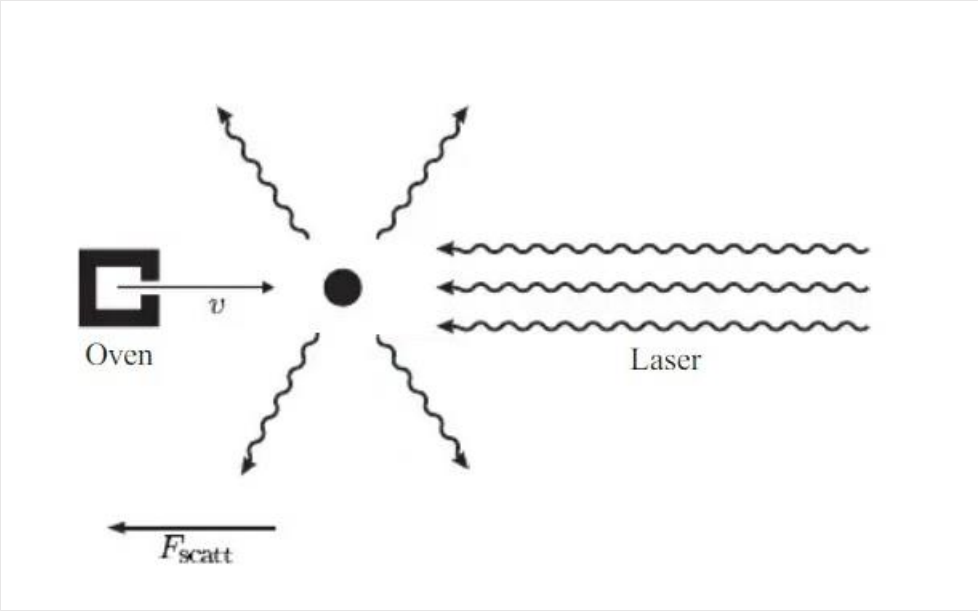
ρ_2: excited state population



$$F_{scatt} = \hbar k \frac{\Gamma}{2} \frac{I / I_{sat}}{1 + I/I_{sat} + 4\delta^2/\Gamma^2}$$

$k=\omega/c=2\pi/\lambda$
I- laser intensity
I_{sat}- saturation intensity
$\delta= \omega- \omega_0$ (detuning)

Deceleration by scattering force



$F_{scatt}=h/\lambda R_{scatt}$

$R_{scatt}=\rho_2/\tau=\rho_2\,\Gamma$ $(\Gamma=1/\tau)$

ρ_2: excited state population

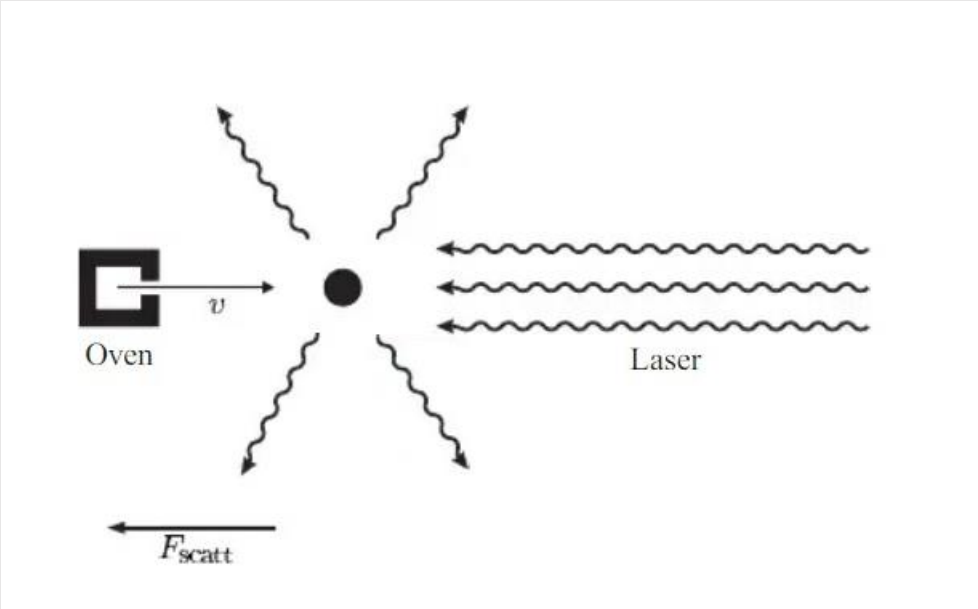
$$F_{scatt}=\hbar k\frac{\Gamma}{2}\frac{I/I_{sat}}{1+I/I_{sat}+4\delta^2/\Gamma^2}$$

(for very high laser intensity, I→ ∞, ρ_2→ ½ . Why?)

Then, we have (highest possible deceleration force)

$$F_{max}=\frac{h\Gamma}{2\lambda}$$

Deceleration by scattering force



$F_{scatt}=h/\lambda R_{scatt}$

$R_{scatt}=\rho_2/\tau=\rho_2\,\Gamma$ $(\Gamma=1/\tau)$

ρ_2: excited state population

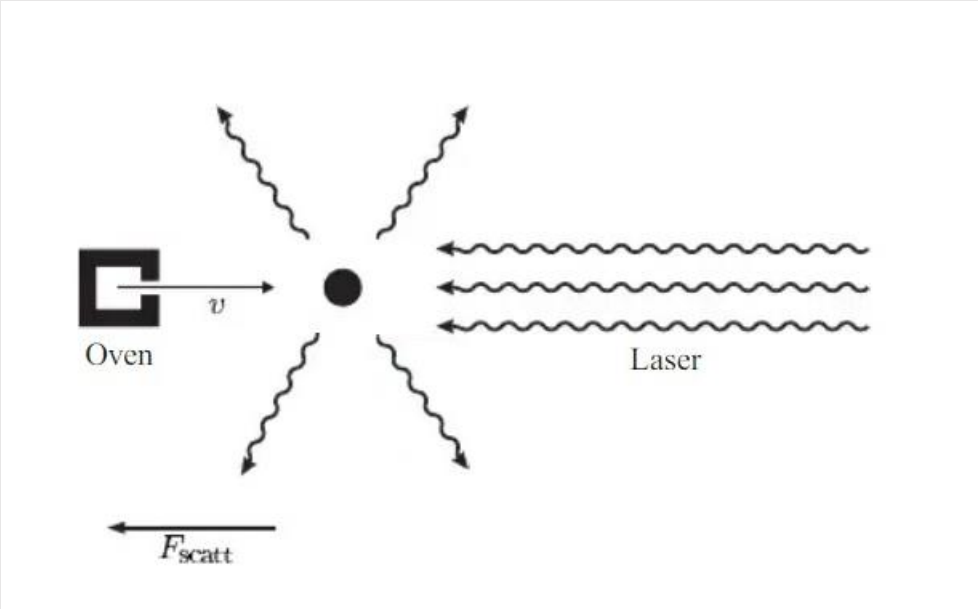
$$F_{scatt}=\hbar k\frac{\Gamma}{2}\frac{I/I_{sat}}{1+I/I_{sat}+4\delta^2/\Gamma^2}$$

(for very high laser intensity, I→ ∞, ρ_2→ ½ . Why? Because stimulated absorption and emission are symmetric, $B_{12}=B_{21}$)

Then, we have (highest possible deceleration force)

$$F_{max}=\frac{h\Gamma}{2\lambda}$$

Deceleration by scattering force



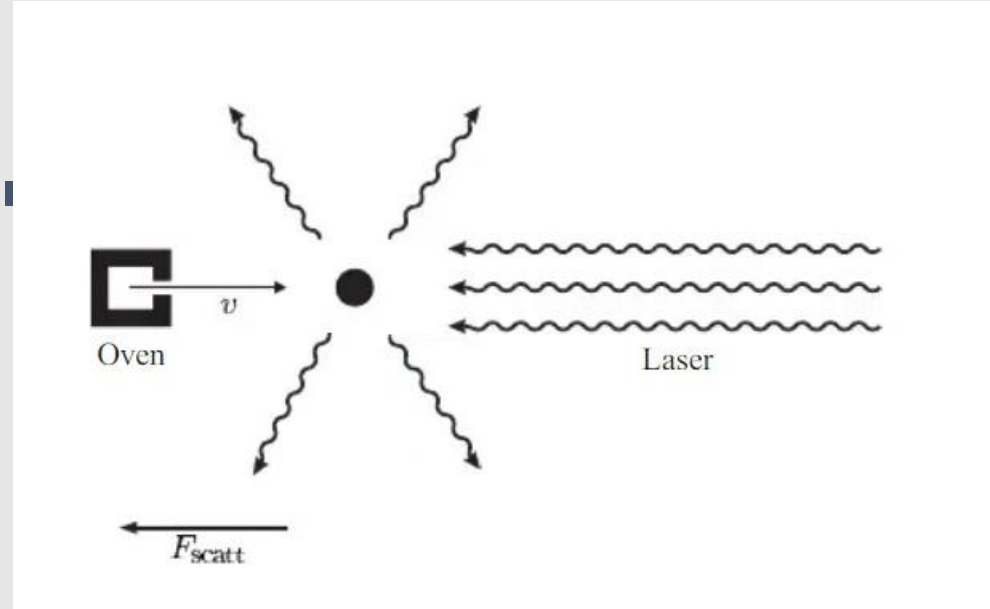
$$F_{max} = \frac{h\Gamma}{2\lambda}$$

$$a_{max} = \frac{F_{max}}{M} = \frac{h\Gamma}{2\lambda M} = \frac{p\Gamma}{M2} = \frac{v_r\Gamma}{2}$$

p

$v_r = \frac{h}{M\lambda}$ - recoil velocity (change in the velocity of the atom due to an absorption or emission of a photon)

Deceleration by scattering force



$$v_r = \hbar k/M \equiv h/(\lambda M)$$

Na $^2S_{1/2} \rightarrow {}^2P_{3/2}$ $\lambda = 589$ nm

$V_r = ?$

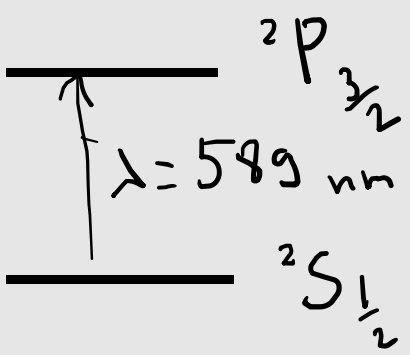
$V_r = \dfrac{h}{589\,nm \cdot M}$

$h = 6{,}626 \times 10^{-34}\ \dfrac{m^2 kg}{s}$

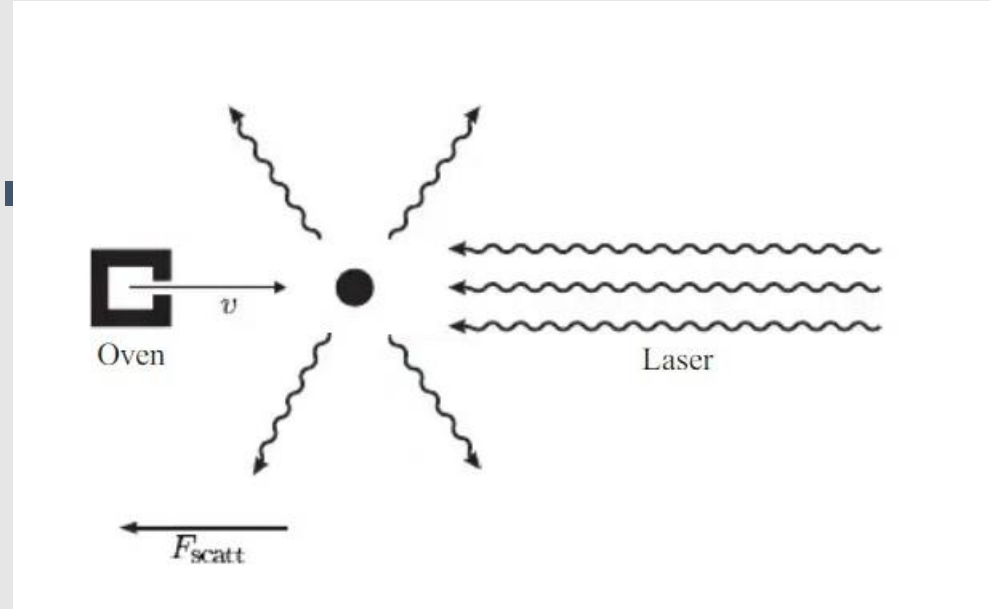
$M = 3{,}8 \times 10^{-26}$ kg

$\lambda = 5{,}89 \times 10^{-7}$ m

$V_r = 0{,}03\ \dfrac{m}{sec}$



Deceleration by scattering force



$$v_r = \hbar k/M \equiv h/(\lambda M)$$

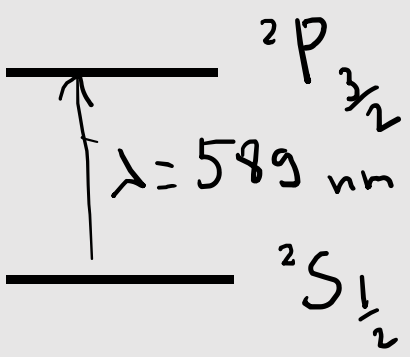
$$a_{max} = \frac{F_{max}}{M} = \frac{\hbar k}{M}\frac{\Gamma}{2} = \frac{v_r}{2\tau}$$

Na $^2S_{1/2} \rightarrow {}^2P_{3/2}$ $\lambda = 589$ nm

$a_{max} = ?$

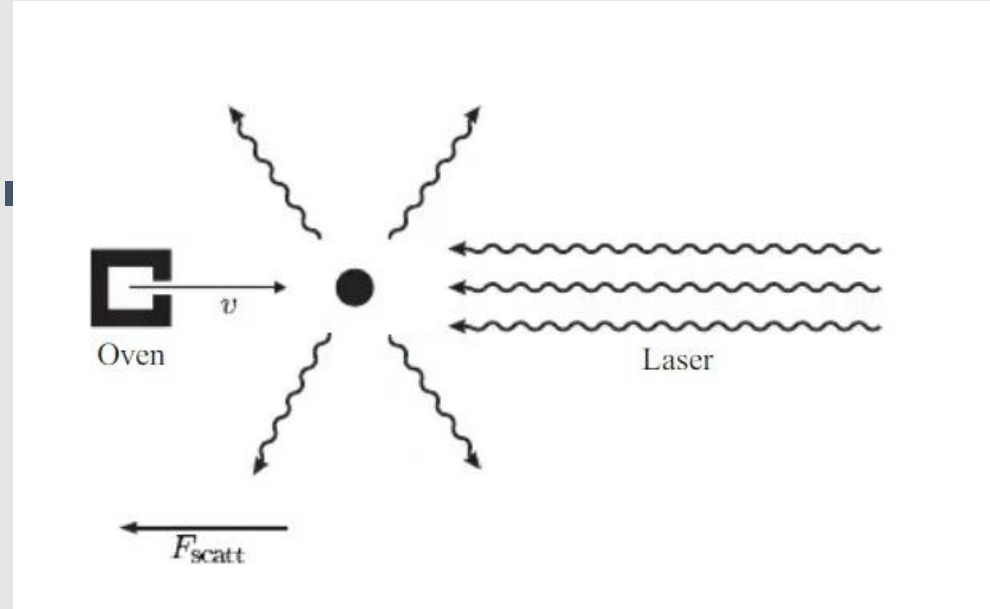
$V_r = 0.03$ m/sec

$\tau = 16$ ns



$$a_{max} = \frac{V_r}{2\tau} \sim 10^6 \text{ m/s}^2$$

Deceleration by scattering force

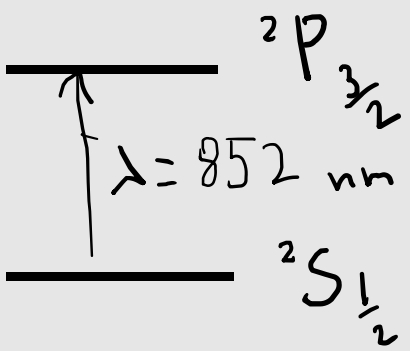


$$a_{max} = \frac{F_{max}}{M} = \frac{\hbar k}{M}\frac{\Gamma}{2} = \frac{v_r}{2\tau}$$

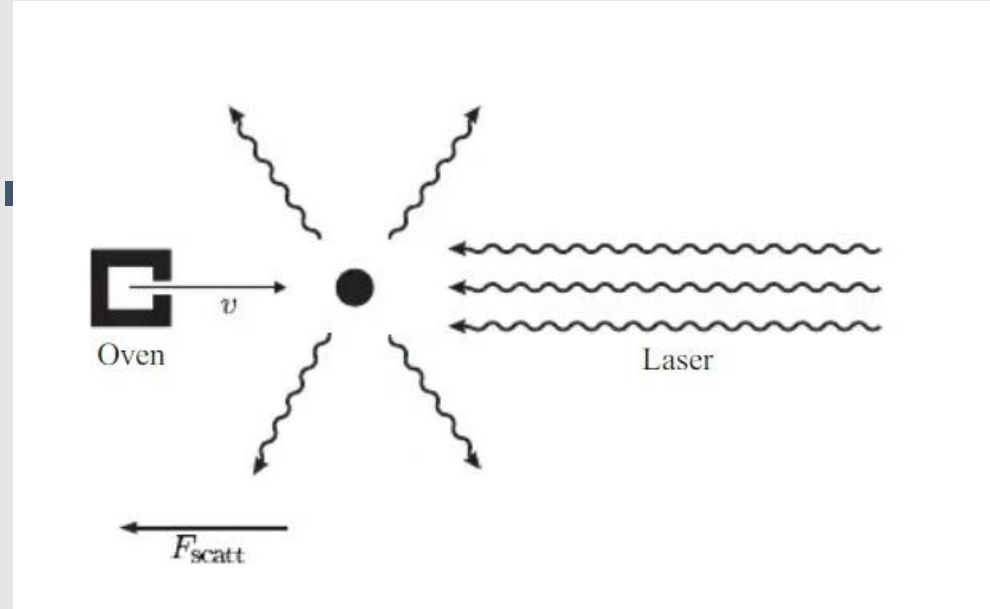
$$v_r = \hbar k/M \equiv h/(\lambda M)$$

Cs $^2S_{1/2} \rightarrow {}^2P_{3/2}$ $\lambda = 852$ nm

$v_r = ?$ $M = 2{,}21 \times 10^{-25}$ kg



Deceleration by scattering force



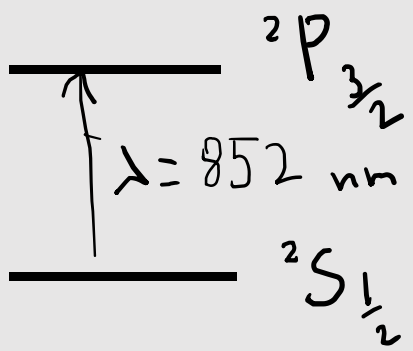
$$a_{max} = \frac{F_{max}}{M} = \frac{\hbar k}{M}\frac{\Gamma}{2} = \frac{v_r}{2\tau}$$

$$v_r = \hbar k/M \equiv h/(\lambda M)$$

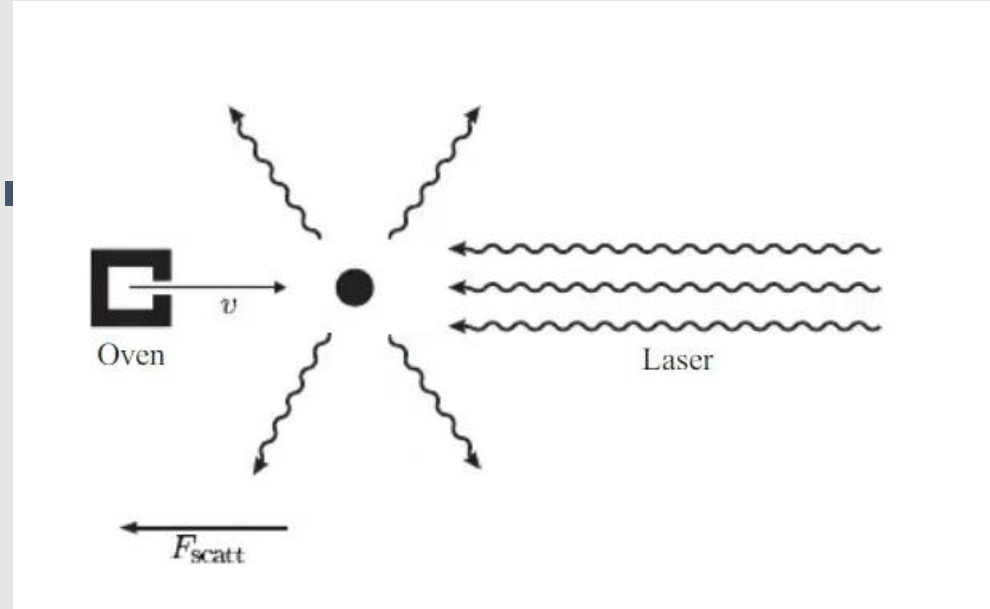
$Cs \quad {}^2S_{1/2} \rightarrow {}^2P_{3/2} \quad \lambda = 852 \text{ nm}$

$M = 2{,}21 \times 10^{-25} \text{ kg}$

$V_r = 0{,}0035 \frac{m}{sec}$



Deceleration by scattering force

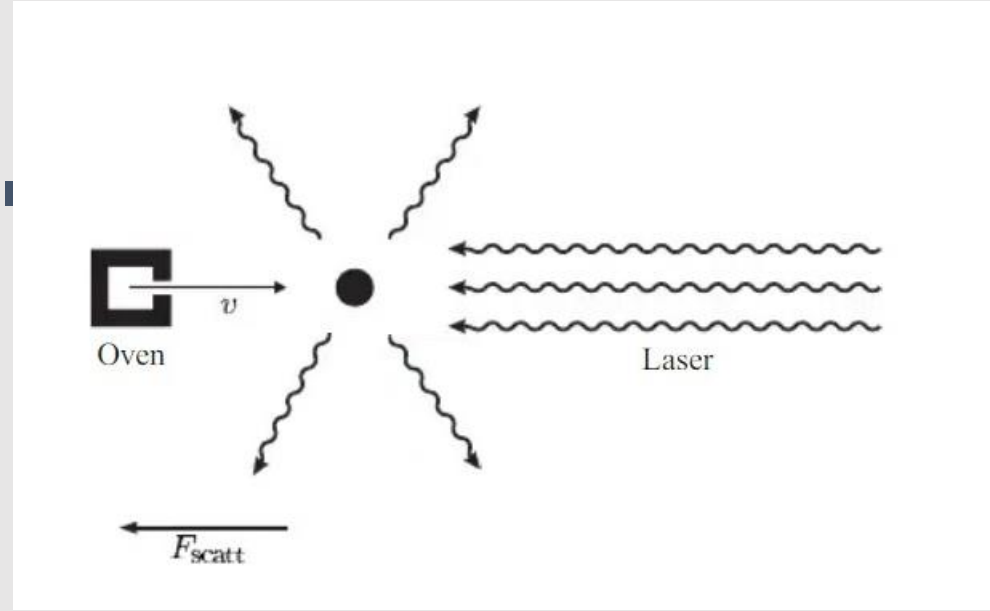


$$a_{max} = \frac{F_{max}}{M} = \frac{\hbar k}{M}\frac{\Gamma}{2} = \frac{v_r}{2\tau}$$

$$v_r = \hbar k/M \equiv h/(\lambda M)$$

Where will the atoms stop?

Deceleration by scattering force



$$a_{max} = \frac{F_{max}}{M} = \frac{\hbar k}{M}\frac{\Gamma}{2} = \frac{v_r}{2\tau}$$

$$v_r = \hbar k/M \equiv h/(\lambda M)$$

Where will the atoms stop?

For Na:

3600 km/h

Most probable velocity in beam ($T = 900\,\mathrm{K}$)	v_0	$1000\,\mathrm{m\,s^{-1}}$
Resonance wavelength	λ	589 nm
Lifetime of the excited state	τ	16ns
Recoil velocity	$v_r = h/(\lambda M)$	$3\,\mathrm{cm\,s^{-1}}$
Stopping distance (at half of the maximum deceleration)	$2v_0^2\tau/v_r$	1.1 m

$$v^2 = v_0^2 + 2z\,a$$

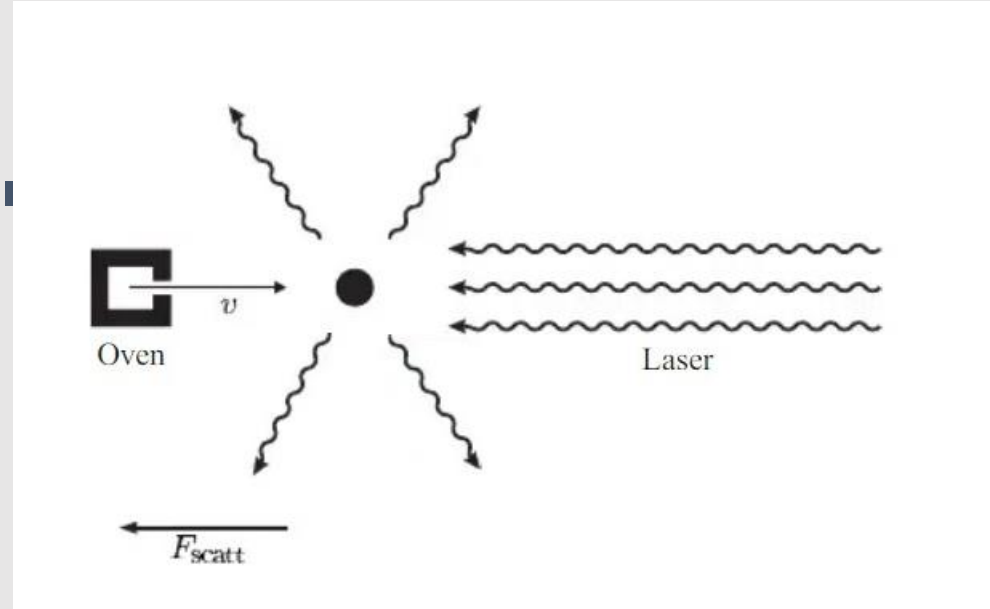
$v_0 = 1000$ m/s

$a = \frac{a_{max}}{2}$

$v = 0$

$z = 1$ m

Deceleration by scattering force



$$a_{max} = \frac{F_{max}}{M} = \frac{\hbar k}{M}\frac{\Gamma}{2} = \frac{v_r}{2\tau}$$

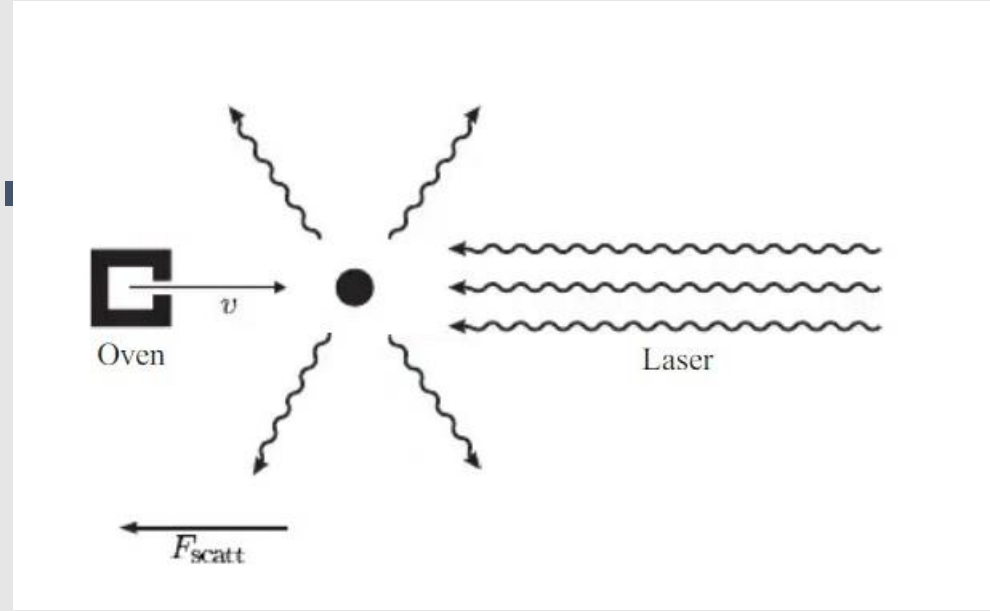
$$v_r = \hbar k/M \equiv h/(\lambda M)$$

How many photons will be scattered?

For Na:

Most probable velocity in beam ($T = 900\,\mathrm{K}$)	v_0	$1000\,\mathrm{m\,s^{-1}}$
Resonance wavelength	λ	$589\,\mathrm{nm}$
Lifetime of the excited state	τ	16ns
Recoil velocity	$v_r = h/(\lambda M)$	$3\,\mathrm{cm\,s^{-1}}$
Stopping distance (at half of the maximum deceleration)	$2v_0^2\tau/v_r$	1.1 m

Deceleration by scattering force



$$a_{max} = \frac{F_{max}}{M} = \frac{\hbar k}{M}\frac{\Gamma}{2} = \frac{v_r}{2\tau}$$

$$v_r = \hbar k/M \equiv h/(\lambda M)$$

1.1 m

500 m/s (on average)

$t = 0{,}0022$ s

$\frac{\Gamma}{2} = \frac{1}{\tau} \cdot \frac{1}{2} = 31\,000\,000\ \frac{1}{s}$

$\tau = 16$ ns

~70000 photons scattered

For Na:

Most probable velocity in beam ($T = 900$ K)	v_0	1000 m s^{-1}
Resonance wavelength	λ	589 nm
Lifetime of the excited state	τ	16ns
Recoil velocity	$v_r = h/(\lambda M)$	3 cm s^{-1}
Stopping distance (at half of the maximum deceleration)	$2v_0^2\tau/v_r$	1.1 m

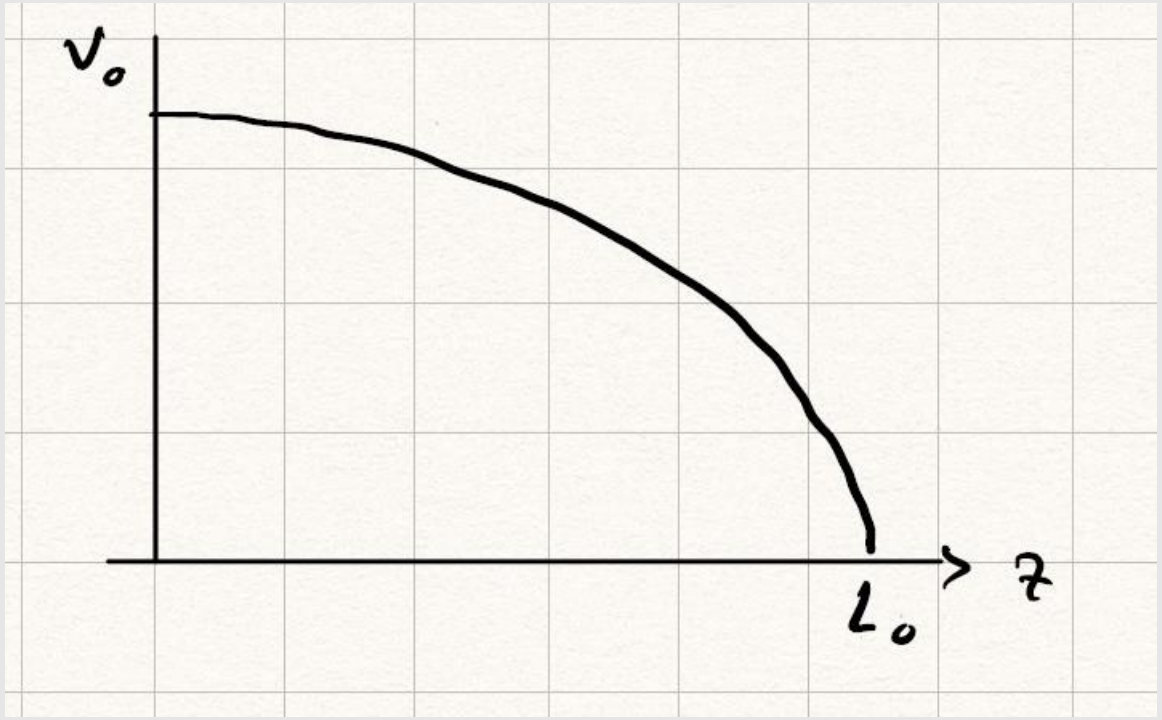
Deceleration by scattering force

- What is wrong with this story?

Deceleration by scattering force

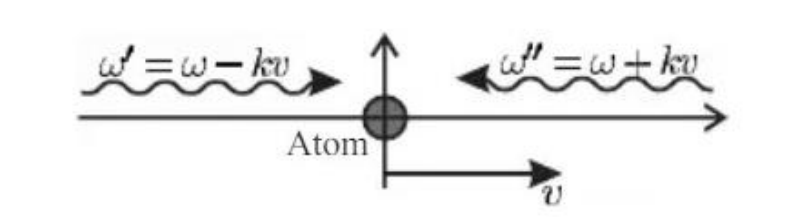
- What is wrong with this story?

$$V = V_0 \sqrt{1 - \frac{z}{L_0}}$$



Doppler shift

- Atoms absorb radiation at frequency ω in the rest frame
- Say we have an atom moving at velocity *v:*



$$k = \frac{\omega}{c} = \frac{2\pi}{\lambda}$$

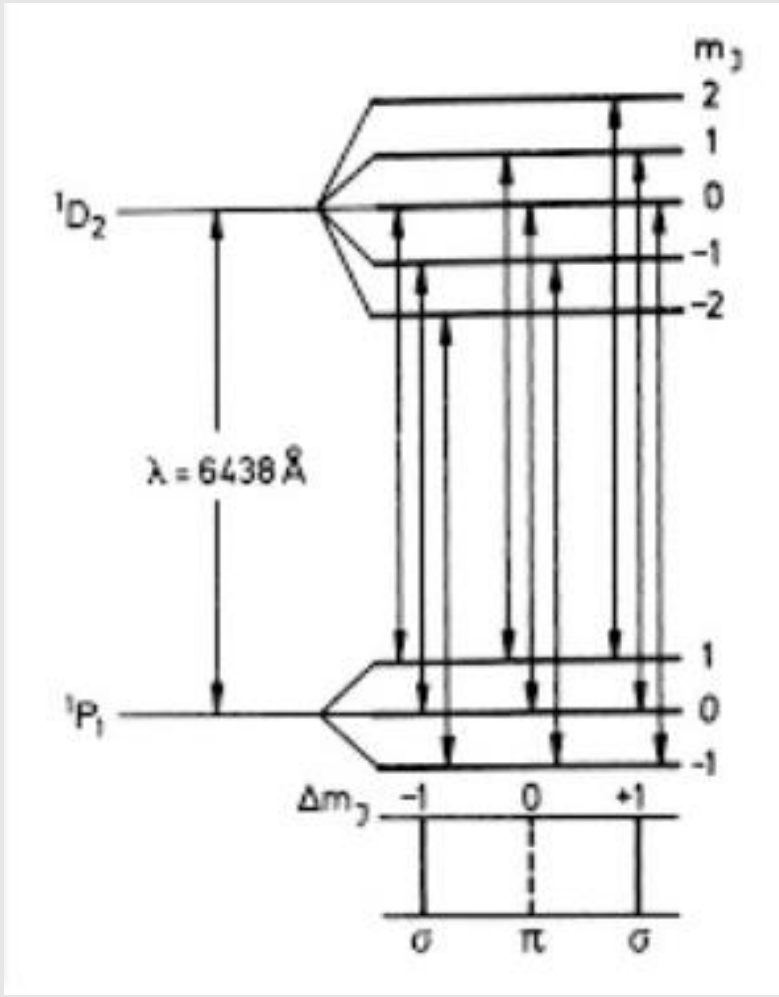
- It will see different frequency radiation, depending on whether it's moving towards or away from its source.
- That means we need to account for the Doppler shift when selecting the laser frequency

[in general, $f = \left(1 + \frac{\Delta v}{c}\right) f_0$]

(Zeeman effect)

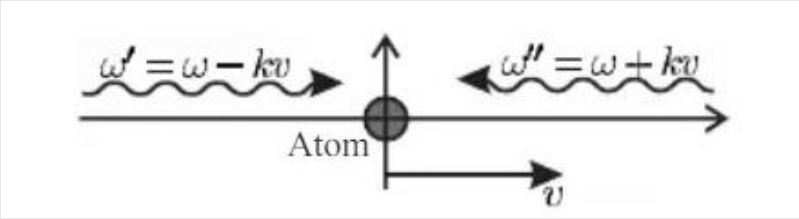
$$E_{\mathrm{ZE}} = g_J \mu_{\mathrm{B}} B M_J$$

$$g_J = \frac{3}{2} + \frac{S(S+1) - L(L+1)}{2J(J+1)}$$



The levels will split according to M_J and shift proportionally to the strength of the applied magnetic field

Zeeman slowing technique



Compensate for the changing Doppler shift by applying a varying magnetic field.

$$v = v_0\sqrt{1 - \frac{z}{L_0}} \quad \Longrightarrow$$

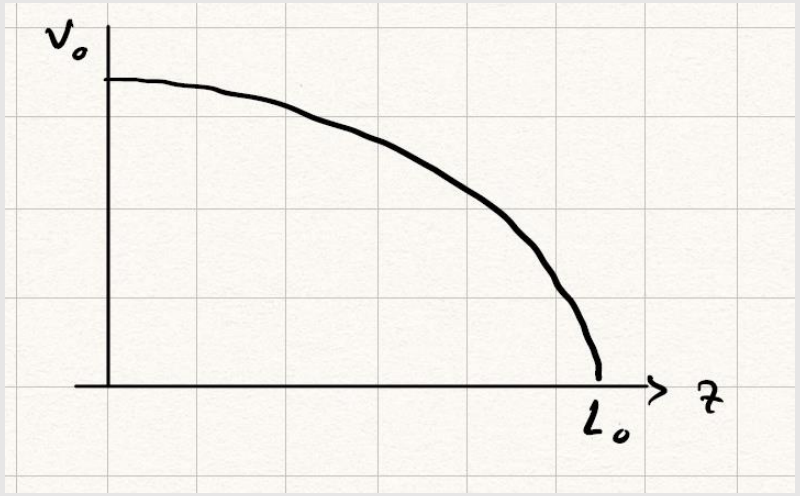
$$\omega_0 + \frac{\mu_B B(z)}{\hbar} = \omega + kv$$

Doppler shift

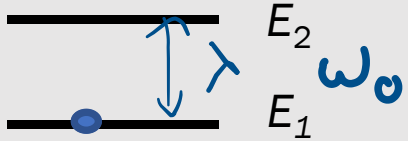
Laser frequency

Resonance frequency

Zeeman shift



$$E_{ZE} = g_J \mu_B B M_J$$

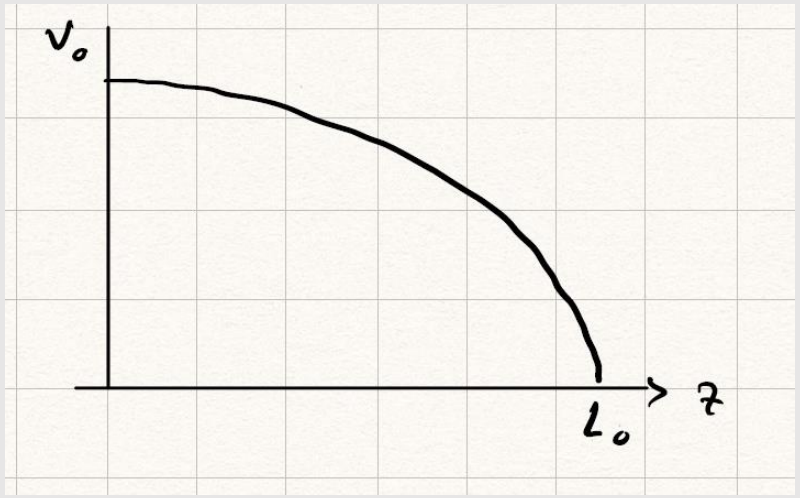


$E = \hbar\omega$

Zeeman slowing technique

Compensate for the changing Doppler shift by applying a varying magnetic field.

$$v = v_0\sqrt{1 - \frac{z}{L_0}} \quad \Longrightarrow \quad \omega_0 + \frac{\mu_B B(z)}{\hbar} = \omega + kv \qquad k = \frac{2\pi}{\lambda}$$



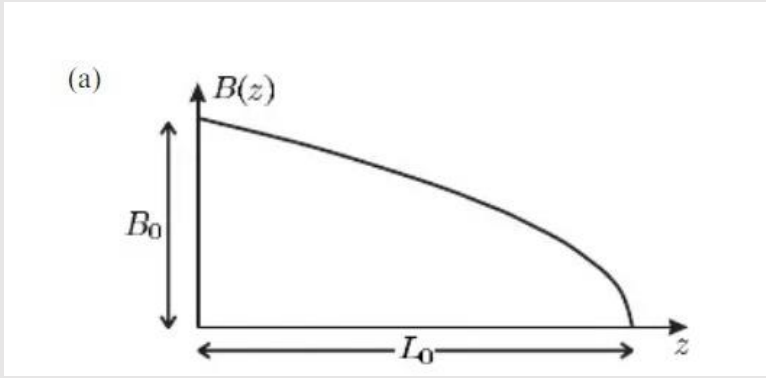
Maximum velocity that can be captured

$$B(z) = B_0\left(1 - \frac{z}{L_0}\right)^{1/2} + B_{\text{bias}}$$

$$B_0 = \frac{\hbar v_0}{\lambda \mu_B}$$

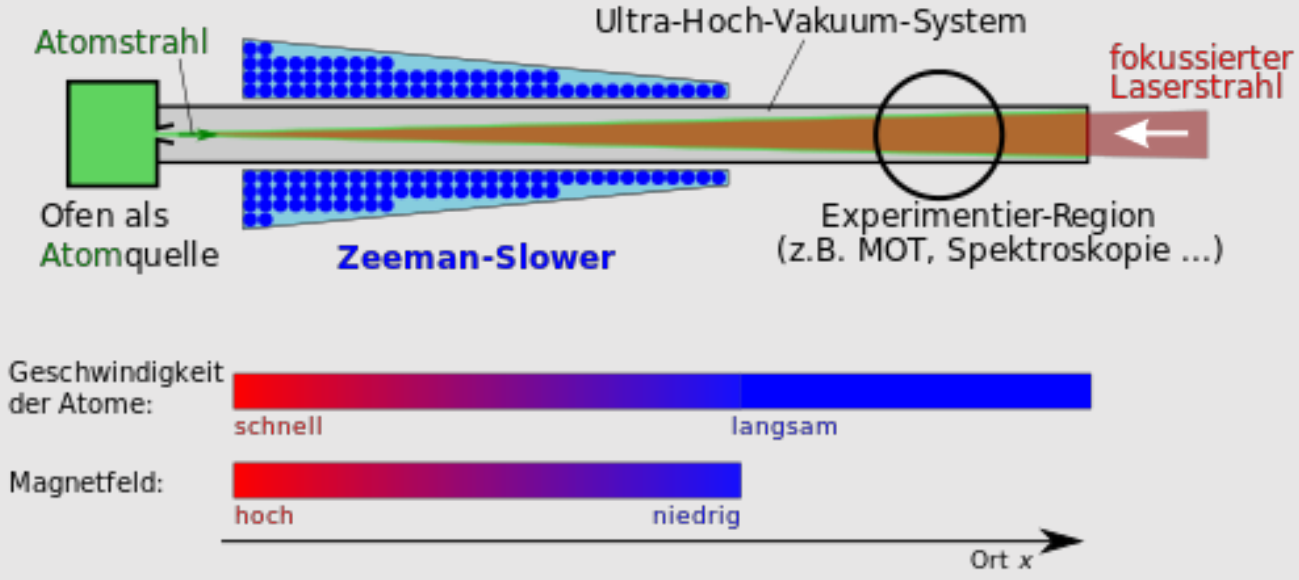
$$\mu_B B_{\text{bias}} = \hbar\omega - \hbar\omega_0$$

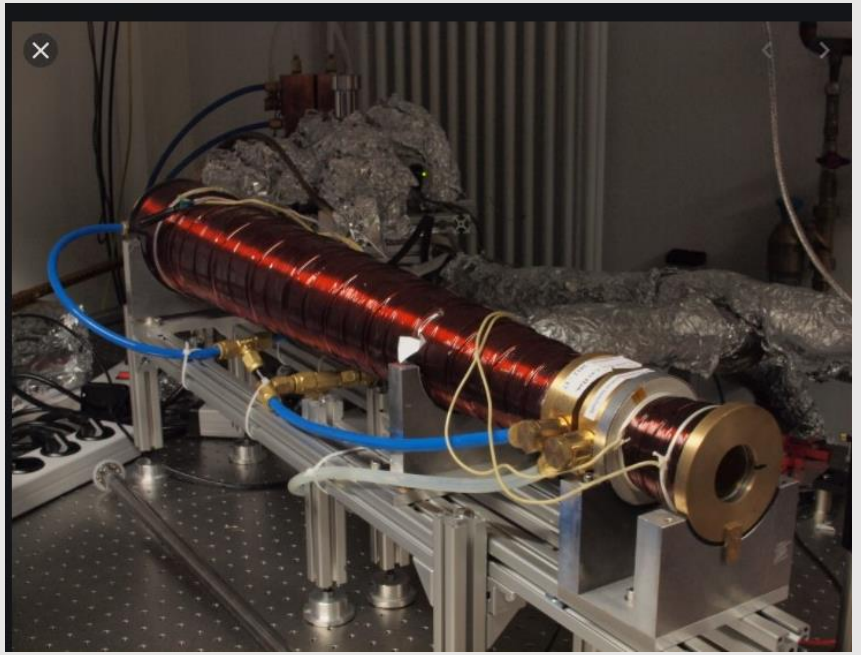
$$E_{\text{ZE}} = g_J \mu_B B M_J$$

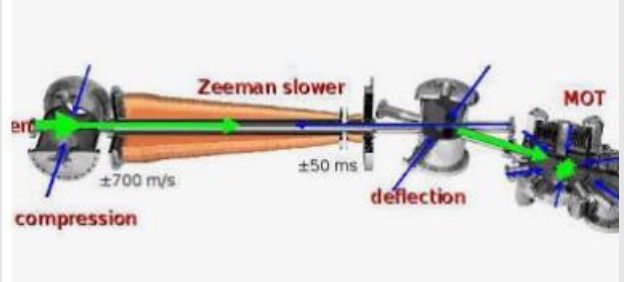


Zeeman slowing technique

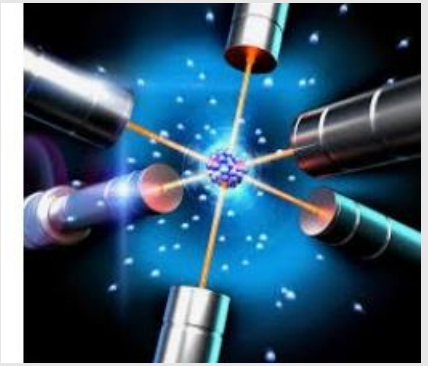
Compensate for the changing Doppler shift by applying a varying magnetic field.

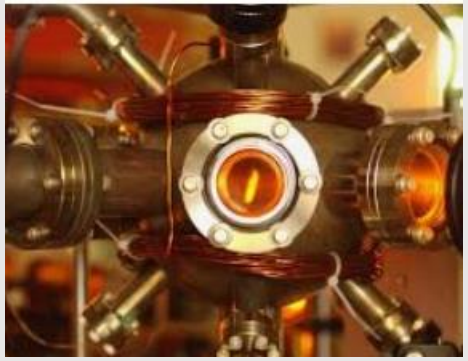


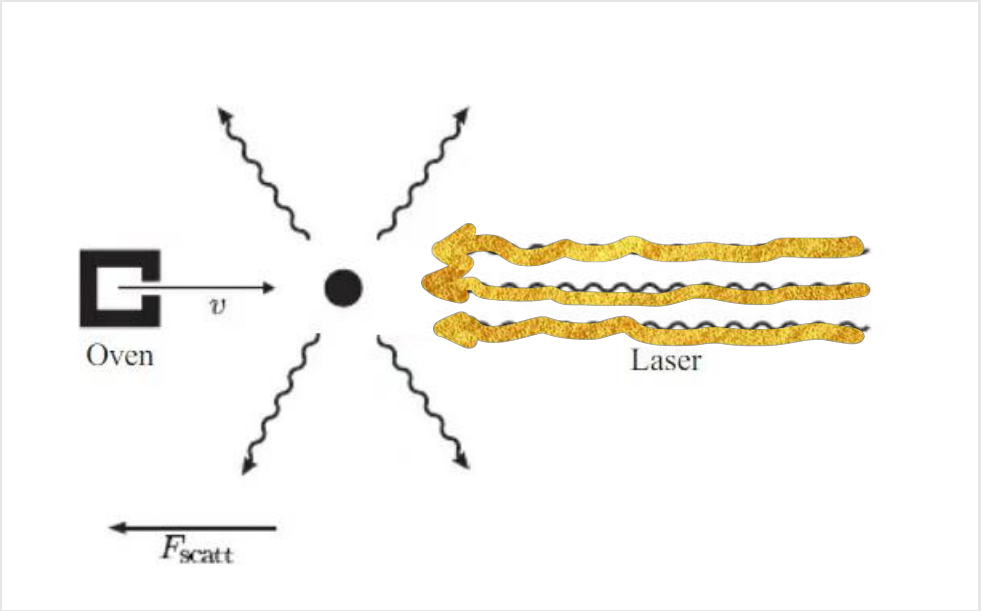


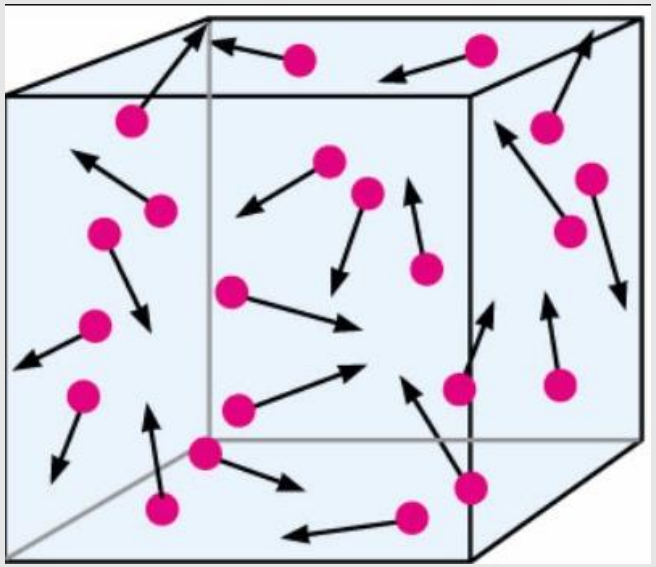


- Slowing
- Cooling
- Trapping

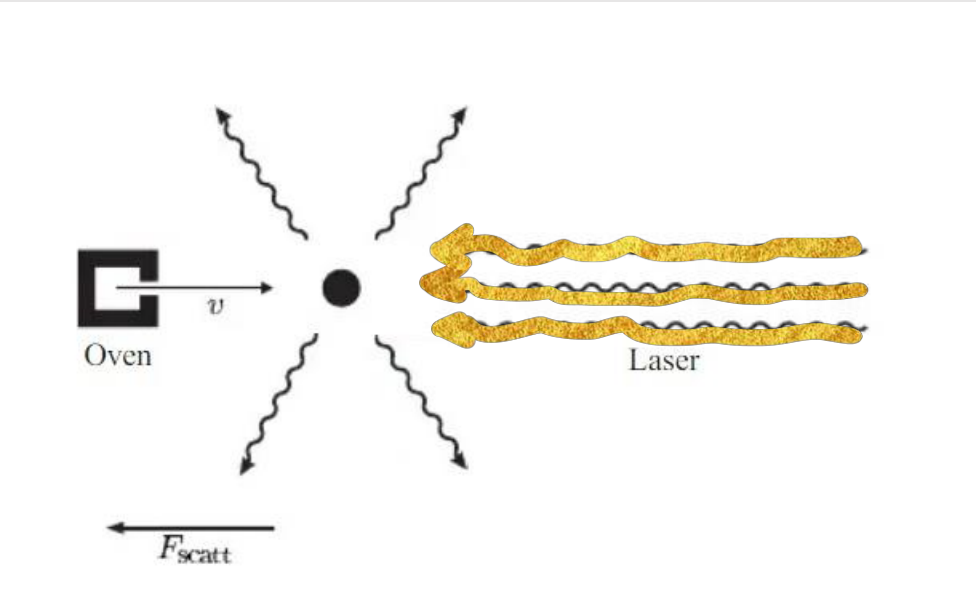


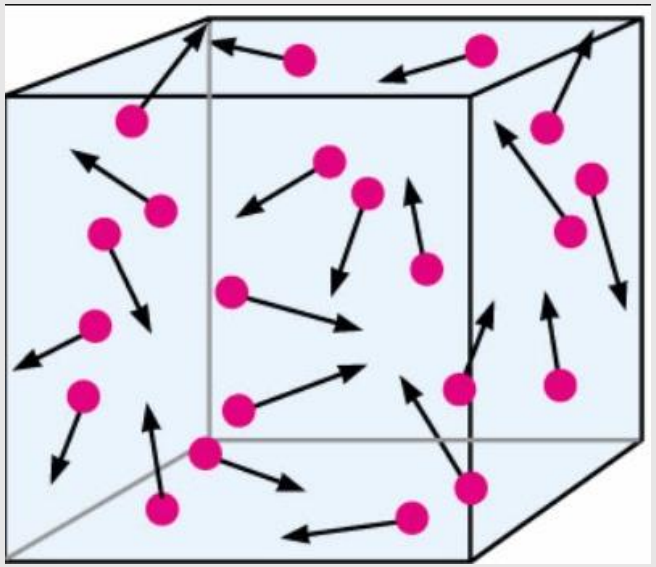



v
Oven
Laser
F_{scatt}



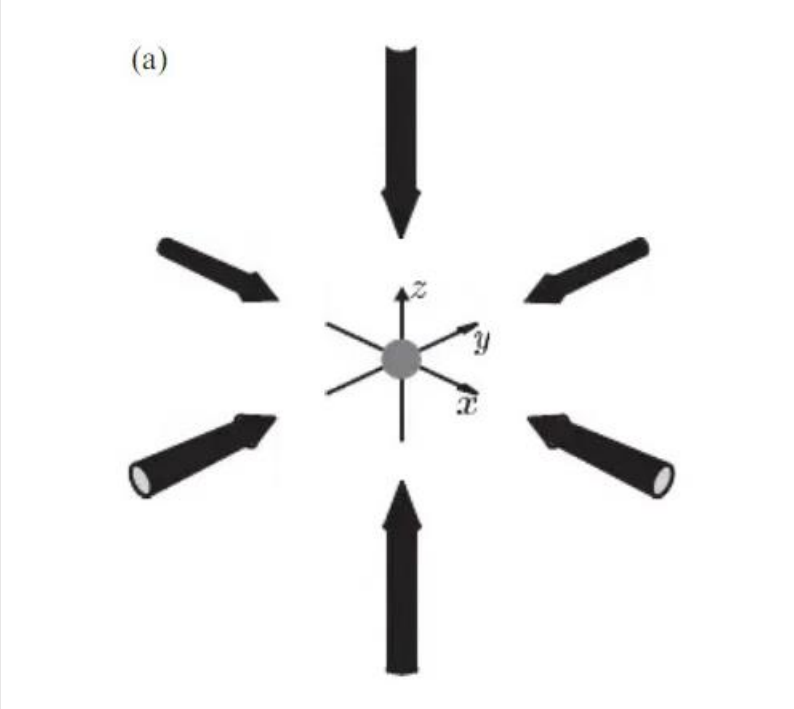
?





?

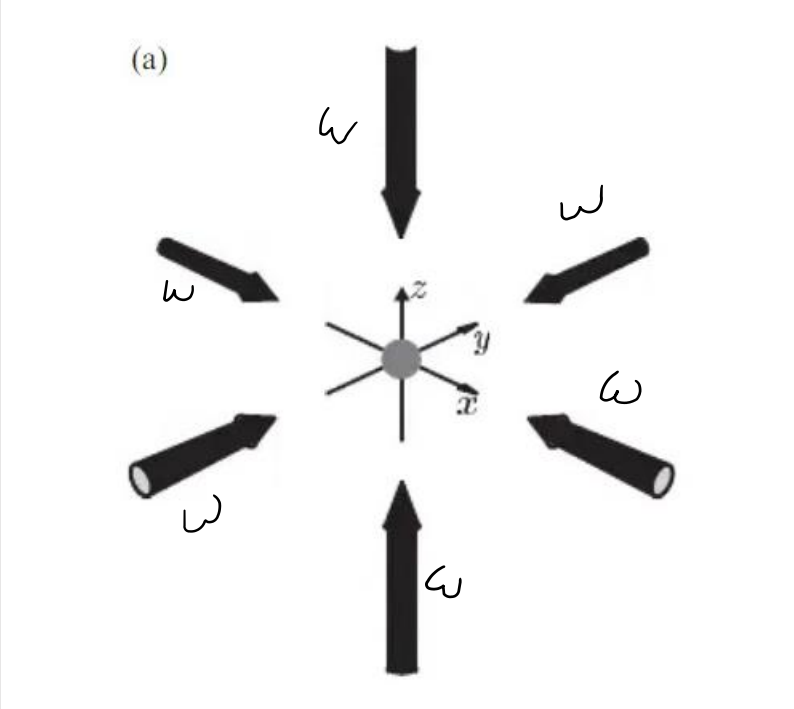
Add more lasers!



Optical molasses

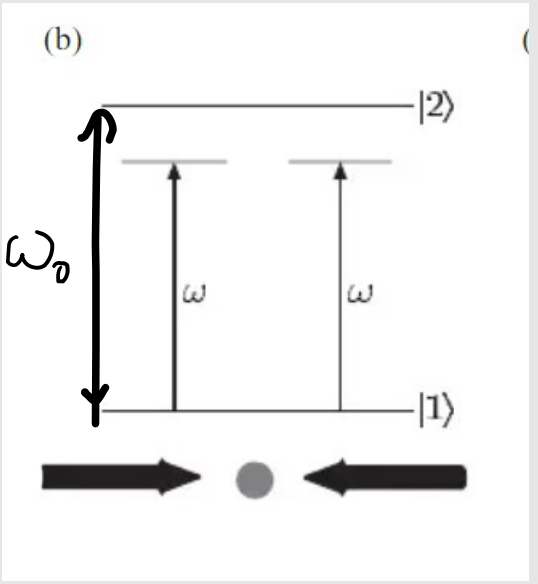


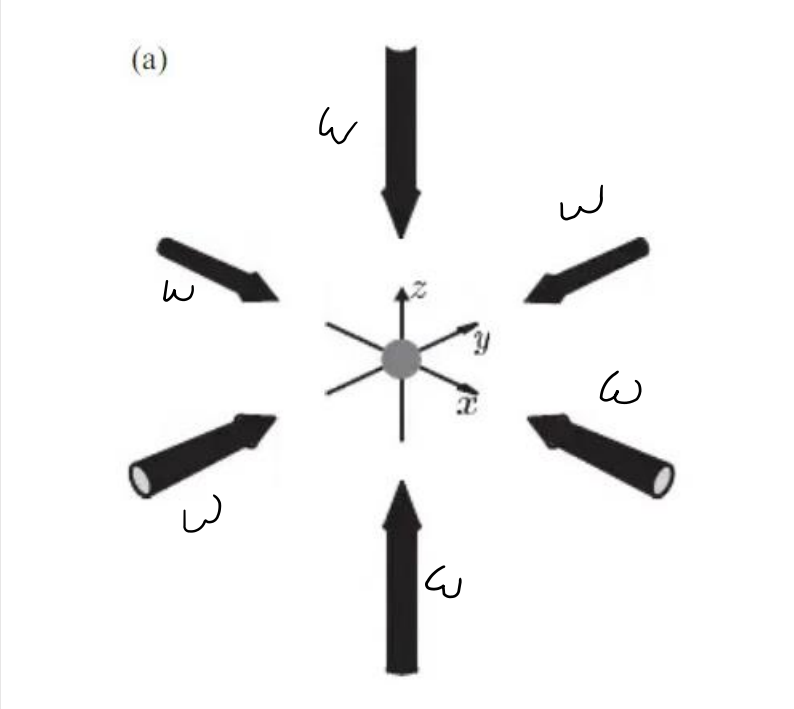




All the lasers have the same frequency, red-detuned from resonance (that is, slightly lower than the transition frequency)

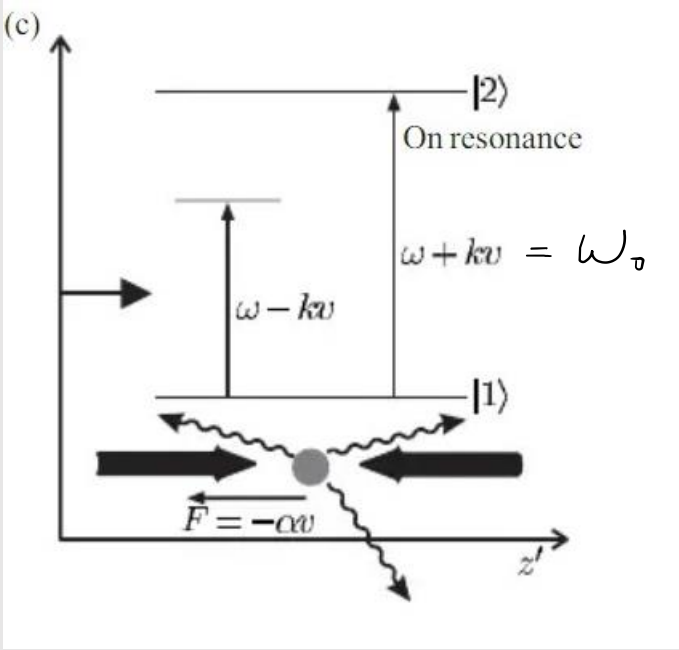
$\omega < \omega_0$

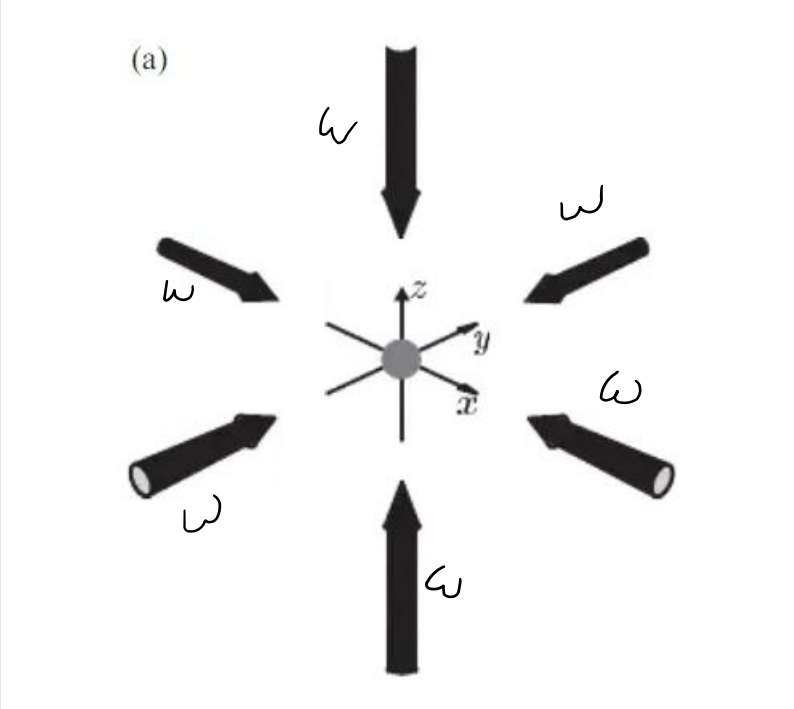




All the lasers have the same frequency, red-detuned from resonance

$\omega < \omega_0$



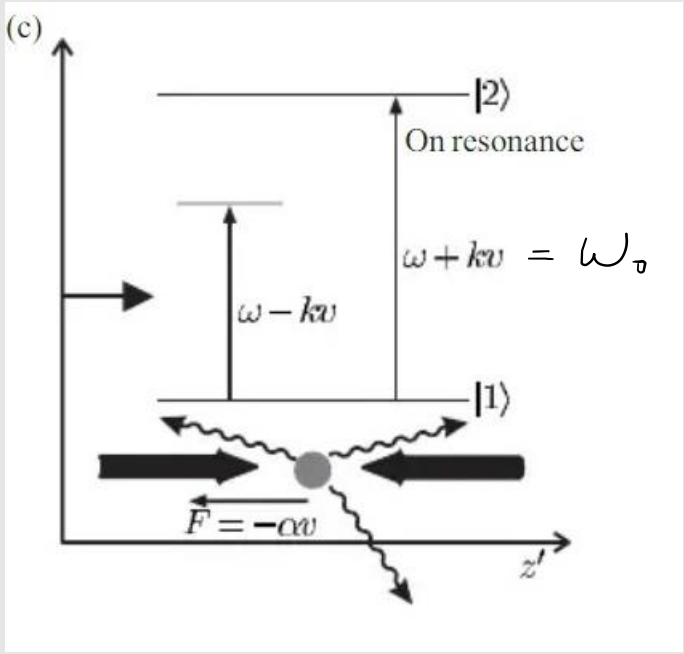


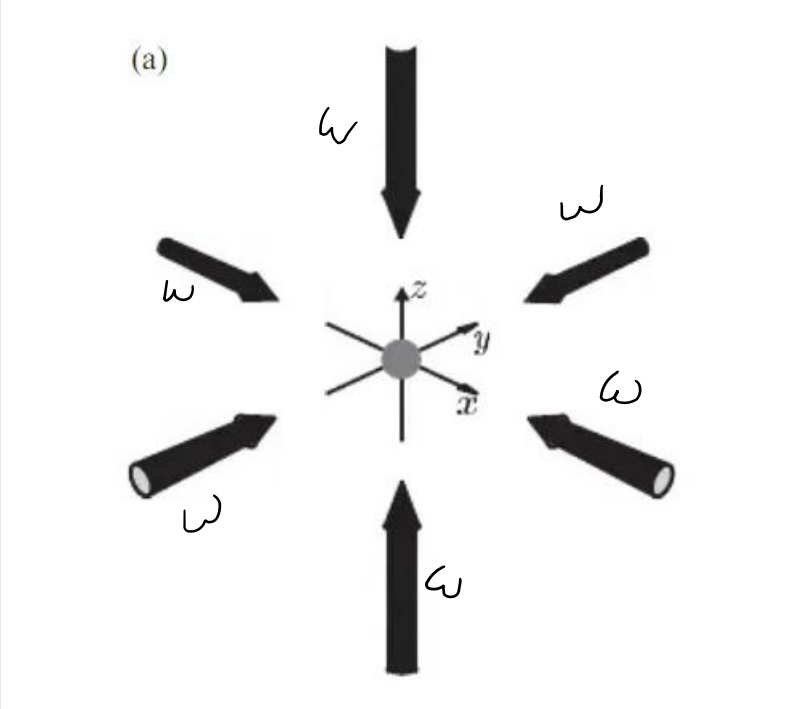
All the lasers have the same frequency, red-detuned from resonance

$\omega < \omega_0$

$k = \frac{\omega}{c} = \frac{2\pi}{\lambda}$

$$F_{\text{molasses}} = F_{\text{scatt}}\left(\omega - \omega_0 - kv\right) - F_{\text{scatt}}\left(\omega - \omega_0 + kv\right)$$

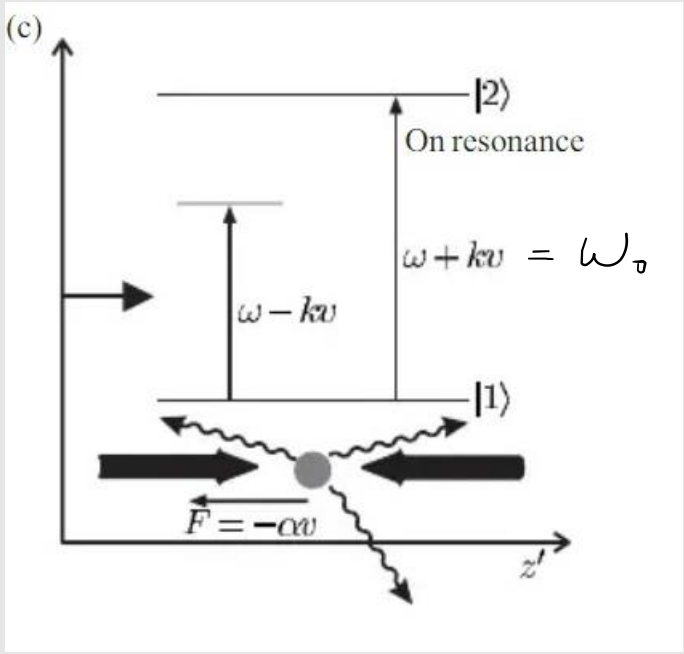




All the lasers have the same frequency, red-detuned from resonance

$$k = \frac{\omega}{c} = \frac{2\pi}{\lambda}$$

$$\omega < \omega_0$$

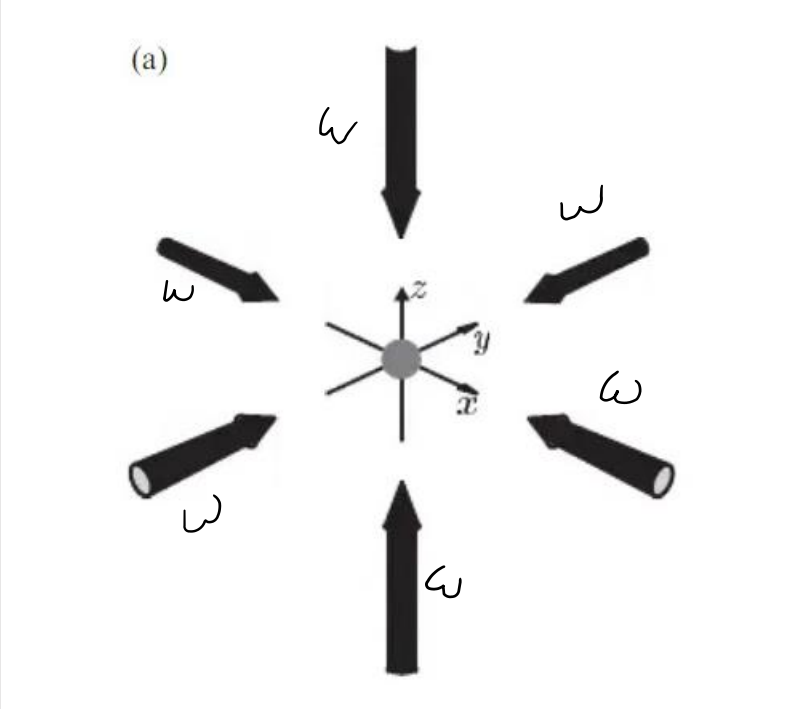


For small *v* (a few m/sec), use Taylor series:

$$f(x \pm h) = f(x) \pm hf'(x) +$$

. . .

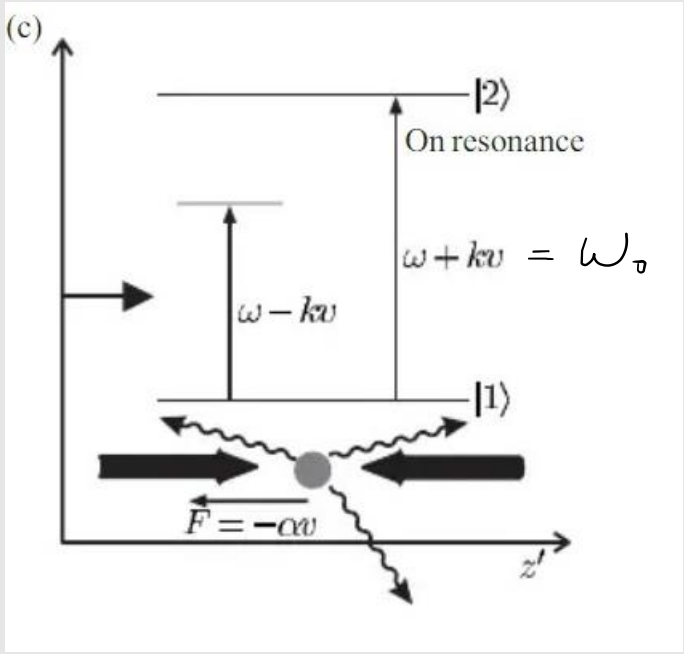
$$F_{\text{molasses}} = F_{\text{scatt}}(\omega - \omega_0 - kv) - F_{\text{scatt}}(\omega - \omega_0 + kv)$$
$$\simeq F_{\text{scatt}}(\omega - \omega_0) - kv\frac{\partial F}{\partial \omega} - \left[F_{\text{scatt}}(\omega - \omega_0) + kv\frac{\partial F}{\partial \omega}\right]$$



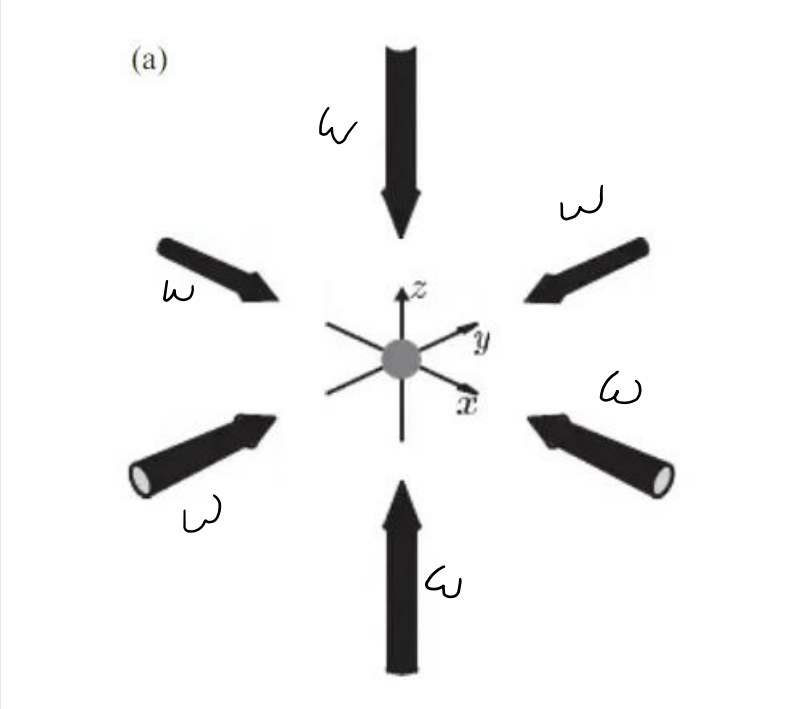
All the lasers have the same frequency, red-detuned from resonance

$\omega < \omega_0$

$k = \frac{\omega}{c} = \frac{2\pi}{\lambda}$

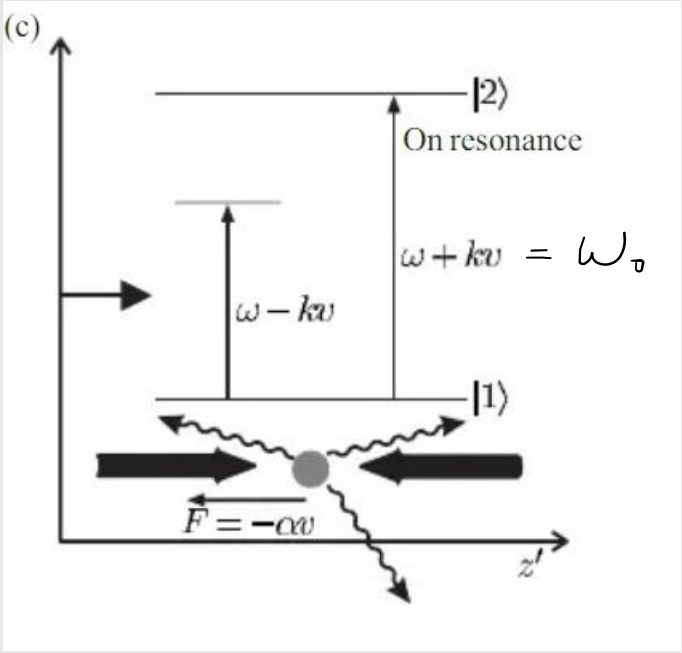


$$
\begin{aligned}
F_{\text{molasses}} &= F_{\text{scatt}}\left(\omega-\omega_0-kv\right)-F_{\text{scatt}}\left(\omega-\omega_0+kv\right) \\
&\simeq F_{\text{scatt}}\left(\omega-\omega_0\right)-kv\frac{\partial F}{\partial \omega}-\left[F_{\text{scatt}}\left(\omega-\omega_0\right)+kv\frac{\partial F}{\partial \omega}\right] \\
&\simeq -2\frac{\partial F}{\partial \omega}kv\,. \qquad (9.15)
\end{aligned}
$$

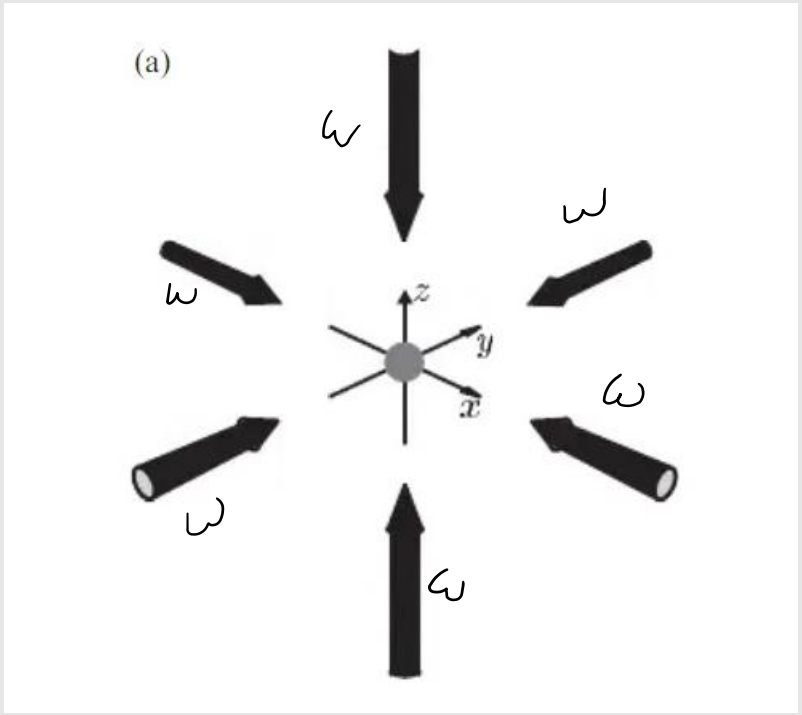


All the lasers have the same frequency, red-detuned from resonance

$\omega < \omega_0$



$$
\begin{aligned}
F_{\text{molasses}} &= F_{\text{scatt}}\left(\omega-\omega_0-kv\right)-F_{\text{scatt}}\left(\omega-\omega_0+kv\right) \\
&\simeq F_{\text{scatt}}\left(\omega-\omega_0\right)-kv\frac{\partial F}{\partial\omega}-\left[F_{\text{scatt}}\left(\omega-\omega_0\right)+kv\frac{\partial F}{\partial\omega}\right] \\
&\simeq -2\frac{\partial F}{\partial\omega}kv \doteq \textbf{-}\alpha v
\end{aligned}
\tag{9.15}
$$



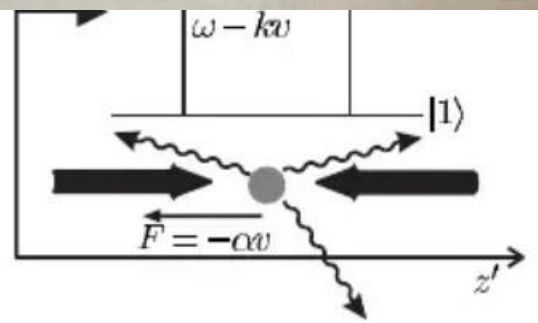


he same frequency, red-detuned from

$\omega < \omega$

12.1.31 A particle of mass m moves in a viscous fluid which resists motion with a force of magnitude $F = c|\vec{v}|$, where $\vec{v}$ is the velocity.

$$F_{\text{molasses}} = F_{\text{scatt}}(\omega - \omega_0 - kv) - F_{\text{scatt}}(\omega - \omega_0 + kv)$$
$$\simeq F_{\text{scatt}}(\omega - \omega_0) - kv\frac{\partial F}{\partial \omega} - \left[F_{\text{scatt}}(\omega - \omega_0) + kv\frac{\partial F}{\partial \omega}\right]$$
$$\simeq -2\frac{\partial F}{\partial \omega}kv \coloneqq -\alpha v \qquad (9.15)$$



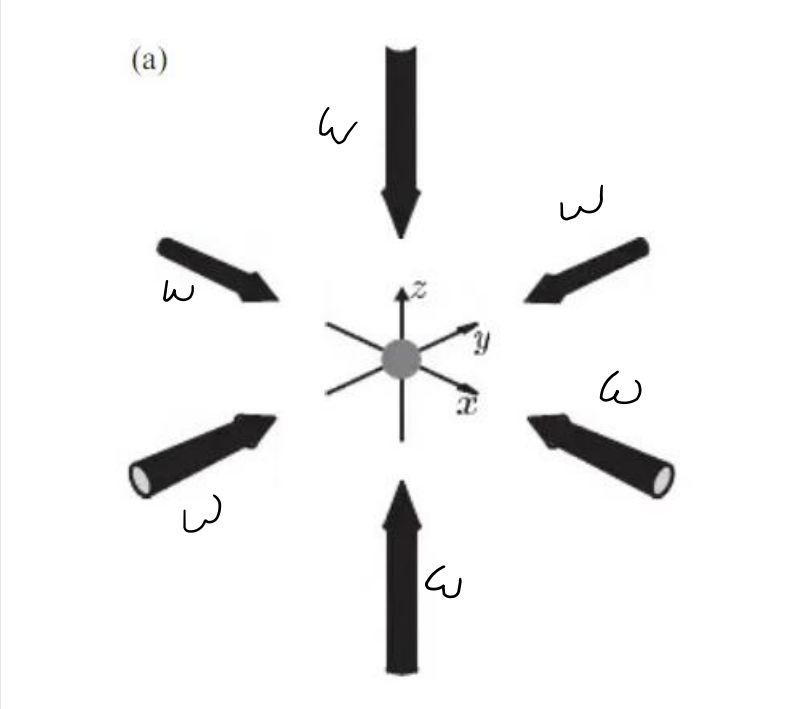
$$F_{\text{scatt}} = \hbar k \frac{\Gamma}{2} \frac{I / I_{\text{sat}}}{1 + I/I_{\text{sat}} + 4\delta^2/\Gamma^2}$$

$k=\omega/c=2\pi/\lambda$

I- laser intensity

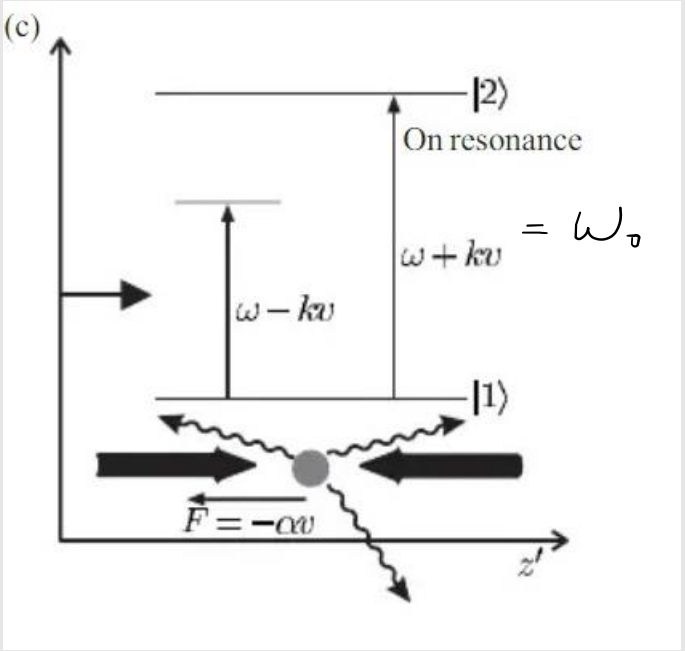
I_{sat}- saturation intensity

$\delta = \omega - \omega_0$ (detuning)

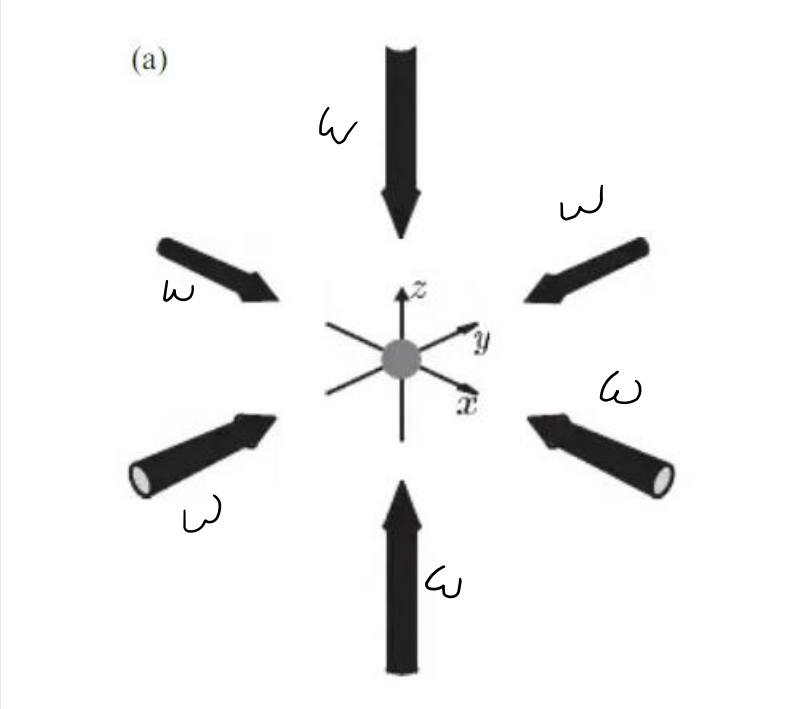


$$\alpha = 2k\frac{\partial F}{\partial \omega} = 4\hbar k^2 \frac{I}{I_{\text{sat}}} \frac{-2\delta/\Gamma}{\left[1 + (2\delta/\Gamma)^2\right]^2}$$

$$\begin{aligned} F_{\text{molasses}} &= F_{\text{scatt}}(\omega - \omega_0 - kv) - F_{\text{scatt}}(\omega - \omega_0 + kv) \\ &\simeq F_{\text{scatt}}(\omega - \omega_0) - kv\frac{\partial F}{\partial \omega} - \left[F_{\text{scatt}}(\omega - \omega_0) + kv\frac{\partial F}{\partial \omega}\right] \\ &\simeq -2\frac{\partial F}{\partial \omega}kv \doteq -\alpha V \end{aligned} \qquad (9.15)$$



$$F_{\text{scatt}} = \hbar k \frac{\Gamma}{2} \frac{I / I_{\text{sat}}}{1 + I/I_{\text{sat}} + 4\delta^2/\Gamma^2}$$

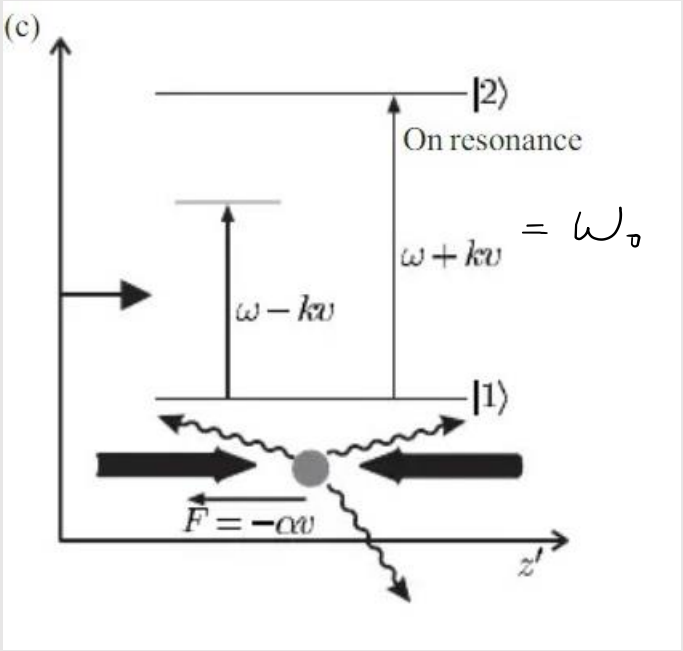


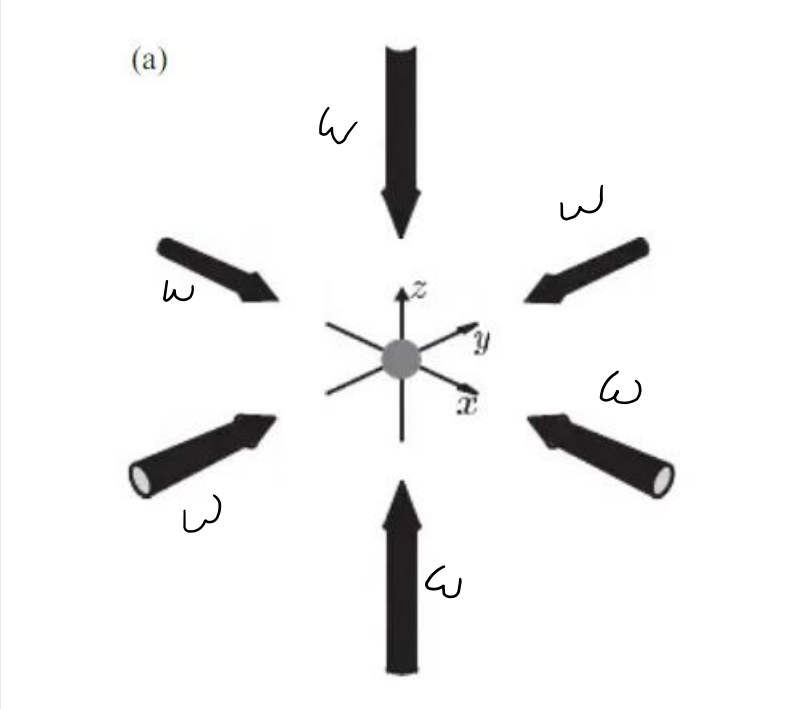
$$\alpha = 2k\frac{\partial F}{\partial \omega} = 4\hbar k^2 \frac{I}{I_{\text{sat}}} \frac{-2\delta/\Gamma}{\left[1 + (2\delta/\Gamma)^2\right]^2}$$

$\frac{I}{I_{sat}} \ll 1$

$\delta = \omega - \omega_0 < 0$

$$\begin{aligned} F_{\text{molasses}} &= F_{\text{scatt}}(\omega - \omega_0 - kv) - F_{\text{scatt}}(\omega - \omega_0 + kv) \\ &\simeq F_{\text{scatt}}(\omega - \omega_0) - kv\frac{\partial F}{\partial \omega} - \left[F_{\text{scatt}}(\omega - \omega_0) + kv\frac{\partial F}{\partial \omega}\right] \\ &\simeq -2\frac{\partial F}{\partial \omega}kv = -\alpha v \end{aligned} \qquad (9.15)$$



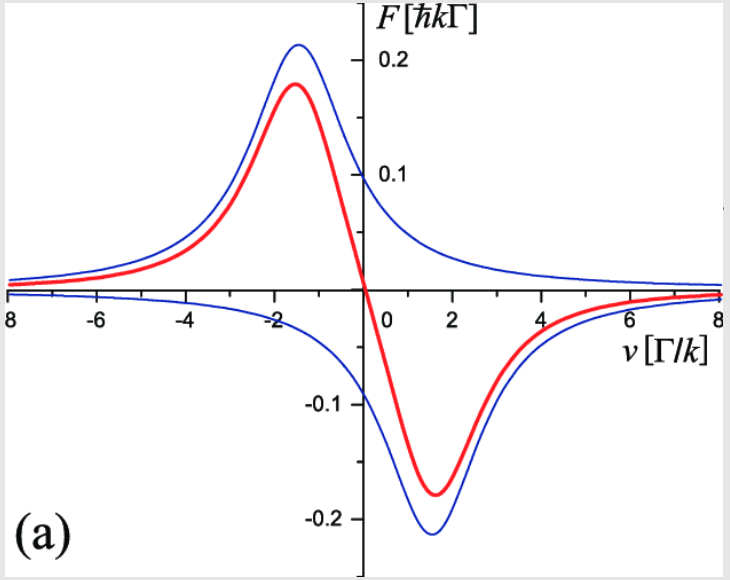


$$F_{\text{scatt}} = \hbar k \frac{\Gamma}{2} \frac{I / I_{\text{sat}}}{1 + I/I_{\text{sat}} + 4\delta^2/\Gamma^2}$$

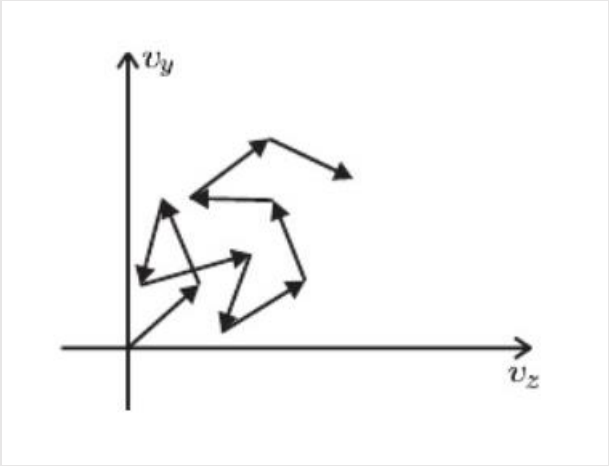
$$\alpha = 2k\frac{\partial F}{\partial \omega} = 4\hbar k^2 \frac{I}{I_{\text{sat}}} \frac{-2\delta/\Gamma}{\left[1 + (2\delta/\Gamma)^2\right]^2}$$

$\frac{I}{I_{sat}} \ll 1$ $\qquad$ $\delta = \omega - \omega_0 < 0$

$F_{molasses} = -\alpha v$



- Doppler cooling limit



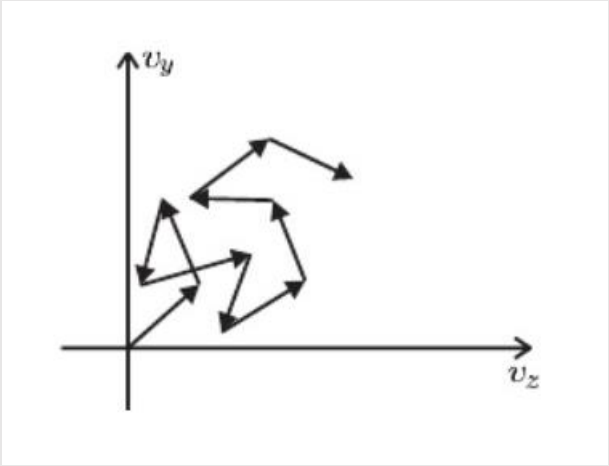
$$k_{\mathrm{B}} T_{\mathrm{D}} = \frac{\hbar \Gamma}{2}$$

$$\left[\frac{1}{2} M \bar{v}^2 = k_B T_D \right]$$

$$(\bar{v}^2)^{1/2} = \sqrt{\frac{k_B T_D \cdot 2}{M}} =$$

$$= \sqrt{\frac{k_B \hbar \Gamma}{M}}$$

- Doppler cooling limit



$$k_B T_D = \frac{\hbar \Gamma}{2}$$

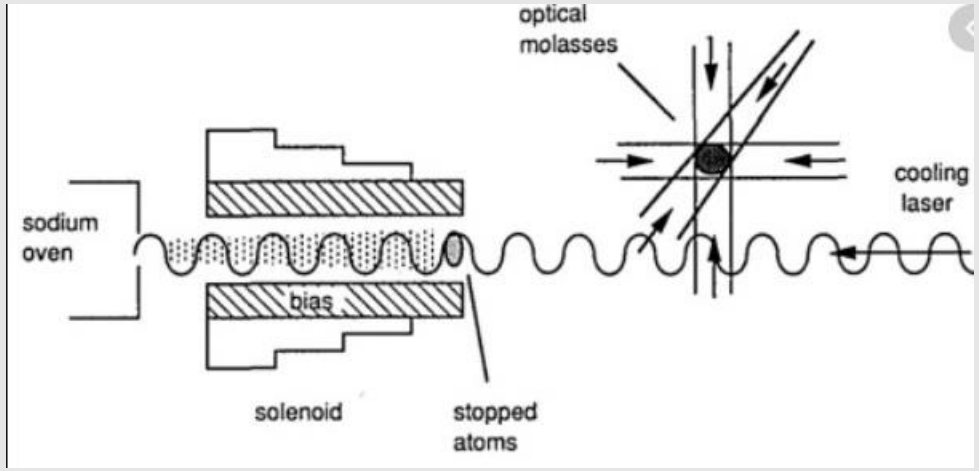
Na: $T_D = 240\ \mu K$

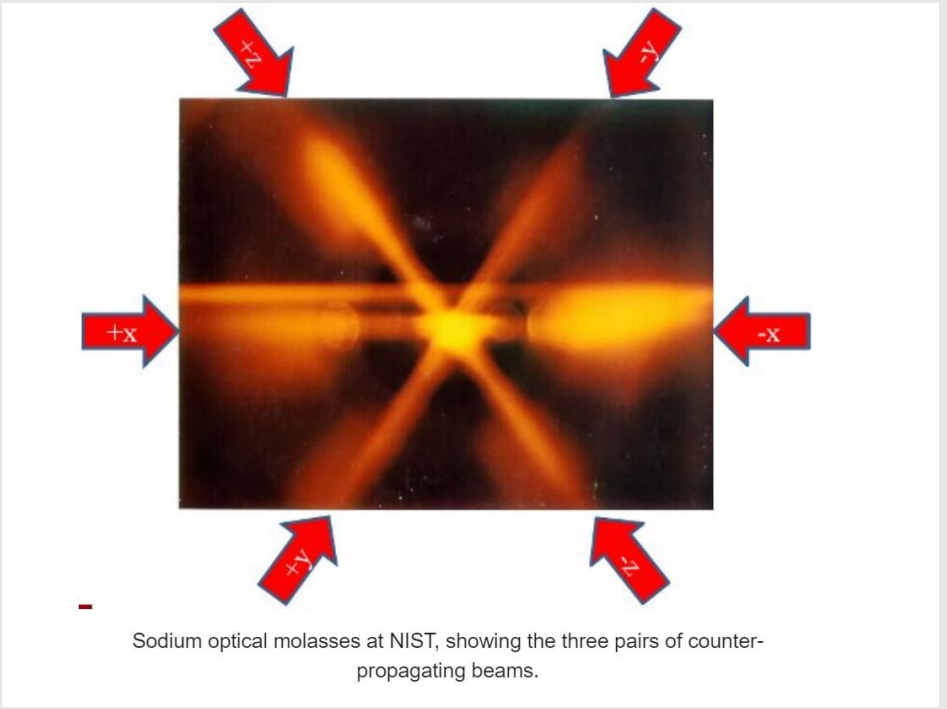
(most probable velocity). $v = 0.5$ m/sec

Xe: $T_D = 120\ \mu K$

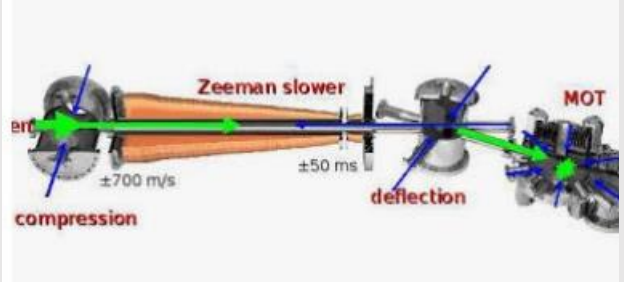
He: $T_D = 38\ \mu K$

K	Boltzmann's constant	1.38×10^{-23} JK^{-1}

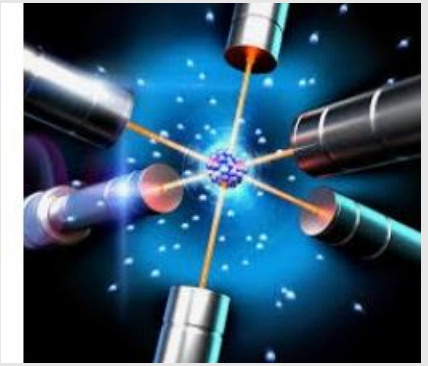


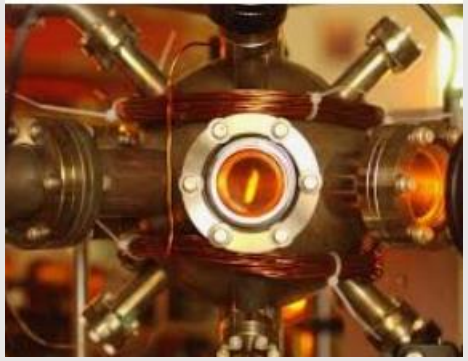


Sodium optical molasses at NIST, showing the three pairs of counter-propagating beams.

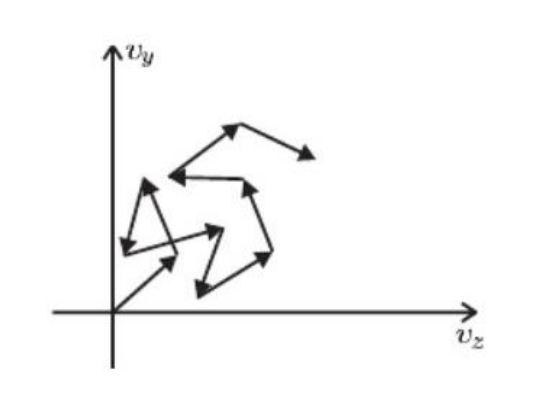


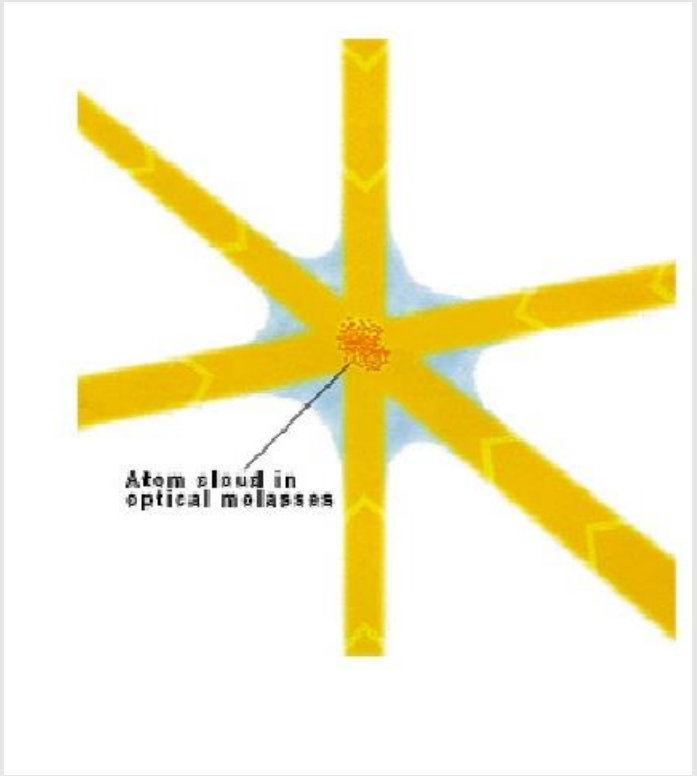
- Slowing
- Cooling
- Trapping



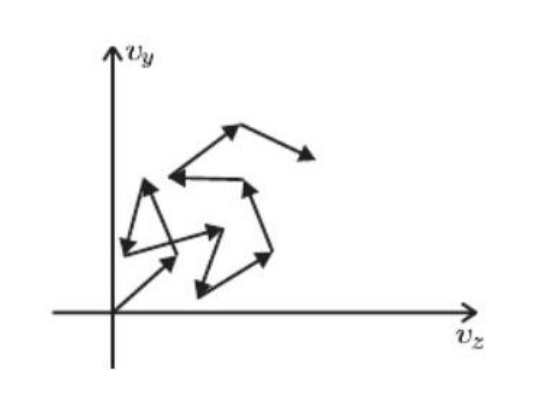


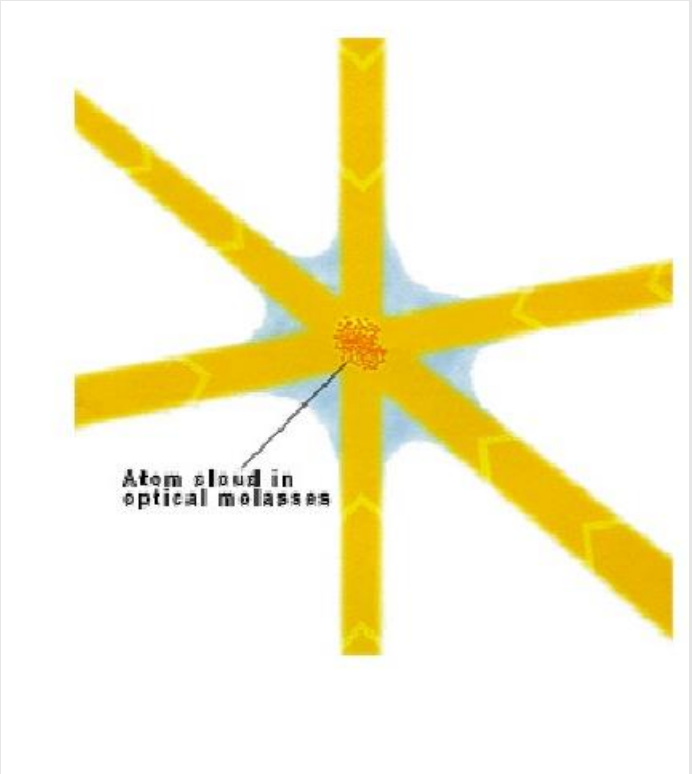
- For beams of 1cm^{2}radius, it takes several seconds for the atoms to diffuse out





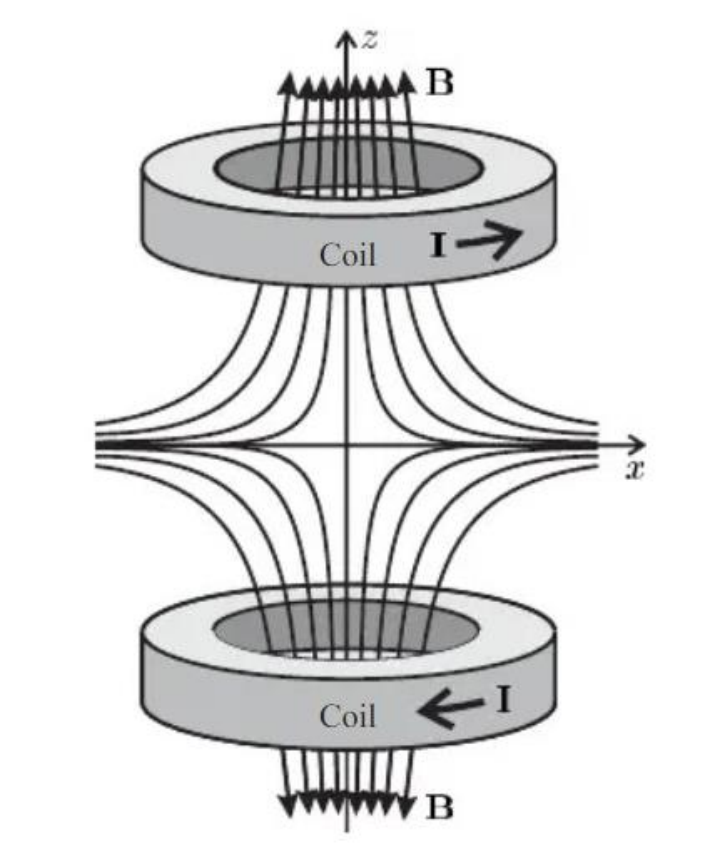
- For beams of $1cm^2$ radius, it takes several seconds for the atoms to diffuse out



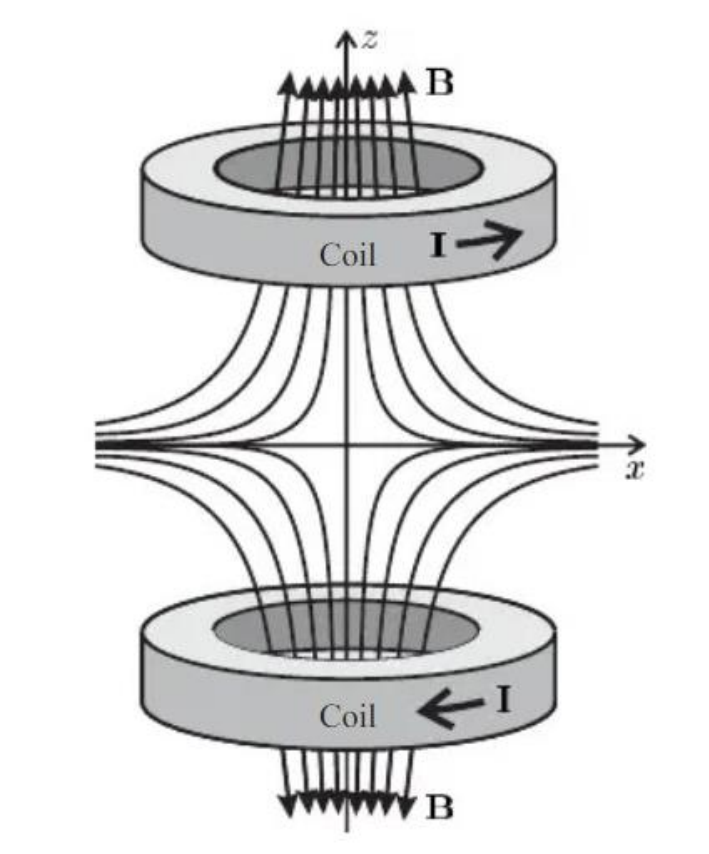


- Apply a force that will depend on the position of the atoms and prevent them from leaving the region

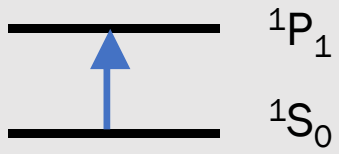
Magneto-optical trap



Magneto-optical trap

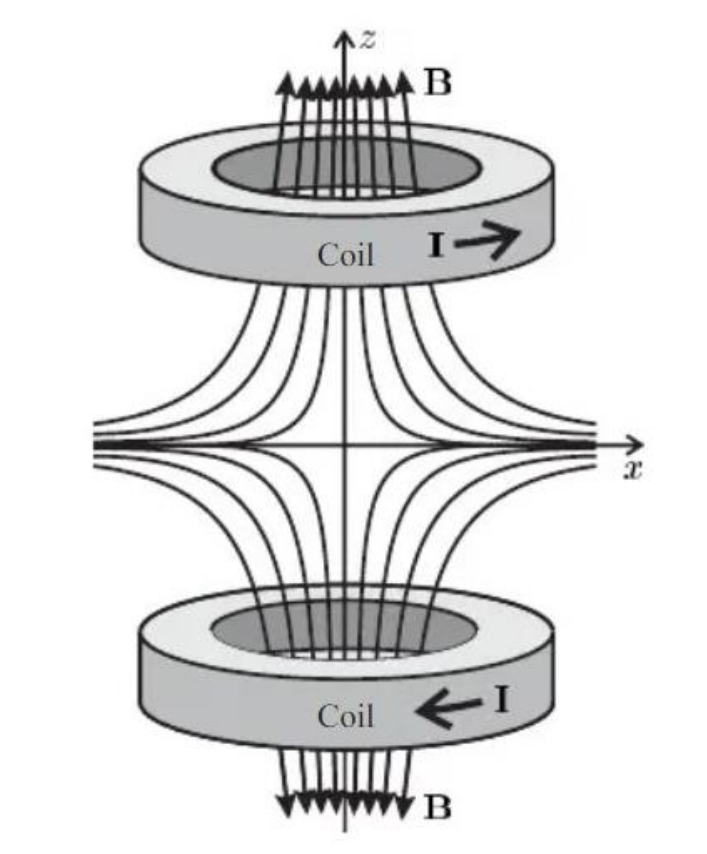


Take a J=0 to J=1 transition (e.g. $4s^2\ {}^1S_0 \rightarrow 4s4p\ {}^1P_1$ in Ca

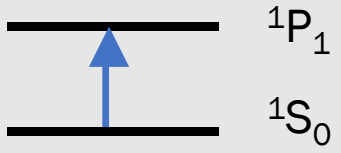


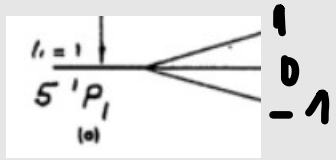
Remember Zeeman effect?

Magneto-optical trap

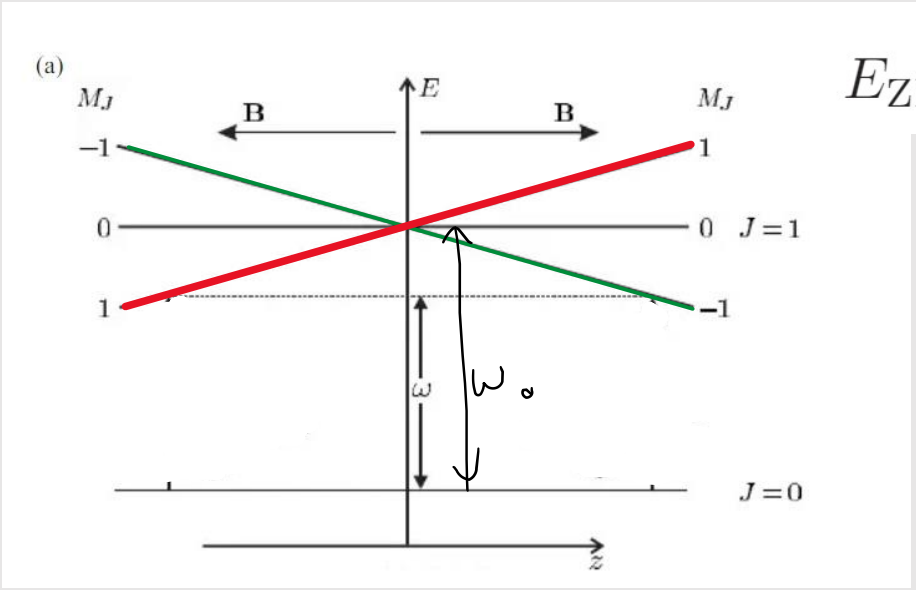


Take a J=0 to J=1 transition (e.g. $4s^2\ {}^1S_0 \rightarrow 4s4p\ {}^1P_1$ in Ca (422 nm))



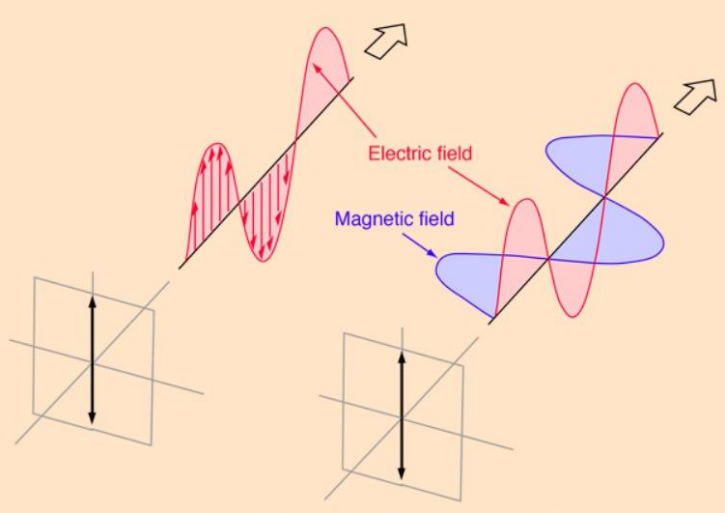


Remember Zeeman effect? $^1P_1$ will split into 3 levels.



$$E_{\mathrm{ZE}} = g_J \mu_{\mathrm{B}} B M_J$$

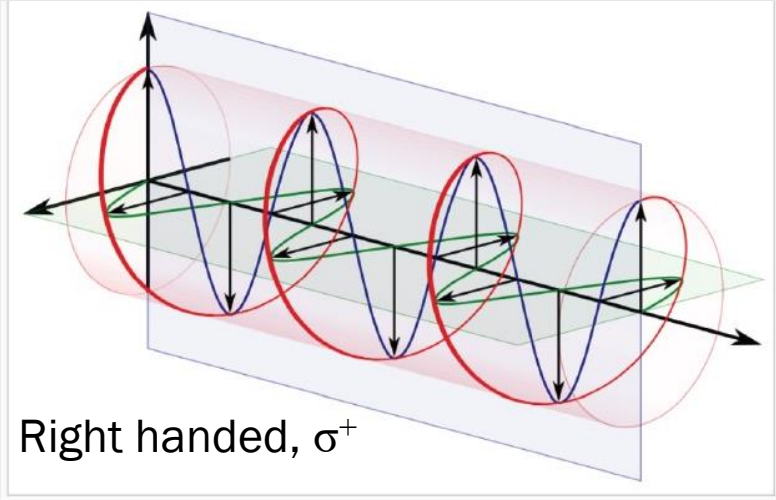
Light polarization



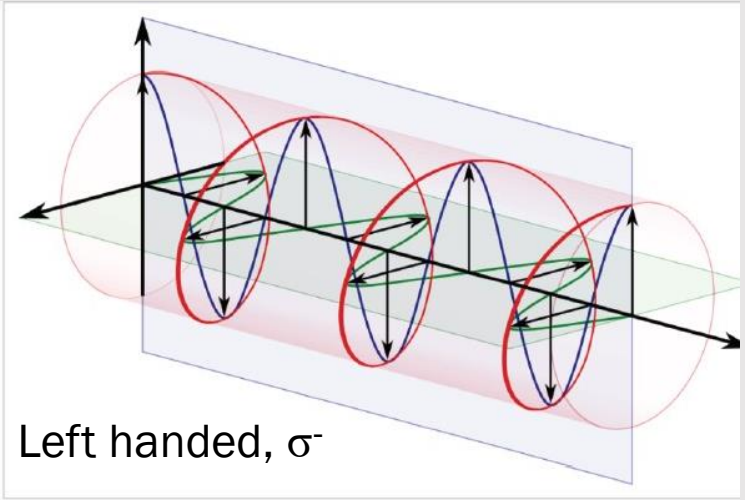
Linear polarization, π

Circular polarization, σ

.

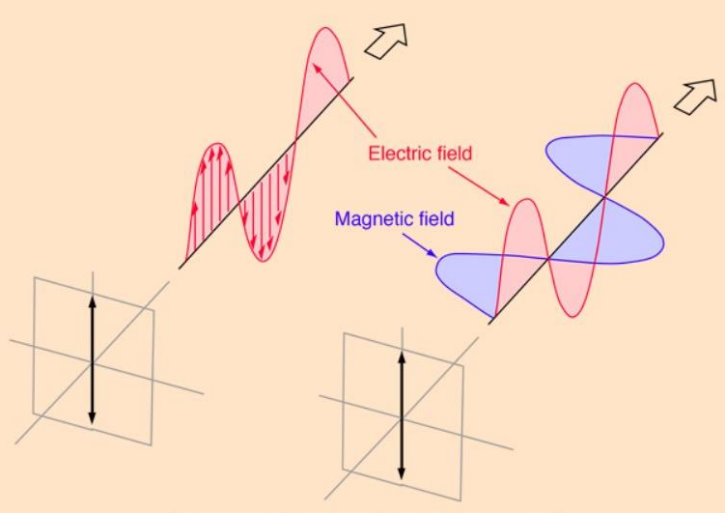


Right handed, σ^+



Left handed, σ^-

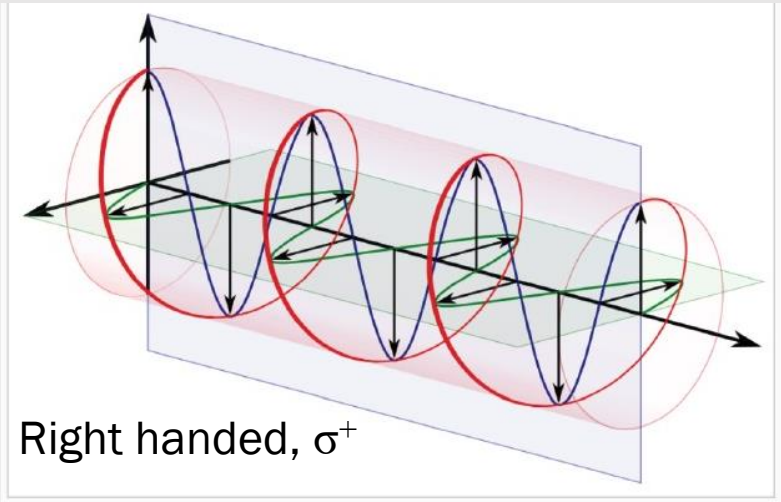
Light polarization



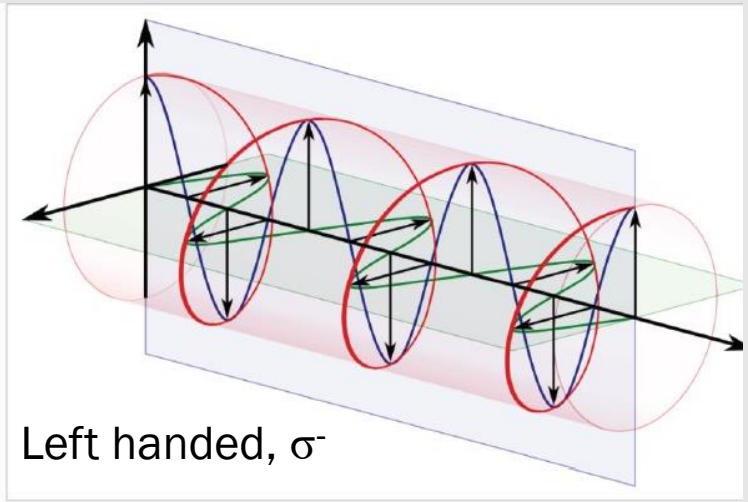
Linear polarization, π

Circular polarization, σ

Selection rules:
π: $0 \leftrightarrow 0$ ($\Delta M_J = 0$)
σ^+: $0 \leftrightarrow 1$ ($\Delta M_J = 1$)
σ^-: $0 \leftrightarrow -1$ ($\Delta M_J = -1$)

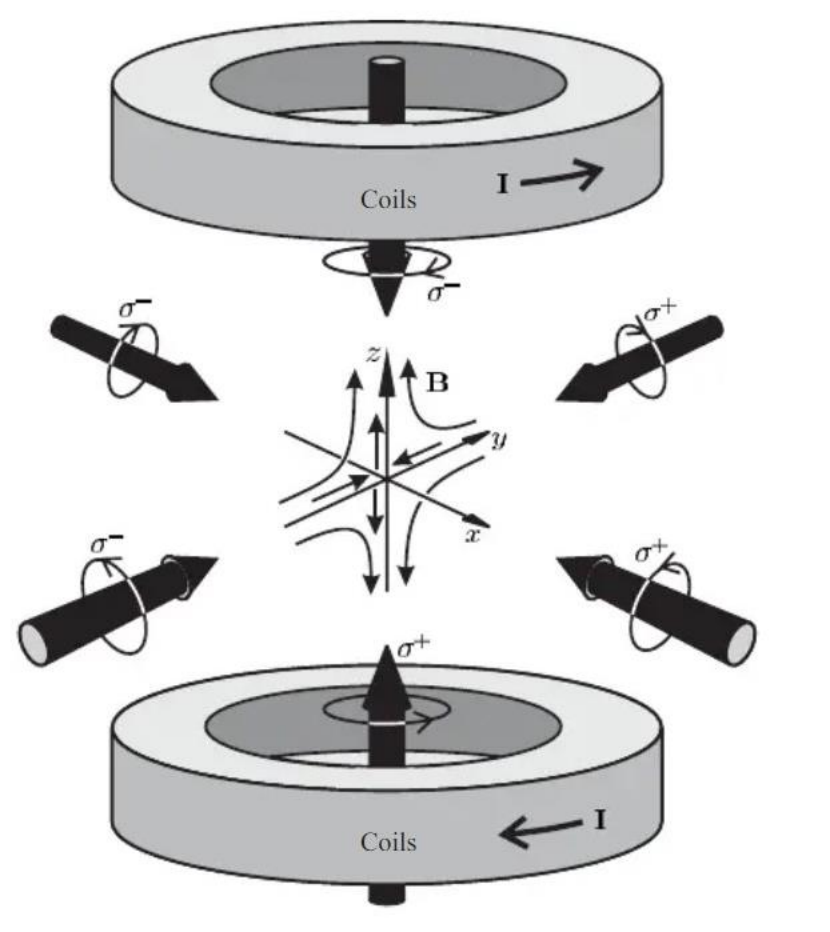


Right handed, σ^+

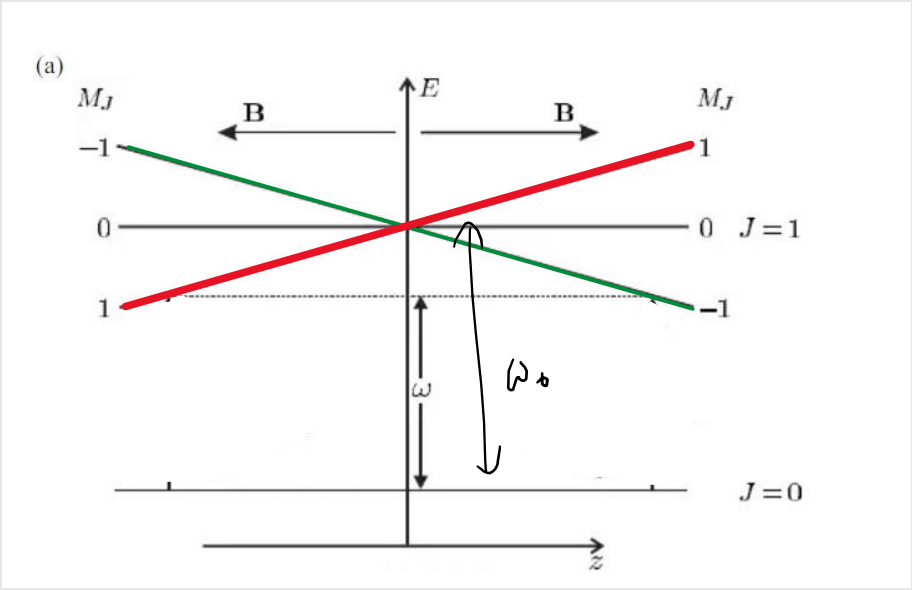


Left handed, σ^-

Magneto-optical trap

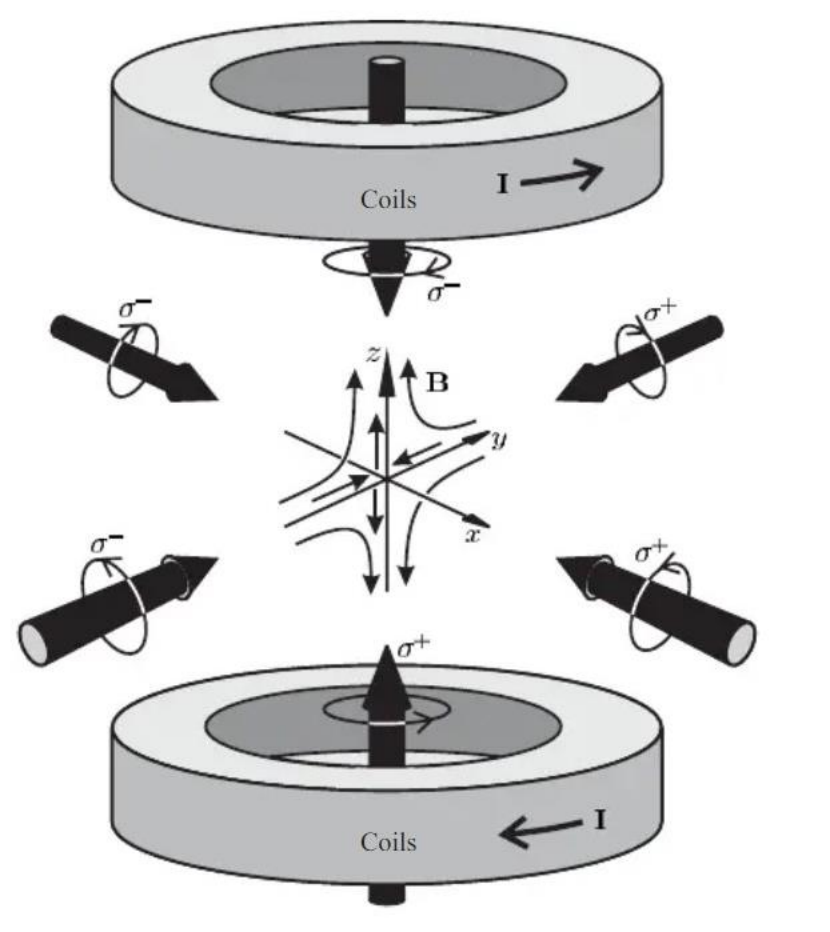


Selection rules:
π: $0 \leftrightarrow 0$ ($\Delta M_J = 0$)
σ^+: $0 \leftrightarrow 1$ ($\Delta M_J = 1$)
σ^-: $0 \leftrightarrow -1$ ($\Delta M_J = -1$)



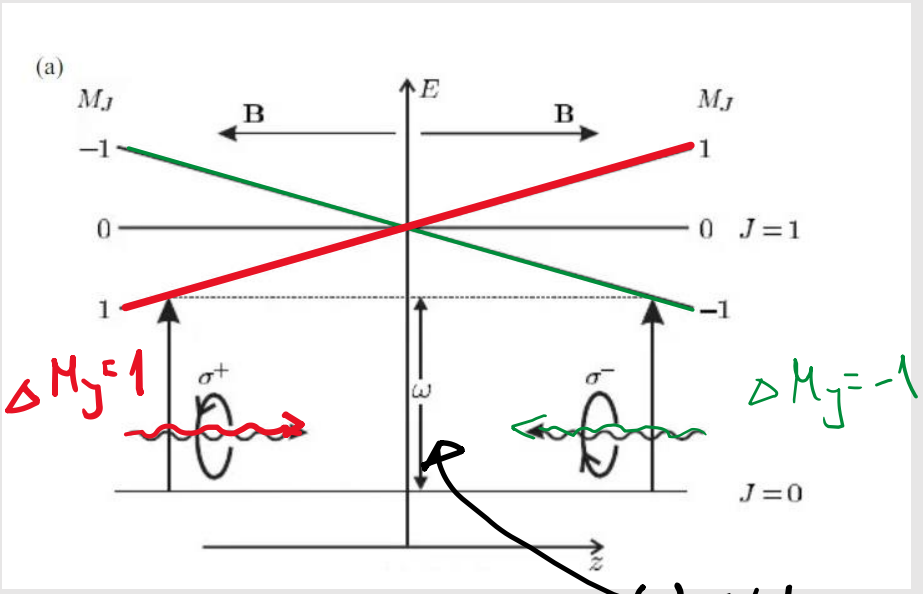
$$E_{\mathrm{ZE}} = g_J \mu_{\mathrm{B}} B M$$

Magneto-optical trap



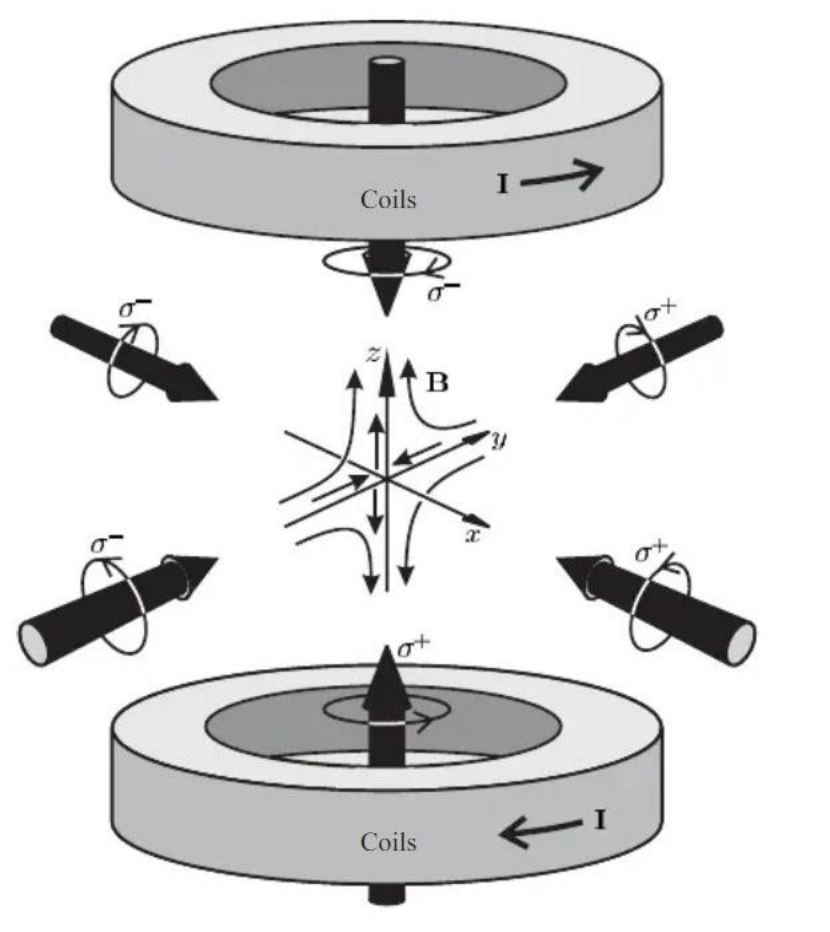
Selection rules:
π: 0←→0 ($\Delta M_J=0$)
σ^+: 0←→1 ($\Delta M_J=1$)
σ^-: 0←→-1 ($\Delta M_J=-1$)

$$E_{\mathrm{ZE}} = g_J \mu_{\mathrm{B}} B M_J$$



Magneto-optical trap

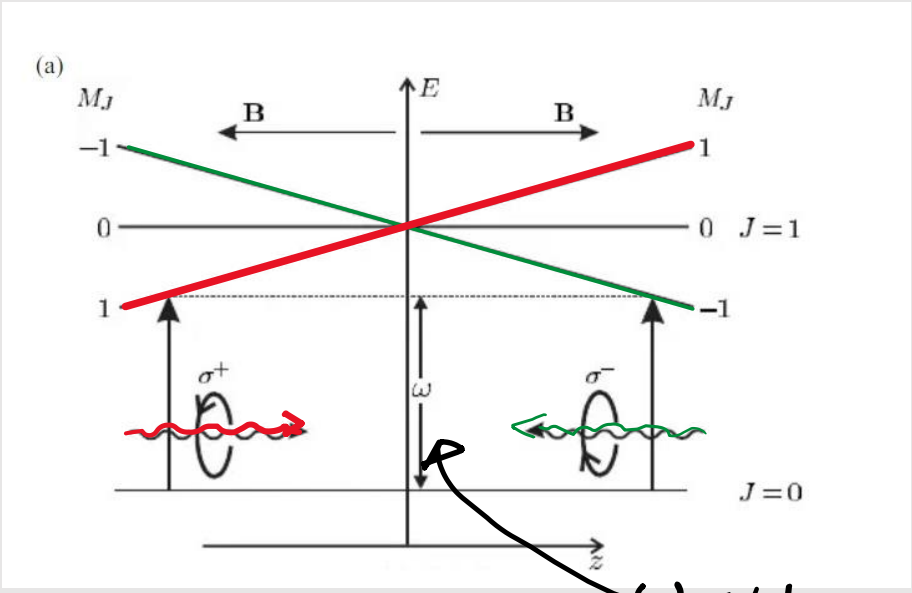
$$F_{\text{molasses}} = F_{\text{scatt}}(\omega - \omega_0 - kv) - F_{\text{scatt}}(\omega - \omega_0 + kv)$$
$$\simeq F_{\text{scatt}}(\omega - \omega_0) - kv\frac{\partial F}{\partial \omega} - \left[F_{\text{scatt}}(\omega - \omega_0) + kv\frac{\partial F}{\partial \omega}\right]$$
$$\simeq -2\frac{\partial F}{\partial \omega}kv\,. \qquad (9.1$$



$$F_{\text{MOT}} = F^{\sigma^+}_{\text{scatt}}(\omega - kv - (\omega_0 + \beta z)) - F^{\sigma^-}_{\text{scatt}}(\omega + kv - (\omega_0 - \beta z))$$
$$\simeq -2\frac{\partial F}{\partial \omega}kv + 2\frac{\partial F}{\partial \omega_0}\beta z\,. \qquad (9.30$$

$\Delta M_J = 1$

$\Delta M_J = -1$



$\omega < \omega_0$

$$E_{\text{ZE}} = g_J \mu_B B M$$

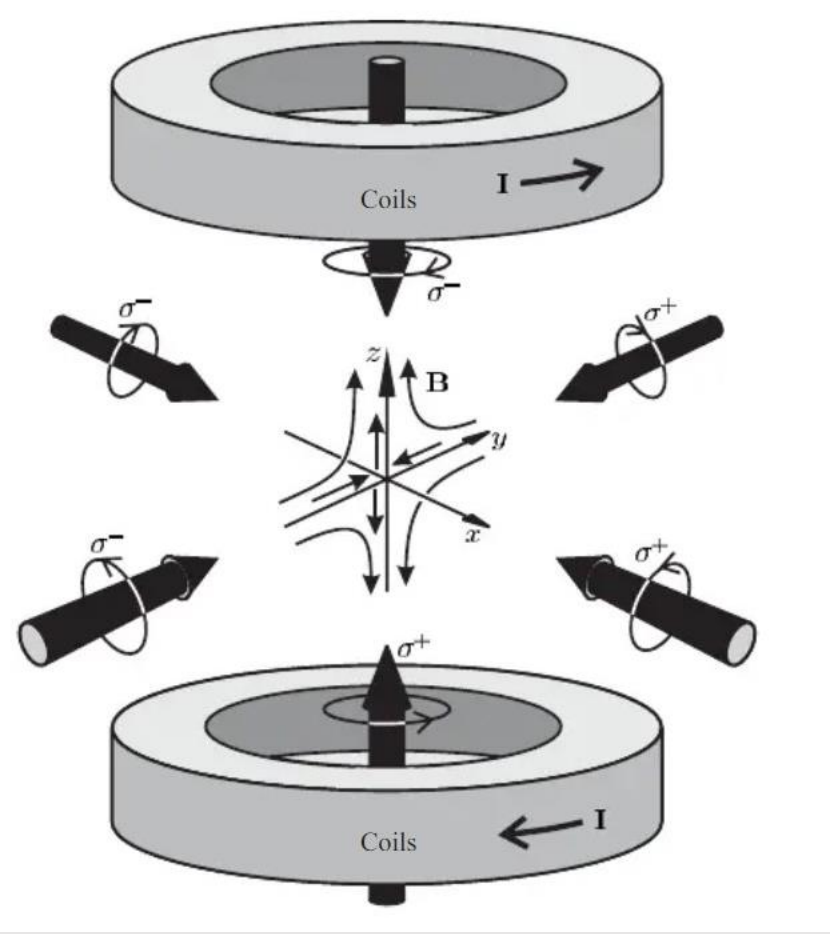
Magneto-optical trap

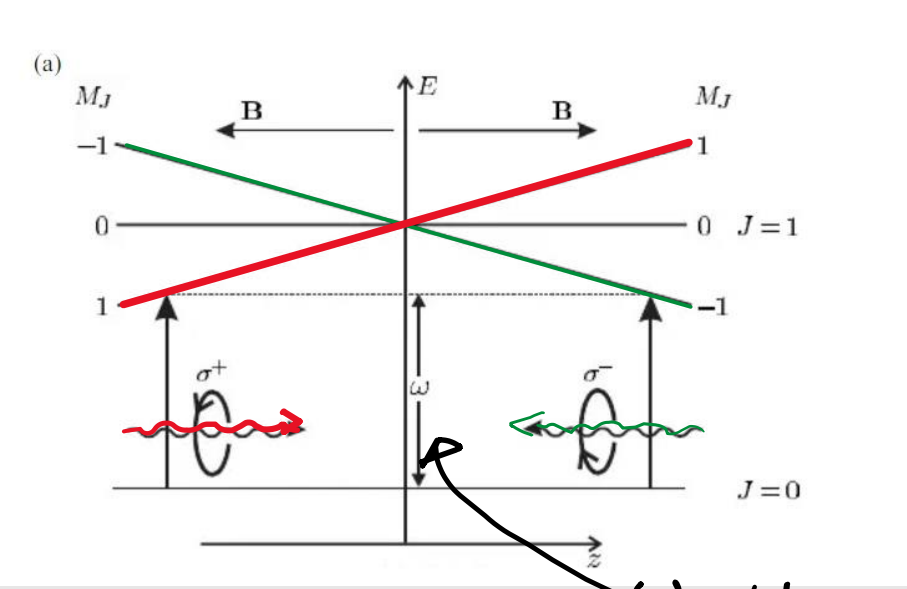
$$F_{\mathrm{MOT}} = F^{\sigma^+}_{\mathrm{scatt}}\left(\omega - kv - (\omega_0 + \beta z)\right) - F^{\sigma^-}_{\mathrm{scatt}}\left(\omega + kv - (\omega_0 - \beta z)\right)$$
$$\simeq -2\frac{\partial F}{\partial \omega} kv + 2\frac{\partial F}{\partial \omega_0}\beta z\,. \qquad (9.30$$

$$\beta z = \frac{g\mu_{\mathrm{B}}}{\hbar}\frac{\mathrm{d}B}{\mathrm{d}z} z$$

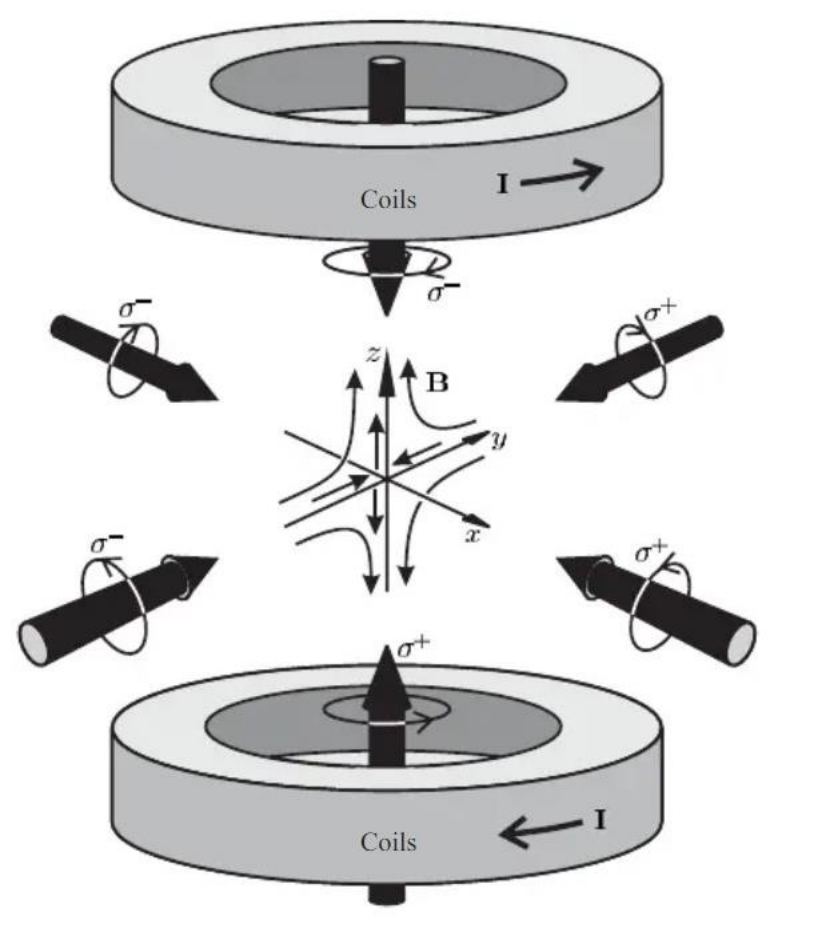
$$E_{\mathrm{ZE}} = g_J \mu_{\mathrm{B}} B M_J$$

E = ħω





Magneto-optical trap

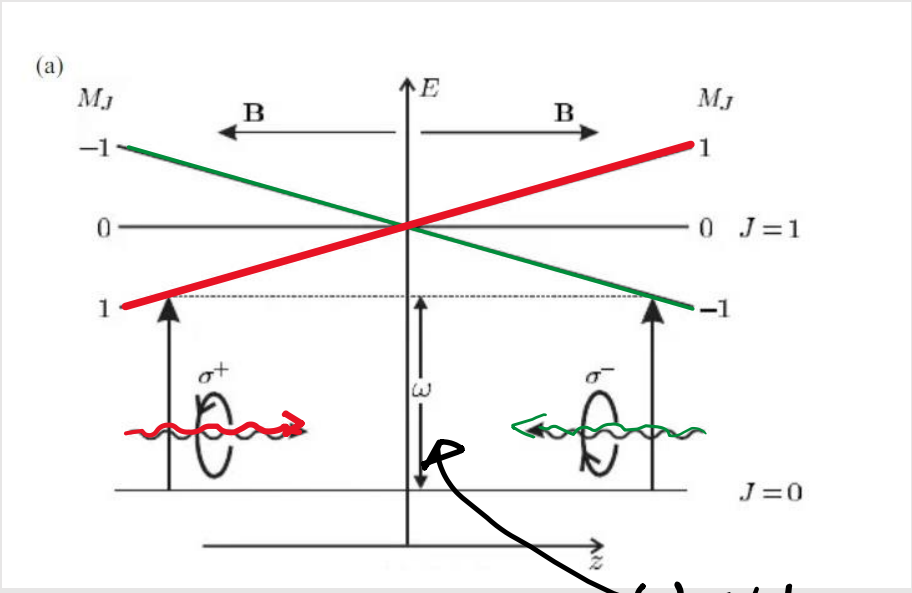


$$F_{\mathrm{MOT}} = F^{\sigma^+}_{\mathrm{scatt}}\left(\omega - kv - (\omega_0 + \beta z)\right) - F^{\sigma^-}_{\mathrm{scatt}}\left(\omega + kv - (\omega_0 - \beta z)\right)$$
$$\simeq -2\frac{\partial F}{\partial \omega} kv + 2\frac{\partial F}{\partial \omega_0}\beta z\,. \qquad (9.30$$

$$\beta z = \frac{g\mu_{\mathrm{B}}}{\hbar}\frac{\mathrm{d}B}{\mathrm{d}z} z$$

$$F_{\mathrm{MOT}} = -2\frac{\partial F}{\partial \omega}(kv + \beta z)$$
$$= -\alpha v - \frac{\alpha\beta}{k} z\,.$$

$$\left(\alpha = 2\cdot\frac{\partial F}{\partial \omega}\cdot k\right)$$



$$E_{\mathrm{ZE}} = g_J \mu_{\mathrm{B}} B M$$

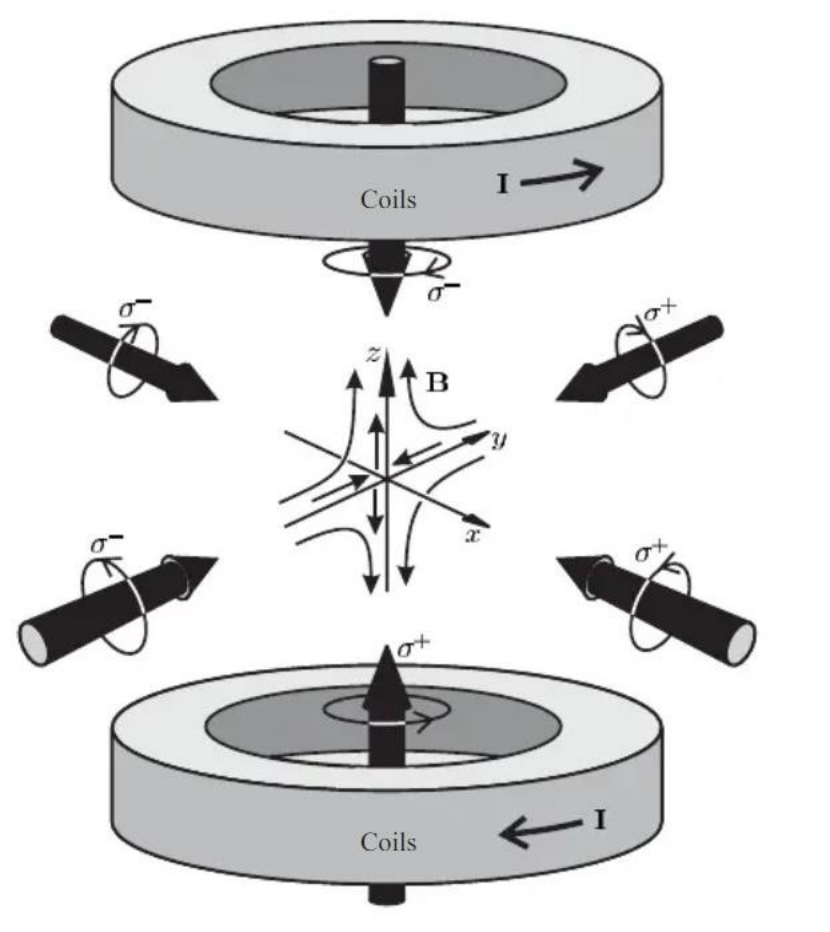
$$\delta = \omega - \omega_0$$

$$\partial F/\partial \omega_0 = -\partial F/\partial \omega$$

Magneto-optical trap

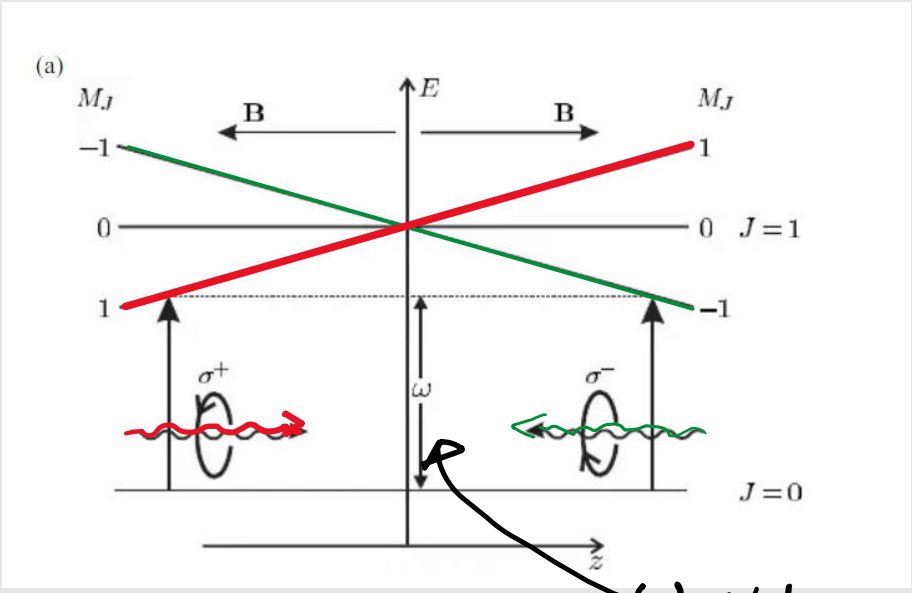
velocity-dependent slowing

position-dependent force

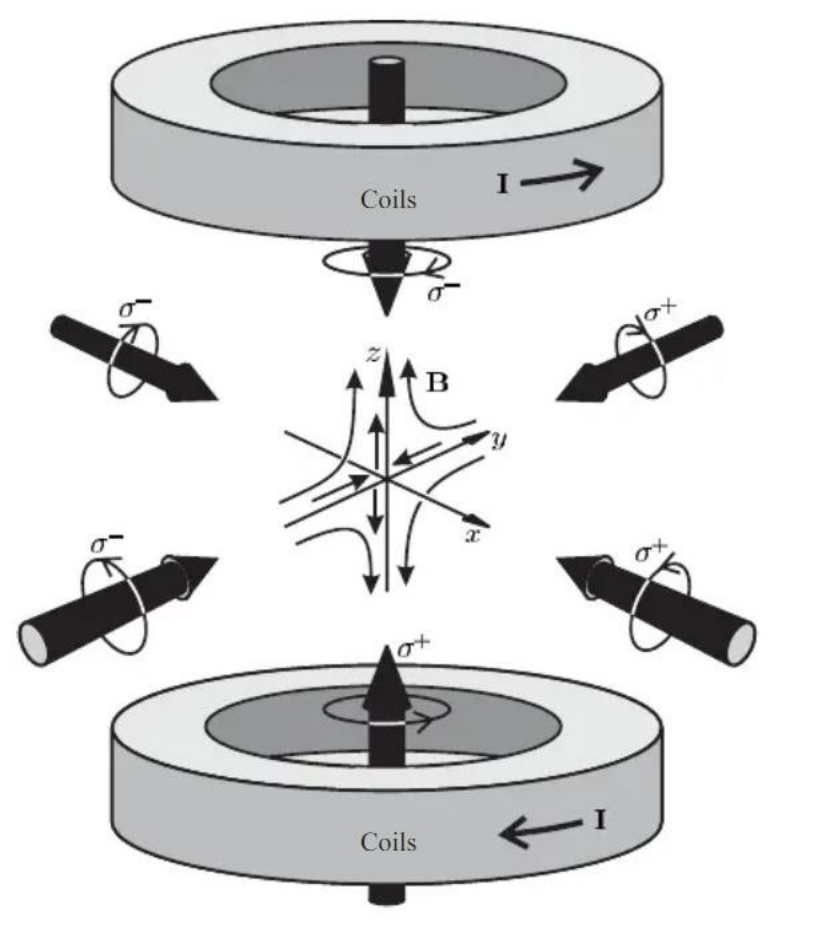


$$\beta z = \frac{g\mu_B}{\hbar}\frac{dB}{dz}z$$

$$F_{MOT} = -2\frac{\partial F}{\partial \omega}(kv + \beta z) = -\alpha v - \frac{\alpha\beta}{k}z.$$



Magneto-optical trap



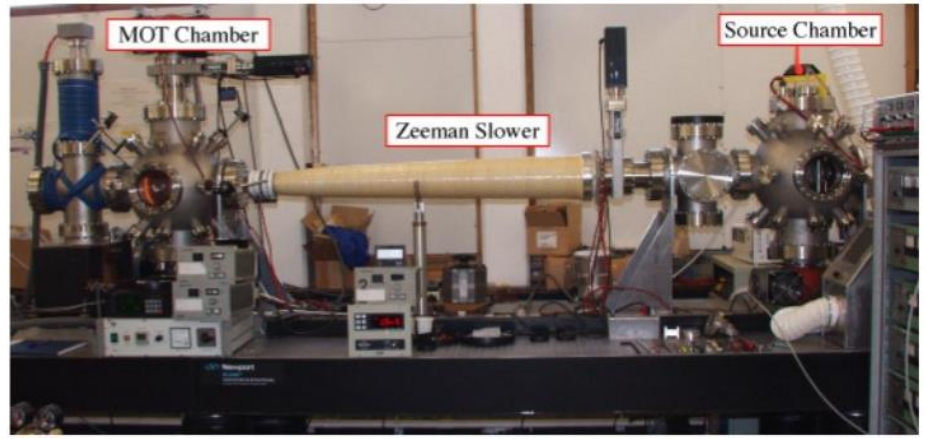
For Na:

v_c(MOT)≈70 m/s

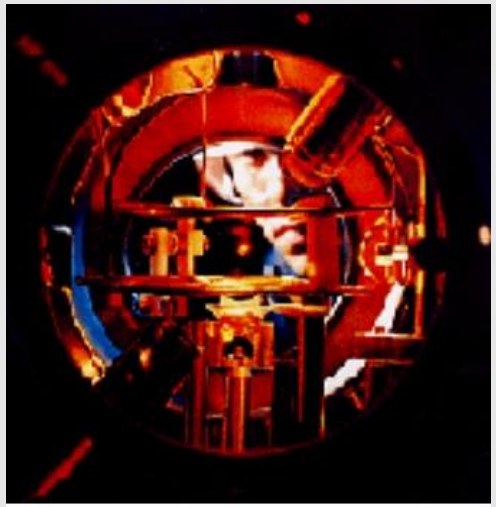
vs.

v(molasses) ≈6 m/sec

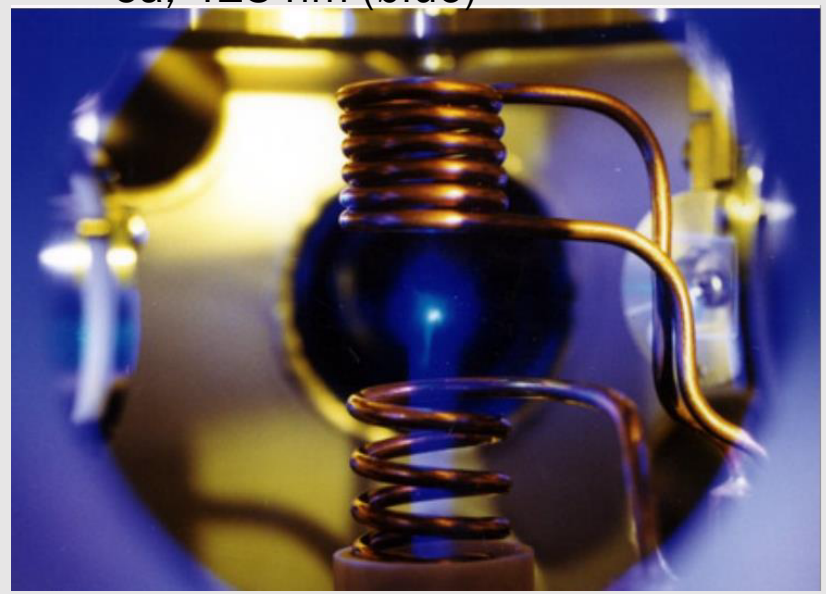
10000000 atoms collected



Na, 589 nm (yellow)

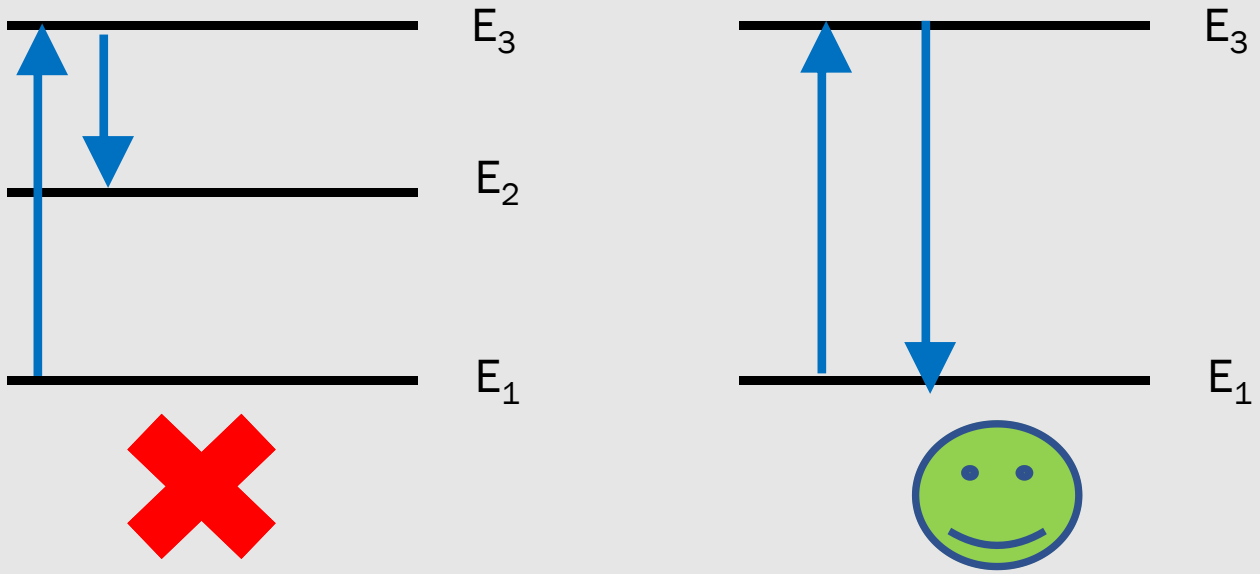


Ca, 423 nm (blue)

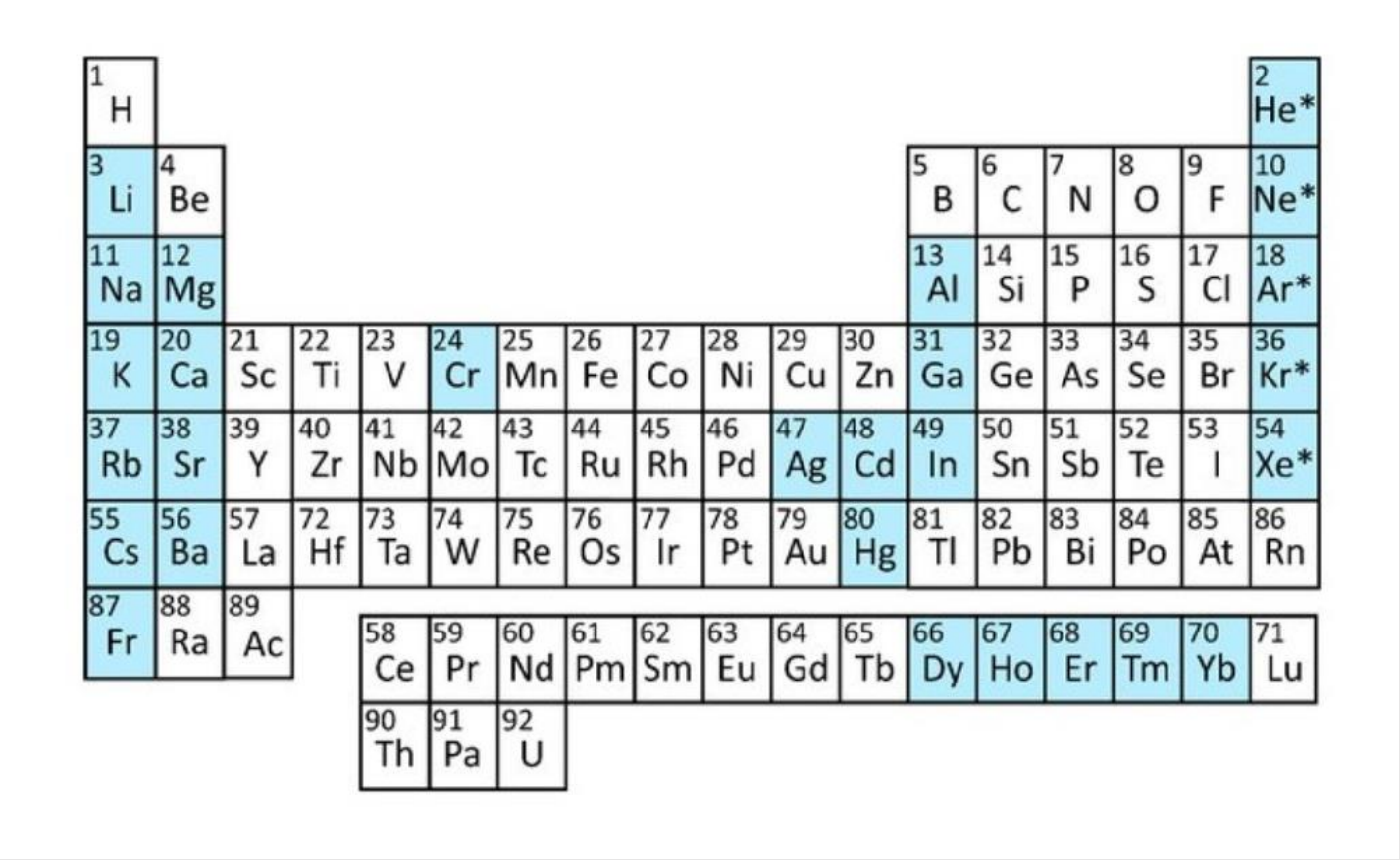


- What makes a good atom (transition) for laser cooling?

- What makes a good atom (transition) for laser cooling?
  - Two level system (effective two-level system), for closed cycling (compare to laser, which requires 3 levels!)
  - Short lifetime of upper state (for efficiency!)
  - Reasonable transition frequency (ideally optical range, where lasers work best)



Atoms that were laser-cooled

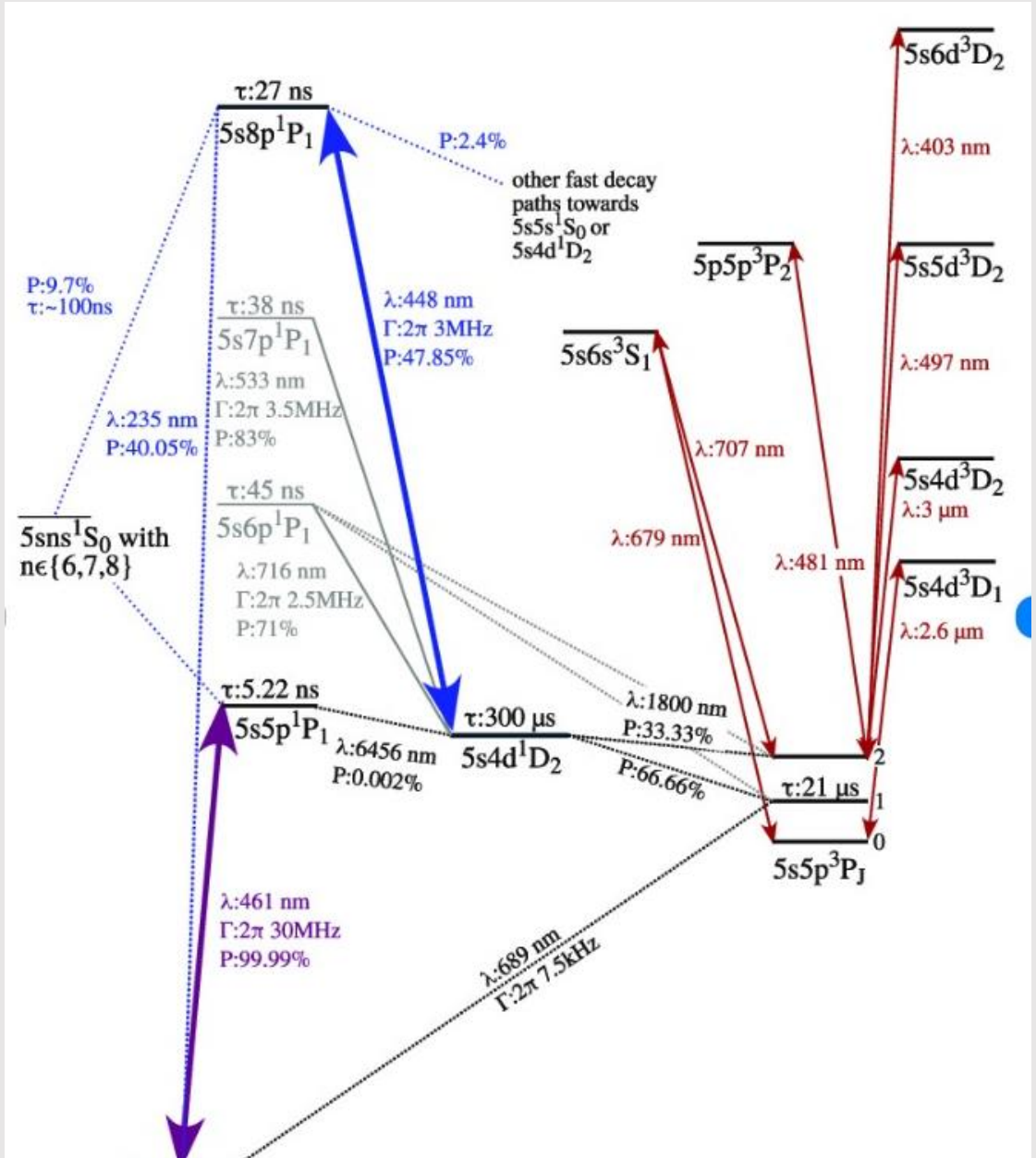


not so simple...

- Here, 2 lasers are needed

- Why do we need the second laser?

Why not repump directly to the excited state used in the cooling?



Optical pumping of $5s4d\,^1D_2$ strontium atoms for laser cooling and imaging

Jens Samland[1,2], Shayne Bennetts[1], Chun-Chia Chen (陳俊嘉)[1], Rodrigo González Escudero[1], Florian Schreck[1,3], and Benjamin Pasquiou[1,3,4,5,*]

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